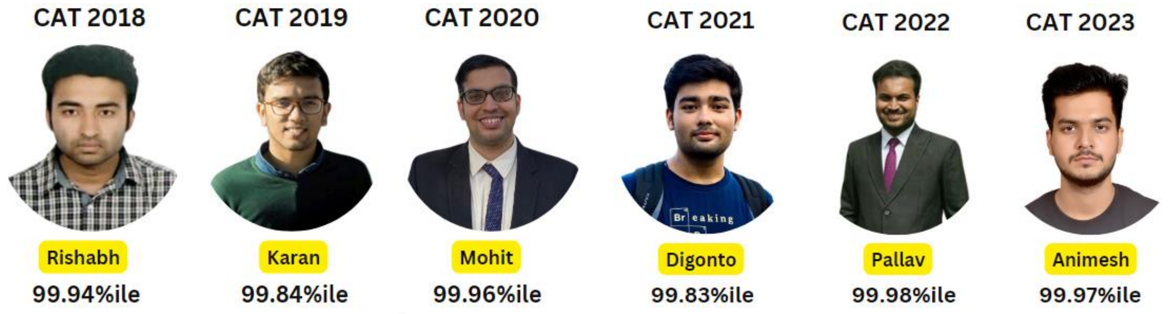
Quantifiers CAT ACADEMY

ALL GEOMETRY NOTES & SHORTCUTS COMPILED



Quantifiers CAT Academy

Started in 2019

Produced 28, 99+ %ilers out of the total ~275 students taught

Managed by the three musketeers Rohit Kanwar, Sahil Arora & Jasneet Dua who are the founders as well as the Expert Teaching facility with over 20+ years of experience

Helped more than 1 lakh aspirants achieve their dreams through their teachings.

Quantifiers is best known for their pedagogy, their emphatic attitude toward every single student, and for the passion with which every single element is driven

Be the part of legacy, join Quantifiers at www.quantifiers.in

QUANTIFIERS CAT ACADEMY

India's Best CAT Academy, now also offline in Chandigarh

India's most trusted online/offline coaching

CAT TOPPERS FROM CHANDIGARH

Rishabh
CAT: 99.94

Ishan
CAT: 99.65

Anubhav
CAT: 99.46

Manshul Garg
CAT: 99.01

Paras
CAT: 98.99

Akshita
CAT: 98.95

Aarushi,PEC
CAT: 98.77

Swarnima, PEC
CAT - 98.60

- *Personal Mentoring Dedication to 24*7 doubt solving*
- *Online Backup of Classes*
- *Test Series with Video Solutions*
- *Includes CAT as well as OMET prep*
- *Smaller Batch Sizes for individuals holistic growth*

Past year results

Pallav
99.98

Amartya
99.91

Anik
99.83

Aman
99.83

Is this your dream scorecard?

Section: Verbal Ability & Reading Comprehension		Section: Data Interpretation & Logical Reasoning		Section: Quantitative Ability		Total	
Scaled Score	Percentile	Scaled Score	Percentile	Scaled Score	Percentile	Overall Scaled Score	Overall Percentile
42.17	99.46	38.36	99.90	48.82	99.93	129.35	99.98

Book free demo for yourself
Call/WA - 9988656560

Join our WhatsApp FREE 24/7 Doubt Solving Group (if group is full, follow us on Instagram @quantifiers_cat for the new link)
[Click here to join WhatsApp group](https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA)
https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA

Important links for all aspirants

ALL QA FORMULAS, all sections –

Click here to watch playlist of entire QA

Complete QA Concepts by Quantifiers -

Click here to Watch Video Concepts

350+ DILR Sets -

Click here to watch playlist of DILR

VA Concepts by Quantifiers -

Click here to Watch Video Concepts

Free TESTS for all aspirants -

Free QA/VA Tests

RC 99 Video Solutions -

Click here to Watch Video Concepts

How to score 99%ile in DILR?

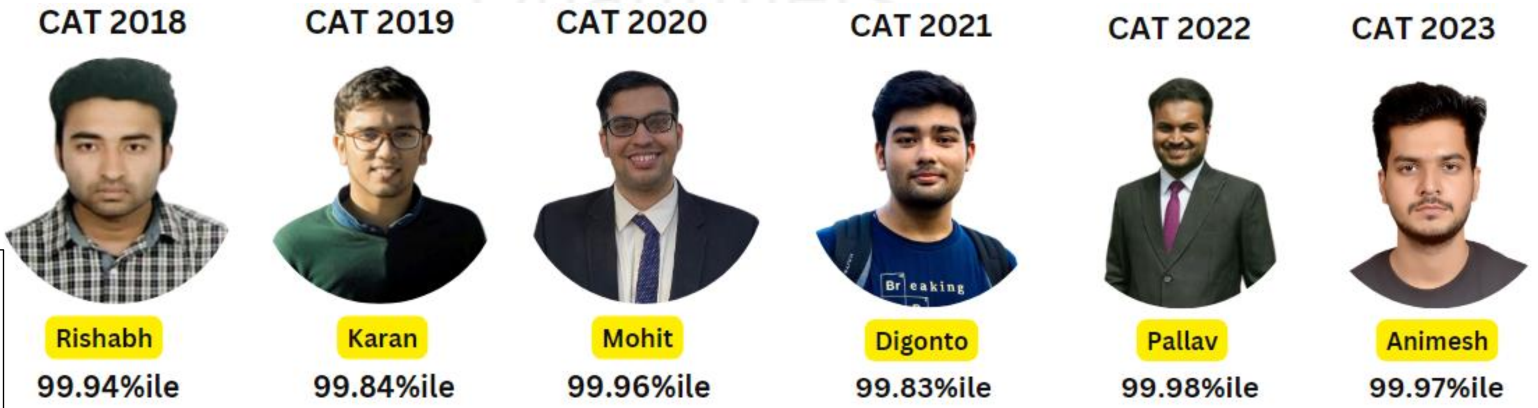
Click here to watch the video

Get Daily Targets on your WhatsApp
or Copy paste this URL:
https://forms.gle/oSih8jQCermJq71q9

BASIC DEFINITION

- Point:- Point is a circle with 0 radius.
- Line:- line is a set of point having only length in a fixed direction. Minimum two point is required to draw a line.

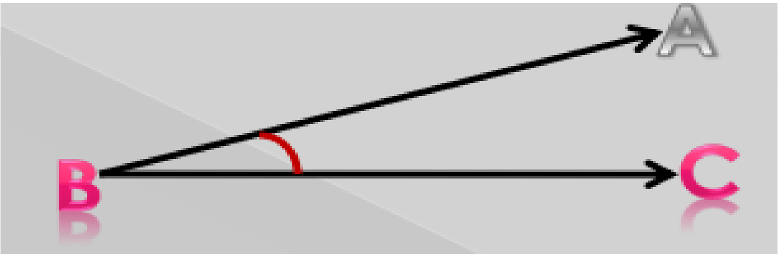
- Ray:- Ray has a fixed starting point but has not a fixed end point.



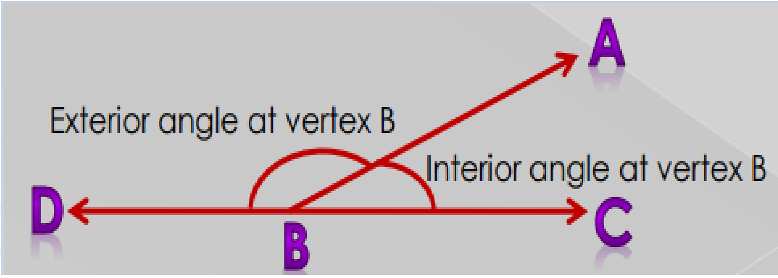
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

ANGLE

- Inclination between two lines is called angle

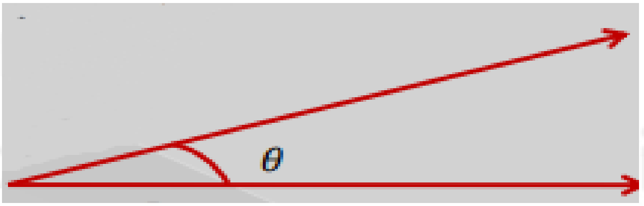


- The sum of interior and exterior angle at a line is 180°

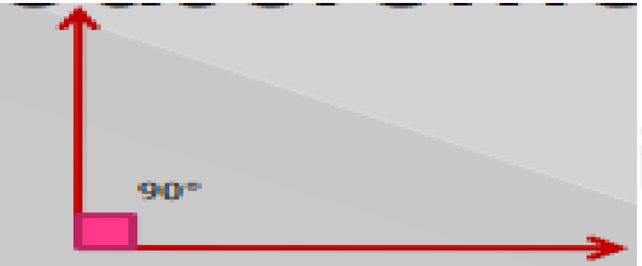


TYPES OF ANGLE

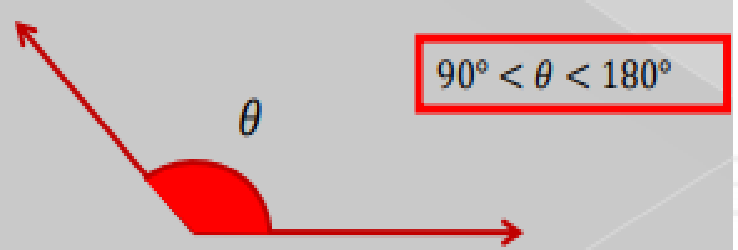
- Acute angle:- It is greater than 0 and smaller than 90.

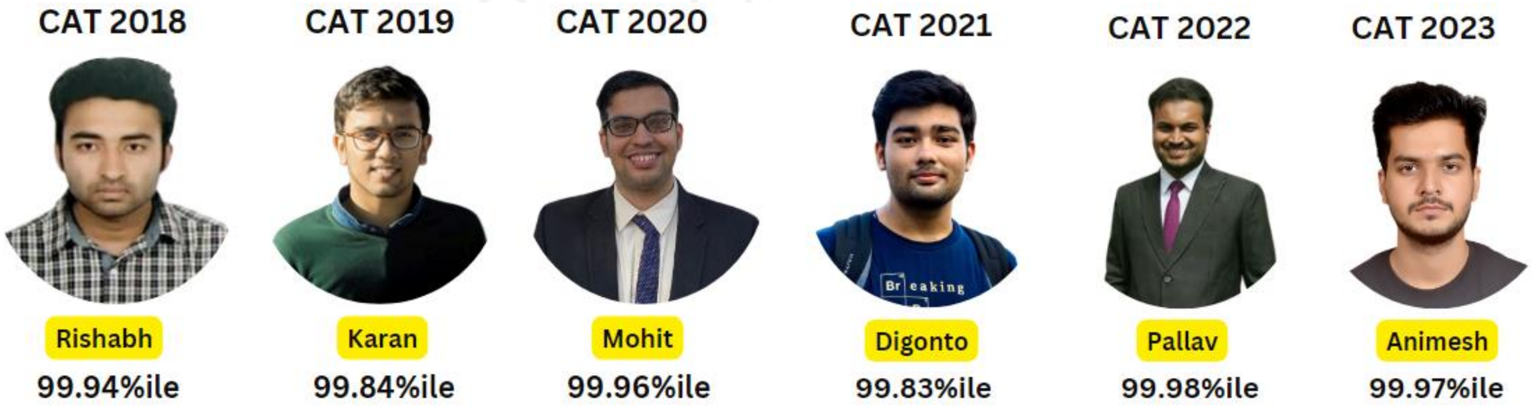


- Right-Angle:- The measurement of right angle is 90 .



- Obtuse Angle:- It is greater than 90 and smaller than 180 .

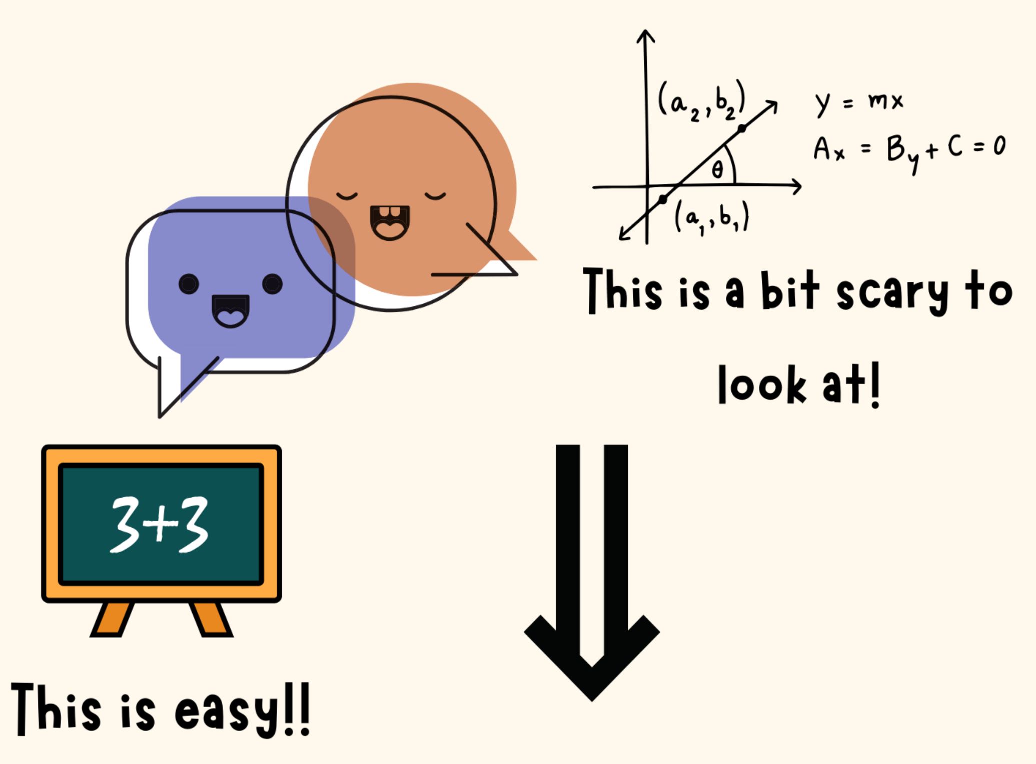




QUIZ TIME

Join Our FB Group for Daily updates, reads & Practice

[Click here to join our Facebook Group]



Complete QA Course to cover basics

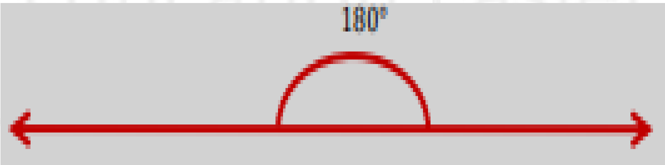
Complete CAT Quant Course:
https://www.youtube.com/playlist?list=PL1glN7tXDkWgkElsdG7za3cxcomVc9Y5e

Advance level QA Questions
QA Quiz Playlist:
https://www.youtube.com/playlist?list=PL1glN7tXDkWhRYY_0EuzQlCG44BpQZa3g

Some XAT level questions
XAT Series:
https://www.youtube.com/playlist?list=PL1glN7tXDkWj7E3p09aN3ZEpwrb3EPI8d

TYPES OF ANGLE

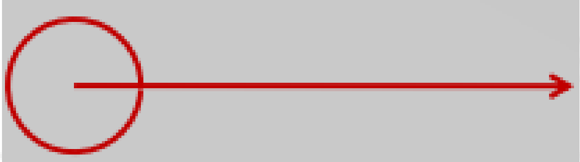
- Straight Angle:- The measurement of straight angle is 180 .

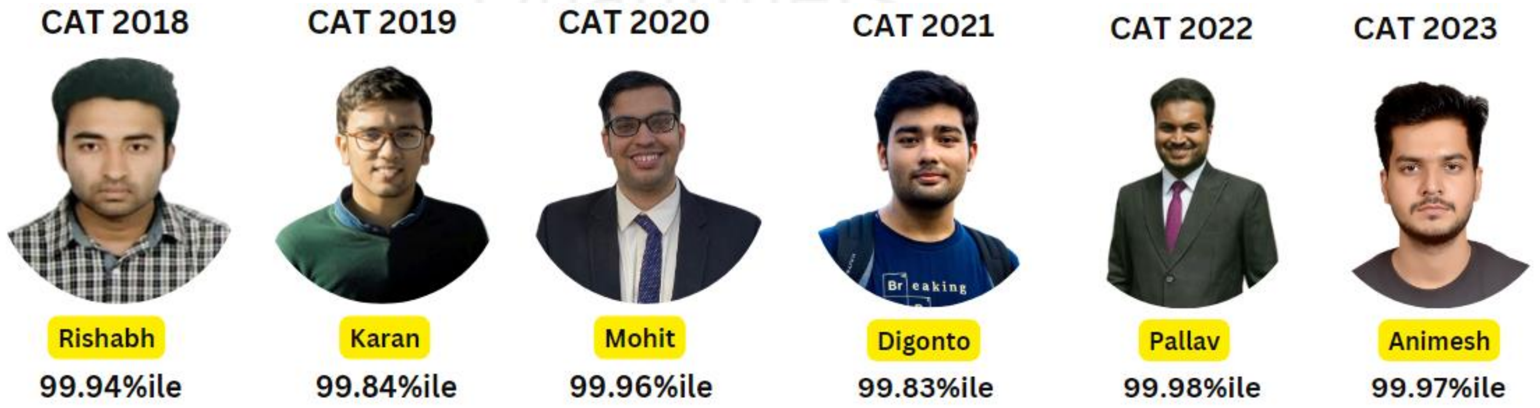


- Reflex Angle:- It is greater than 180° and less than 360° .

$$\boxed{180° < \theta < 360°}$$

- Complete Angle:- The measurement of complete angle is 360° .

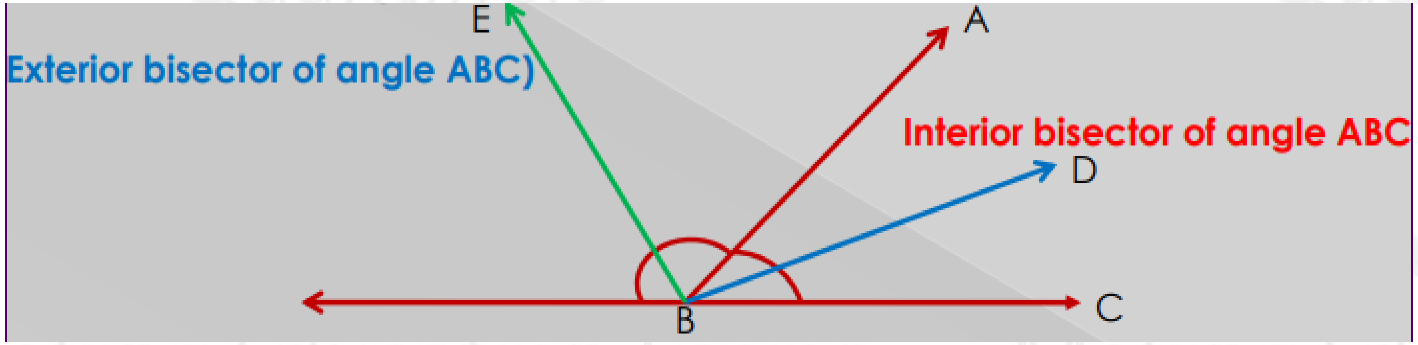




Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

ANGLE BISECTOR

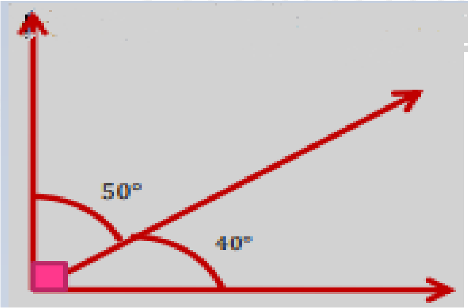
- A line which bisect the angle in equal part is called angle bisector.



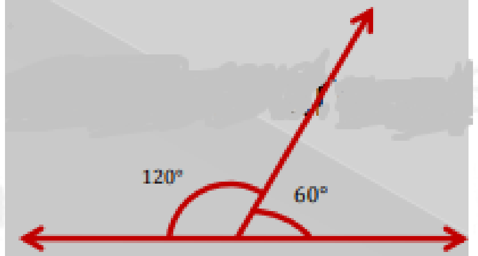
- The angle EBD made between internal bisector and external bisector is always 90° .

COMPLEMENTARY AND SUPPLEMENTARY ANGLE

- Two such angles which sum is 90 degree are called complementary angle.



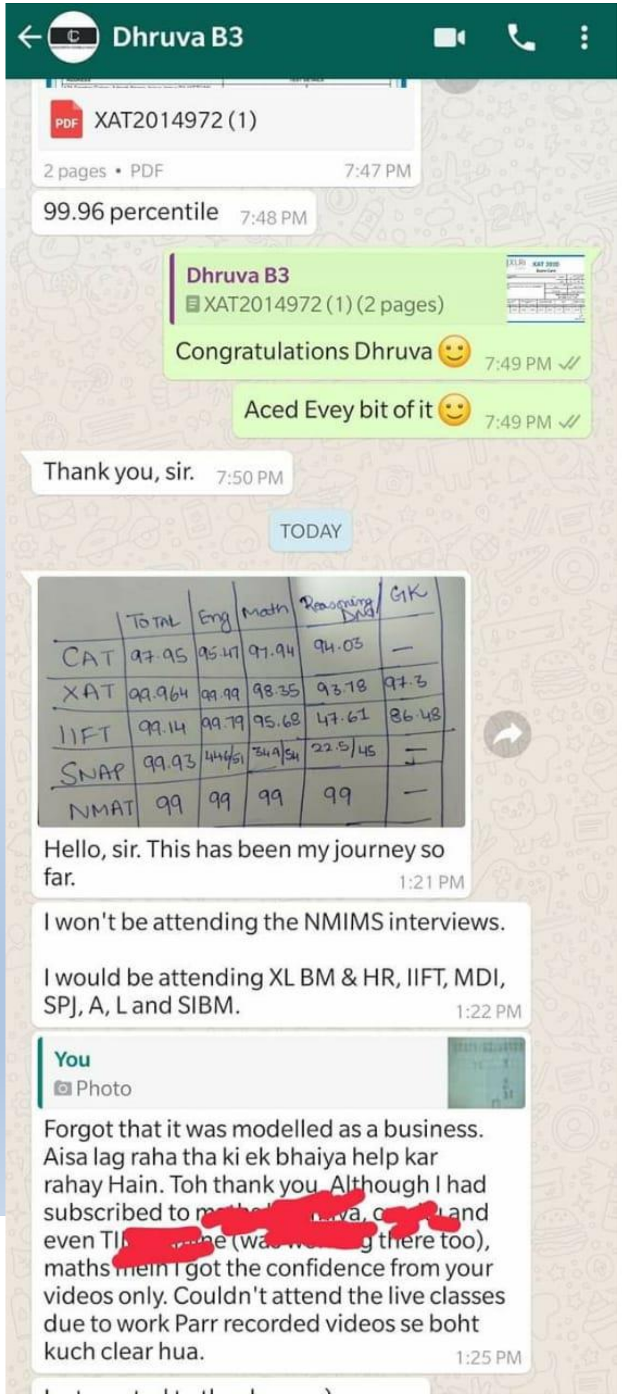
- Two such angles which sum is 180 degree are called supplementary angle.



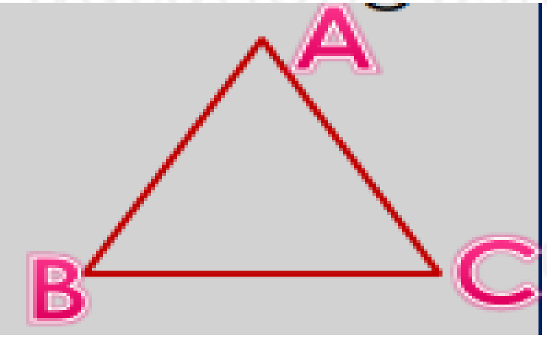
If CAT is an ART, Quantifiers is the Artist

Dhruva Sareen
IIM A
CAT: 97.95
IIFT: 99.14
NMAT:236
XAT: 99.96
SNAP: 99.93
Lawyer

*Well, I'm not asking you to believe my words but go and witness yourself. I was a member of Quantifiers Whatsapp group, I was simply taken aback by the way every doubt was solved instantaneously, how can someone be awake 24*7. I have personally seen doubts getting solved at 2/3/4 in the night. I took up the course in June and my journey was never the same, I can ensure you the amount of personal attention you get there, you can't get anywhere. I mentioned it to Sahil Sir too that I never felt that there is a teacher teaching at the other side, it was always like, an elder brother, who has been guiding/scolding me, the motivational sessions, the workshops every aspect that you can think of was taken care to perfection.QA was my biggest fear, I remember scoring in single digits in my initial mocks, but thanks to Quantifiers for believing in me, making me realize that nothing is impossible. It will not be an injustice to say that these guys worked harder than us, pushing us every single day, giving us regular tasks, DILR sets, RCs.The most important part is, we were never taught the questions which were irrelevant from exam perspective, sometimes I found some questions in mocks which were alien to me, but thanks to the mentoring, I was always aligned to what actually came in CAT.QA is a tough section for non-engineers they say, I disapprove that, but trust me, the more classes I attended, the less I believed the statement, we were always told that "P&C nahi ayega aisa, Binomial me time waste naai kro ………" Special focus on Arithmetic, Geometry and Algebra played a pivotal role in my scores, The booster sessions were cherry on the cake, it was like every concept knocked our brain and entered.DILR focus was a different level, the variety we practiced made sure that we have covered everything that could be, as a result I was able to solve 4 sets on D-day. Finally, I will again ask you to just join Quantifiers without a doubt and witness the magic.*
Thanking Quantifiers
Adios



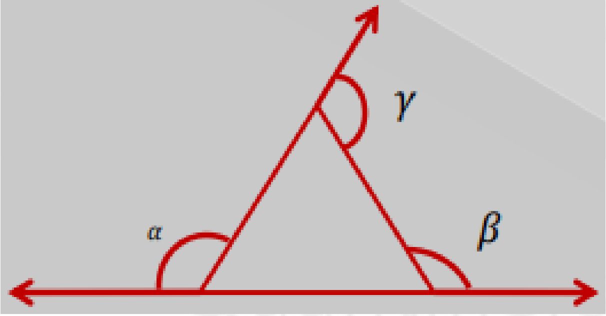
TRIANGLE



- A figure surrounded by three sides is called triangle.

$$\angle A + \angle B + \angle C = 180$$

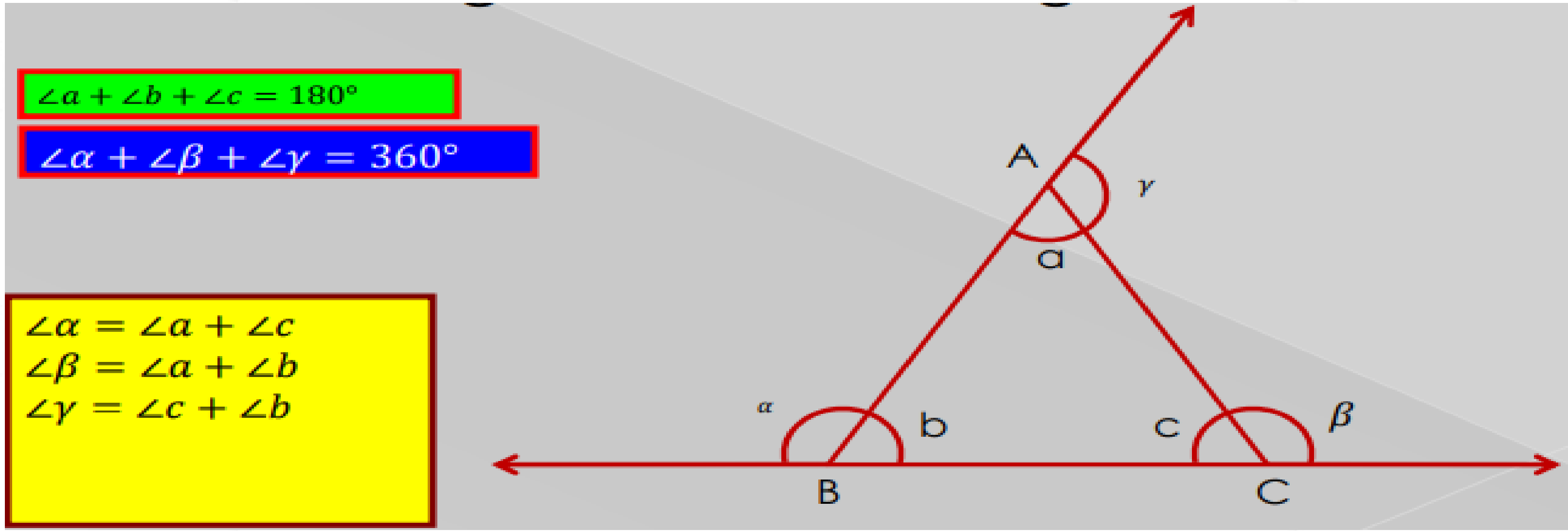
- Triangles have three vertex (A, B & C).
- It has three sides (AB, BC & CA).
- It has three interior angles ∠A ∠B ∠C .
- The sum of three interior angle is 180 degree.
- The sum of three exterior angle of a triangle is 360 degree.



$$\angle\alpha + \angle\beta + \angle\gamma = 360°$$

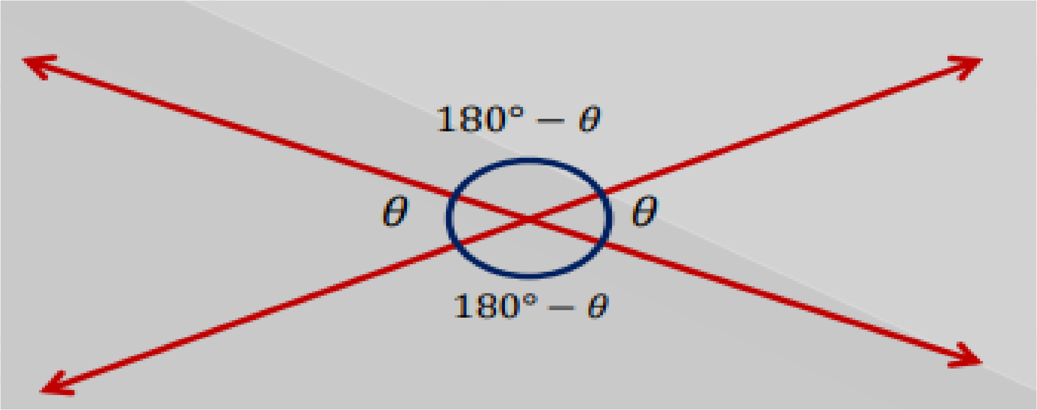
INTERIOR AND EXTERIOR ANGLE OF TRIANGLE

- The value of the exterior angle of a triangle is equal to the sum of remaining two interior angles.



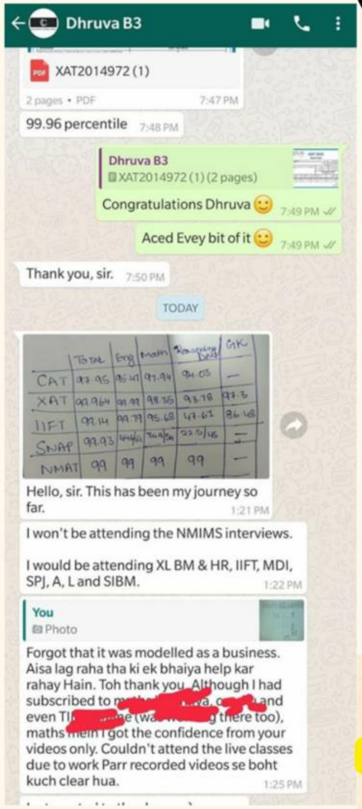
VERTICALLY OPPOSITE ANGLE

- When two lines intersect, the opposite angles formed by them are called vertically opposite angle.



#PROBLEM - MAJOR PROBLEM

Speed!, I study the topic - I understand it, but when I practice I am not able to solve questions, I self-doubt and a lot more, am I even worth IIMs?



Comrade, we at Quantifiers follow a three-layer system. Layer 1 - complete the lecture and on the same night/next day, revise it, after that, take a day gap and do the same lecture again to grasp the concept, after that easy questions - tough questions, chapter wise test, it will get you there!!

Dhruva - a non-engineer and a full time lawyer cracked every possible exam with us, got into IIM Ahmedabad - so can you! Just strategize

1 MyTH

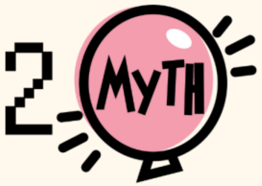
I need to attempt a maximum number of questions to get 99.99%ile

You don't have to attempt all the questions! Yes, there are a total of 66 questions, the key here is to attempt a specific amount for example but in CAT 2023, an attempt of total

Manali (our student) cracked CAT Exam despite getting 4 correct QA questions and got into IIM C. So can you, believe in yourself.

Section		Section		Section		Total	
Verbal Ability & Reading Comprehension		Data Interpretation & Logical Reasoning		Quantitative Ability			
Scaled Score	Percentile	Scaled Score	Percentile	Scaled Score	Percentile	Overall Scaled Score	Overall Percentile
51.10	99.87	23.99	98.25	11.74	87.99	86.83	99.60

Fill this Google form to receive Daily targets in your WhatsApp
https://forms.gle/oSih8jQCermJq71q9



2 MyTH

CAT is meant for Engineers and only engineer make it to IIMs

False, There is a diversity factor that every college needs to maintain! Cherry on top, Even from a non-engineer background, you can still ace the CAT exam.

Pallav (our student) cracked CAT Exam despite being from commerce background and got into IIM A. So can you, believe in yourself.

Section		Section		Section		Total	
Verbal Ability & Reading Comprehension		Data Interpretation & Logical Reasoning		Quantitative Ability			
Scaled Score	Percentile	Scaled Score	Percentile	Scaled Score	Percentile	Overall Scaled Score	Overall Percentile
42.17	99.46	38.36	99.90	48.82	99.93	129.35	99.98

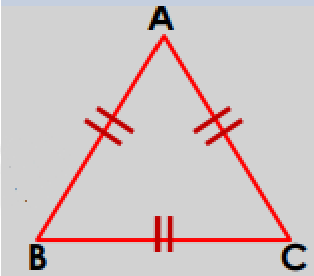
Fill this Google form to receive Daily targets in your WhatsApp
https://forms.gle/oSih8jQCermJq71q9

TYPES OF TRIANGLE (BASED ON SIDES)

- Equilateral triangle:- It has three equal sides. All three angles are equal (60 degree).

AB=BC=CA

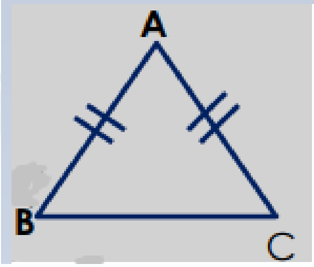
$\angle A = \angle B = \angle C = 60°$



- Isosceles triangle:- It has two equal sides. Angle opposite to equal sides are equal.

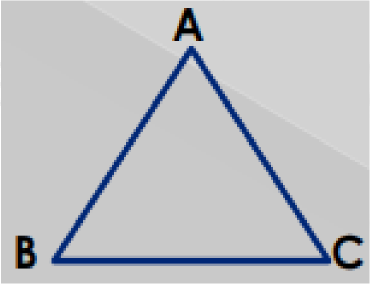
AB=BC≠CA

$\angle B = \angle C \neq \angle A$



- Scalene triangle:- It has no equal sides. No angles are equal.

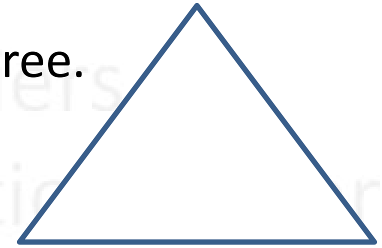
AB≠BC≠CA



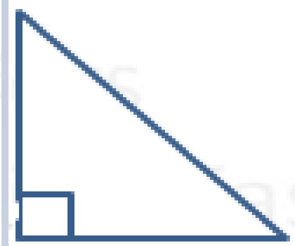
$\angle A \neq \angle B \neq \angle C$

TYPES OF TRIANGLE (BASED ON ANGLE)

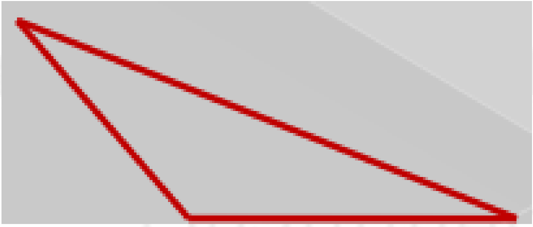
- Acute triangle:- All three angles of an acute triangle are less than 90 degree.

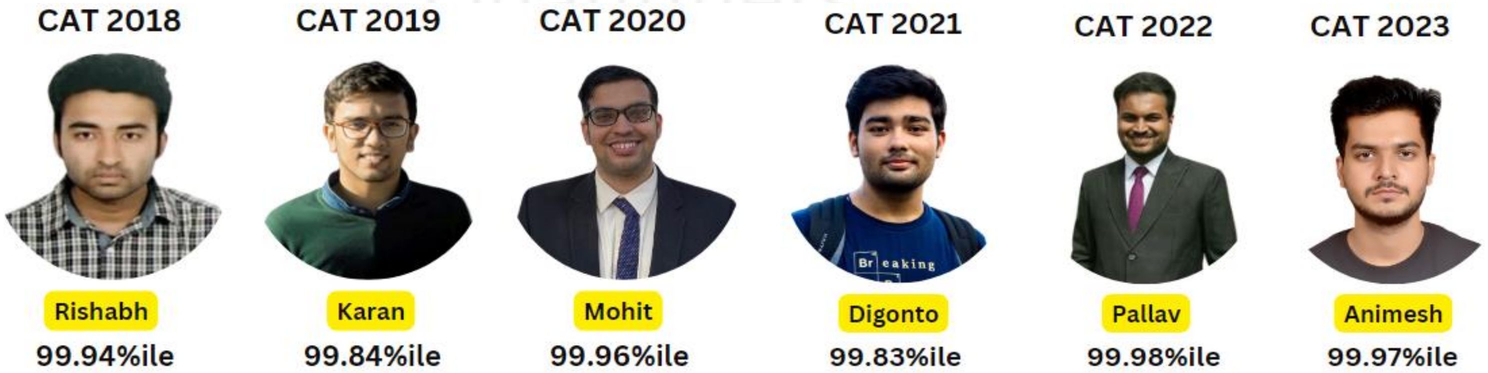


- Right triangle:- One of the three angles of an right triangle is 90 degree.



- Obtuse triangle:- One of the three angles of an obtuse triangle is greater than 90 degree.



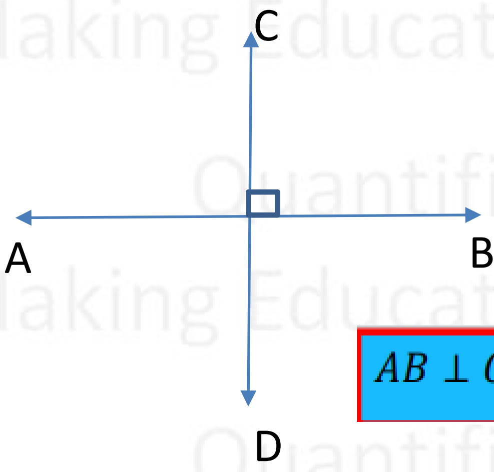


PARALLEL LINES & PERPENDICULAR LINES

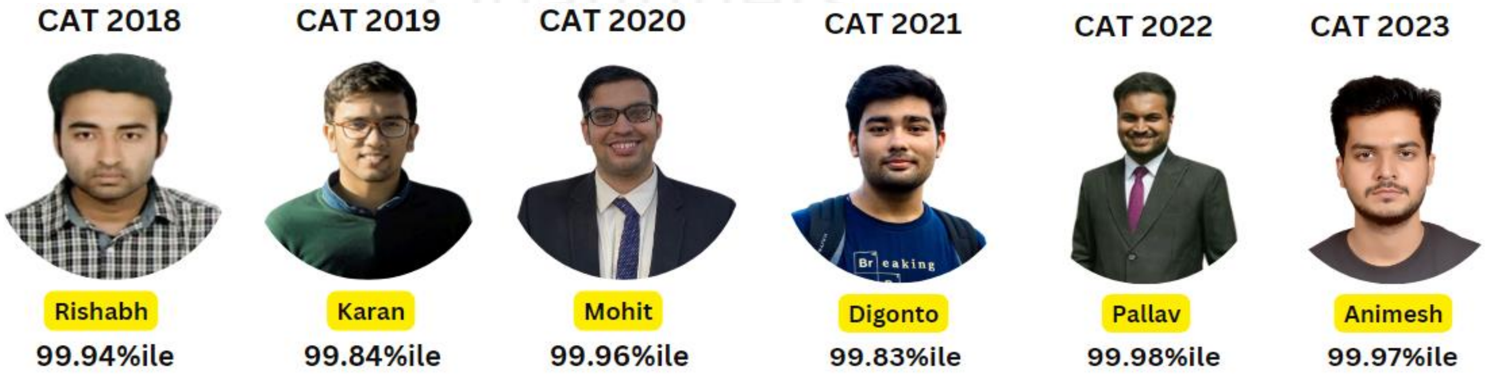
- Two such lines that never meet each other, that is, the distance between them is always the same is called parallel lines.

AB||CD

- If two lines intersect at right angle are called perpendicular lines.



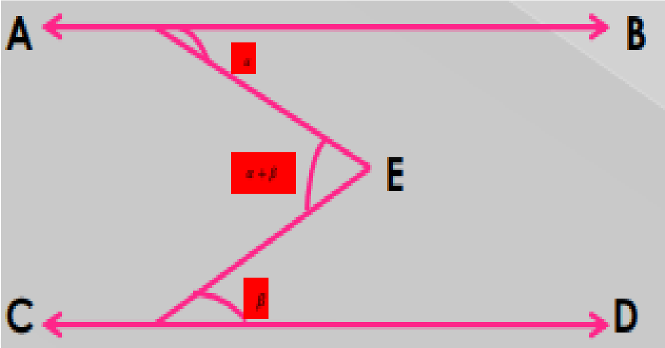
$AB \perp CD$

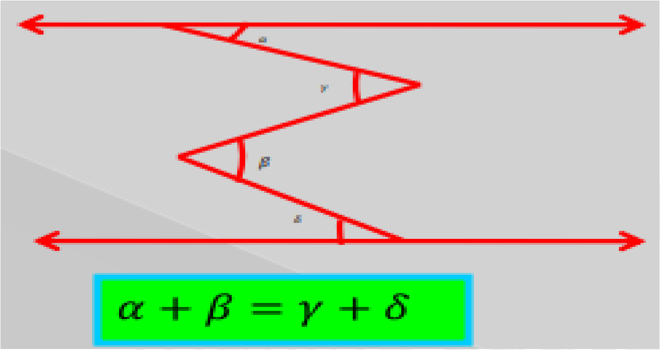


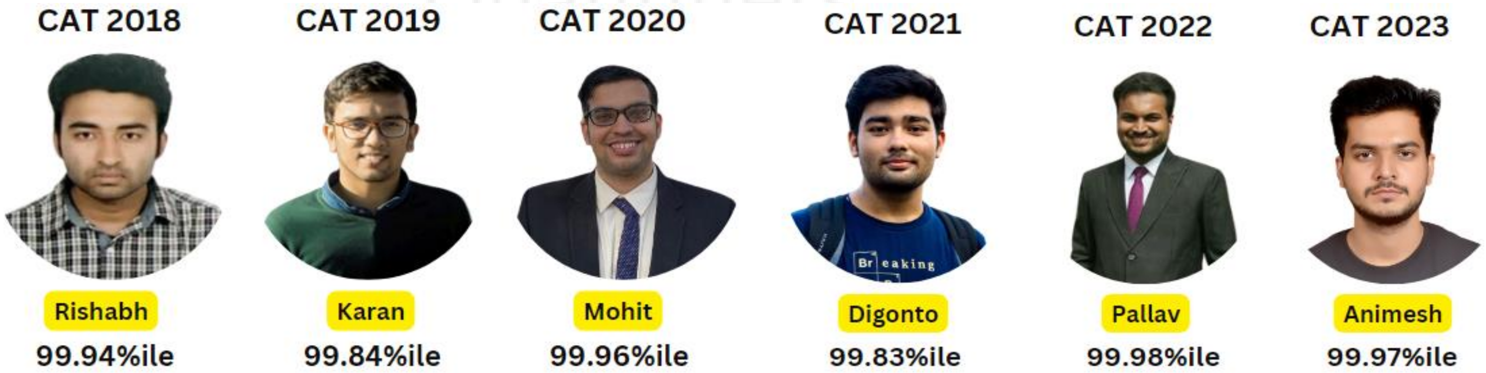
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

IMPORTANT CONCEPT

- If AB and CD are parallel lines.

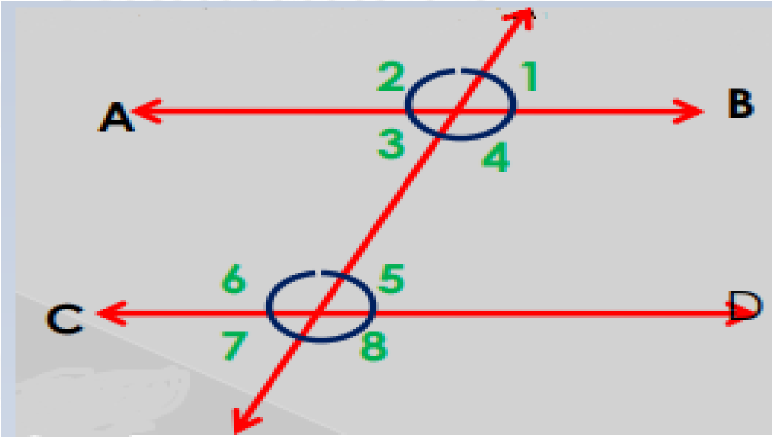






TRANSVERASAL LINES

- A line which intersect parallel lines is called a traversal lines.
- AB and CD are parallel lines.
- MN is traversal lines.



- Corresponding Angles-

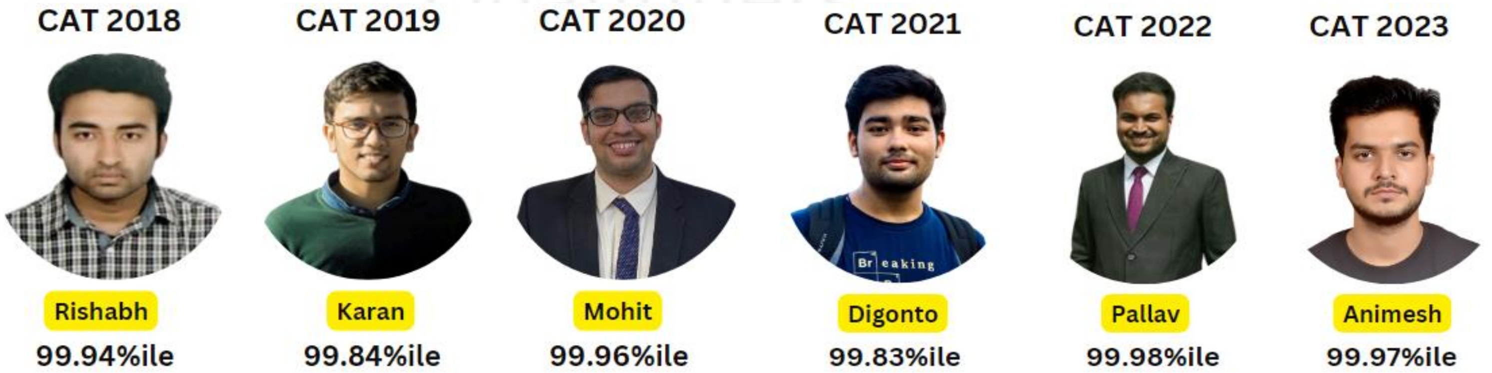
$\angle 1 = \angle 5, \angle 2 = \angle 6, \angle 3 = \angle 7 \& \angle 4 = \angle 8$

- Alternate Angles-

$\angle 3 = \angle 5 \ \& \ \angle 4 = \angle 6$

- Co-interior angles-

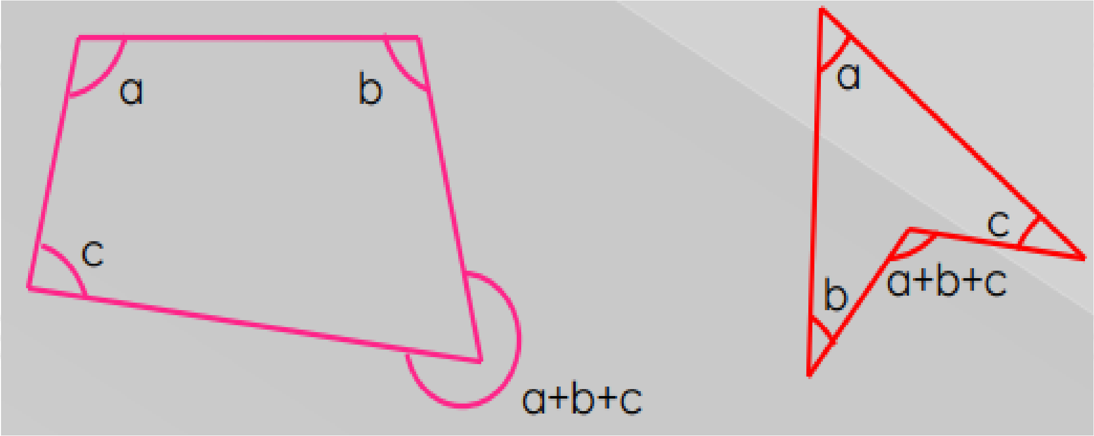
$\angle 4 + \angle 5 = \angle 3 + \angle 6 = 180°$

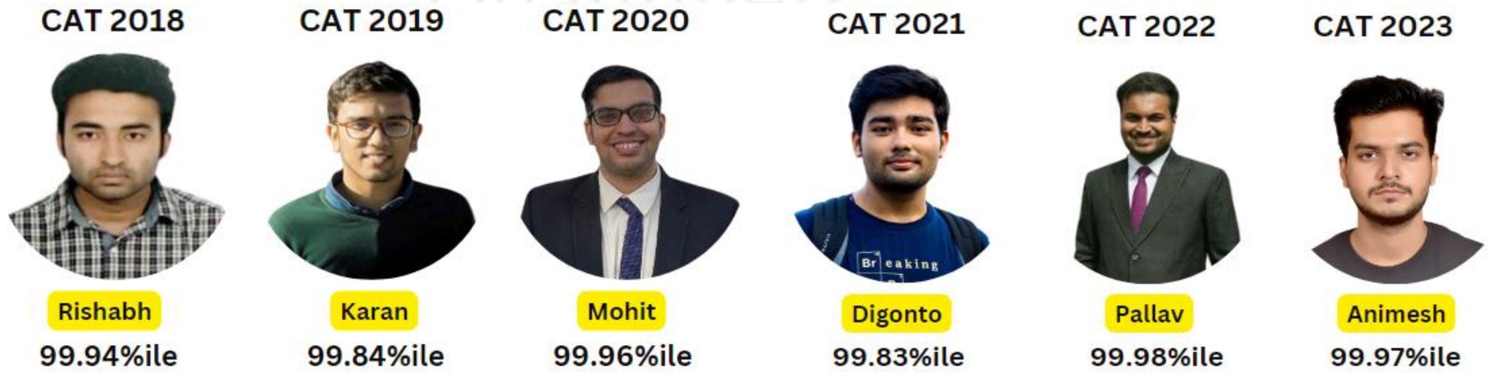


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

IMPORTANT CONCEPT

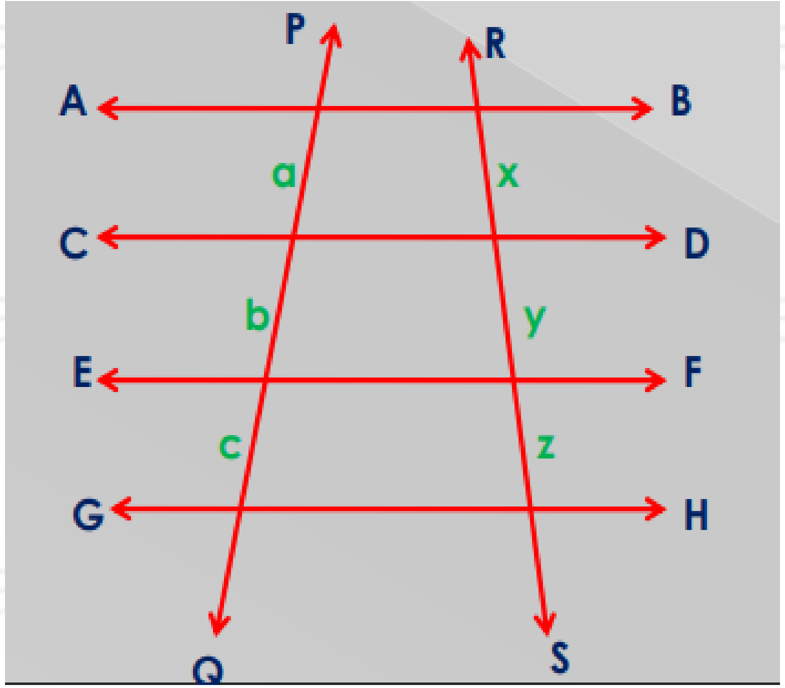
- The sum of angle of a quadrilateral is 360 degree.





BASIC CONCEPT OF PARALLEL LINES

- There are four parallel lines AB, CD, EF & GH and two transversal line PQ and RS.



a: b: c = x: y: z

QUANTIFIERS CAT ACADEMY

Sonia IIM L

**Sonia made it to
IIM Lucknow with
Quantifiers CAT Academy
so can you!!!!**

QUANTIFIERS CAT ACADEMY

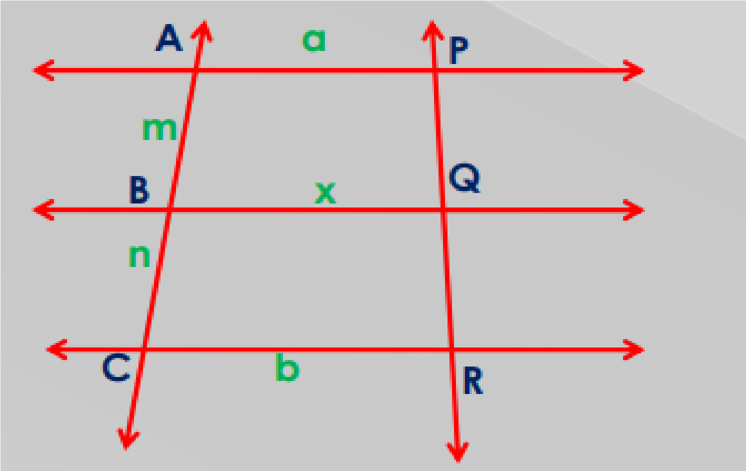
Bhavna IIM Ranchi

**Bhavna made it to
IIM Ranchi with
Quantifiers CAT Academy
so can you!!!!**

Join our WhatsApp FREE 24/7 Doubt Solving Group (if group is full, follow us on Instagram @quantifiers_cat for the new link)
https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA

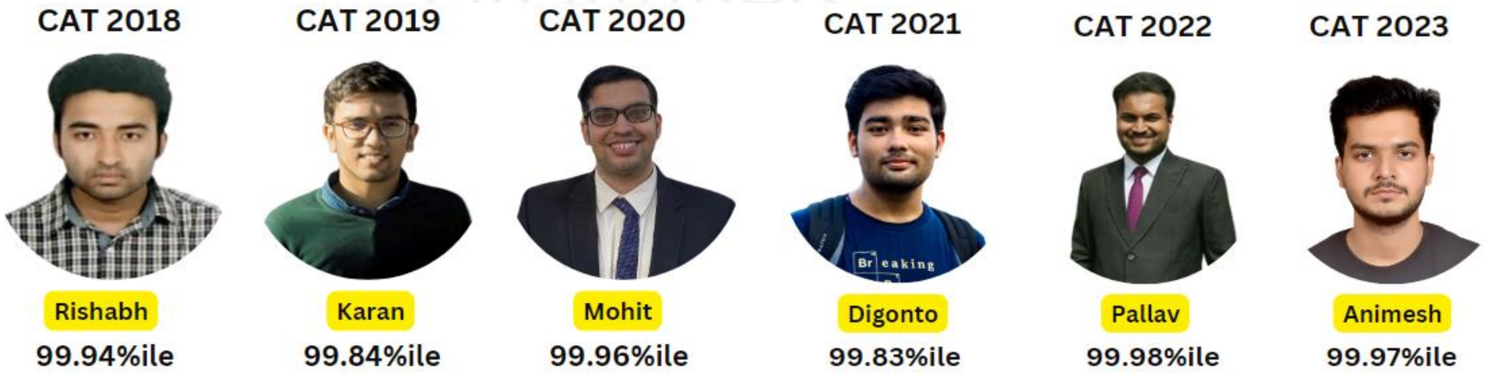
IMPORTANT CONCEPTS

- Three parallel lines and two transversal lines.



$$AB:BC = PQ:QR = m:n$$

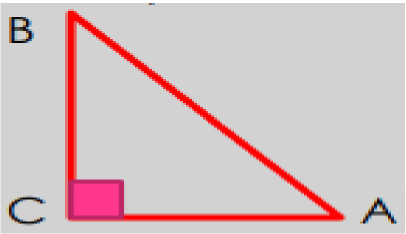
$$BQ = \frac{an + bm}{m + n}$$



ANGLE RATIO

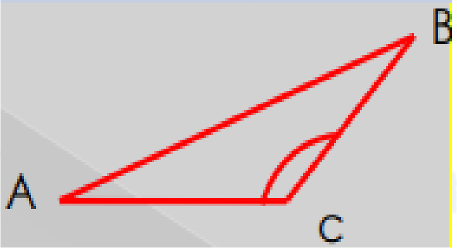
- In a right angle triangle, largest angle is equal to the sum of two remaining angles.

$\angle A + \angle B = 90° = \angle C$



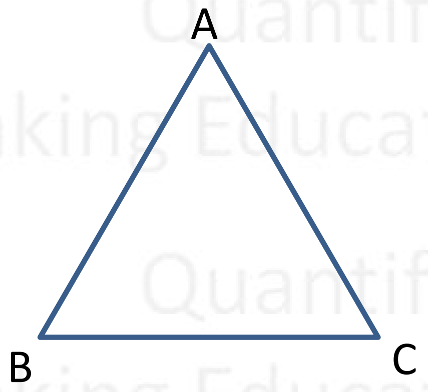
- In a obtuse angle triangle, largest angle is greater than sum of remaining two angles.

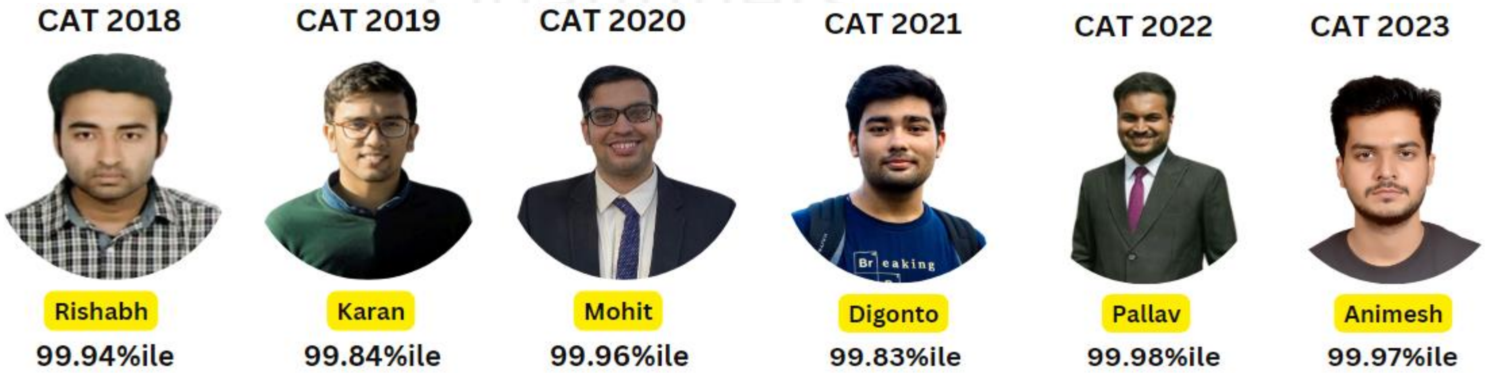
$\angle C > \angle A + \angle B$



- In an acute angle triangle, largest angle is smaller than sum of remaining two angles.

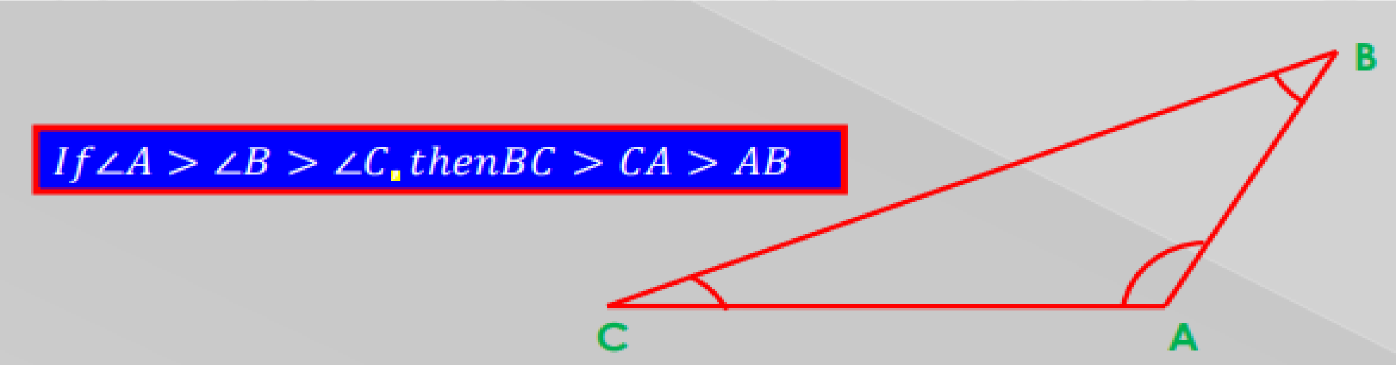
$\angle A < \angle B + \angle C$





CONCEPT OF TRIANGLE

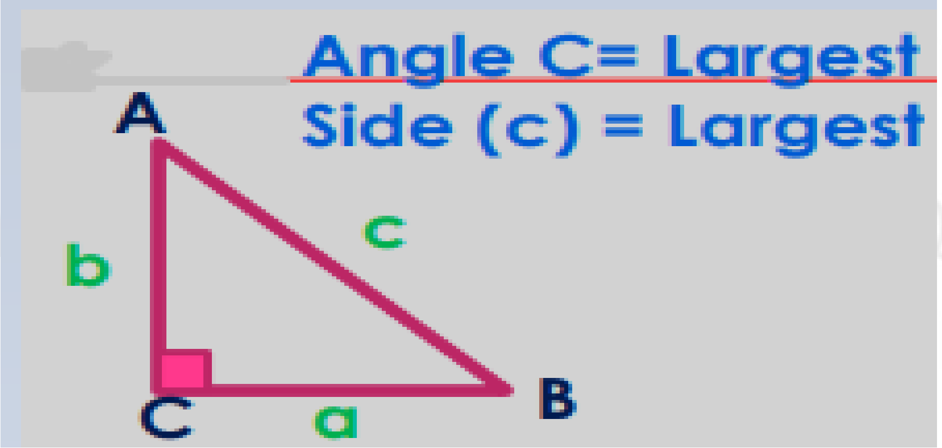
- In any triangle, the side opposite the largest angle will be larger and the side opposite the smallest angle will be smaller.

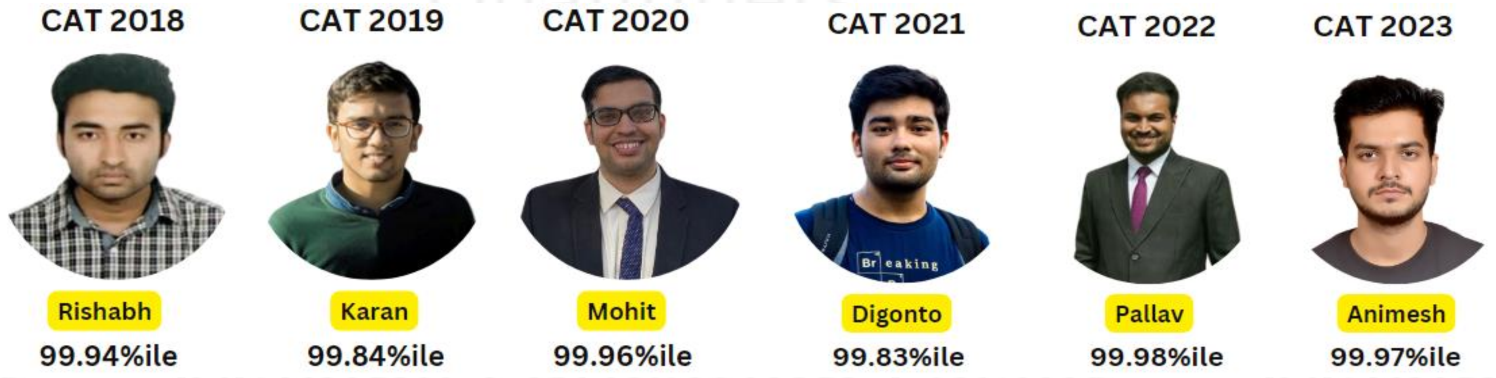


CONCEPT OF SIDE OF TRIANGLE

- In a right angle triangle square of larger side is equal to the sum of squares of remaining two sides

$c^2 = a^2 + b^2$

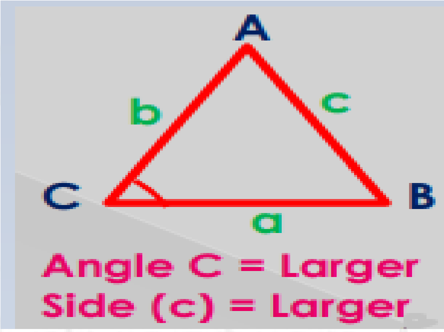




CONCEPT OF SIDE OF TRIANGLE

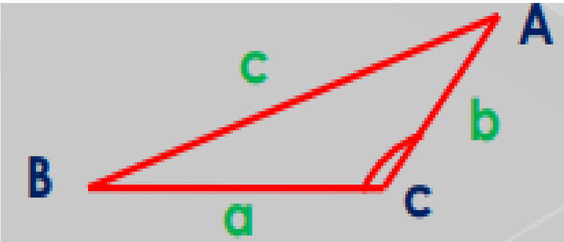
- In an acute angle triangle square of larger side is less than the sum of squares of remaining two sides.

$c^2 < a^2 + b^2$



- In an obtuse angle triangle square of larger side (c) is greater than the sum of squares of remaining two sides.

$c^2 > a^2 + b^2$

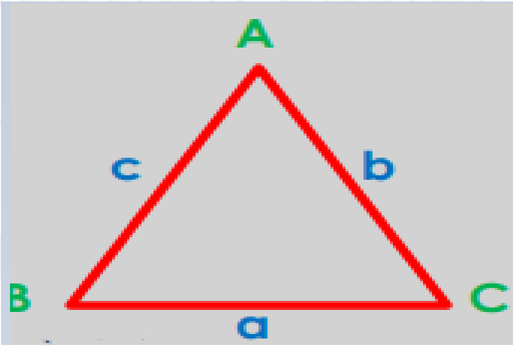


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

AREA OF TRIANGLE

- If three sides of triangle are given then we used Heron's formula.

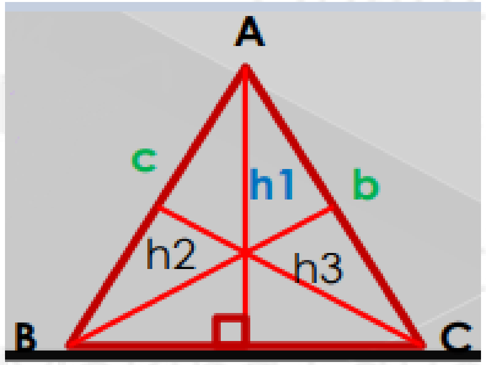
A = √s(s-a)(s-b)(s-c)
Underroot is on entire formula
Also, s = semiperimeter = (a+b+c)/2



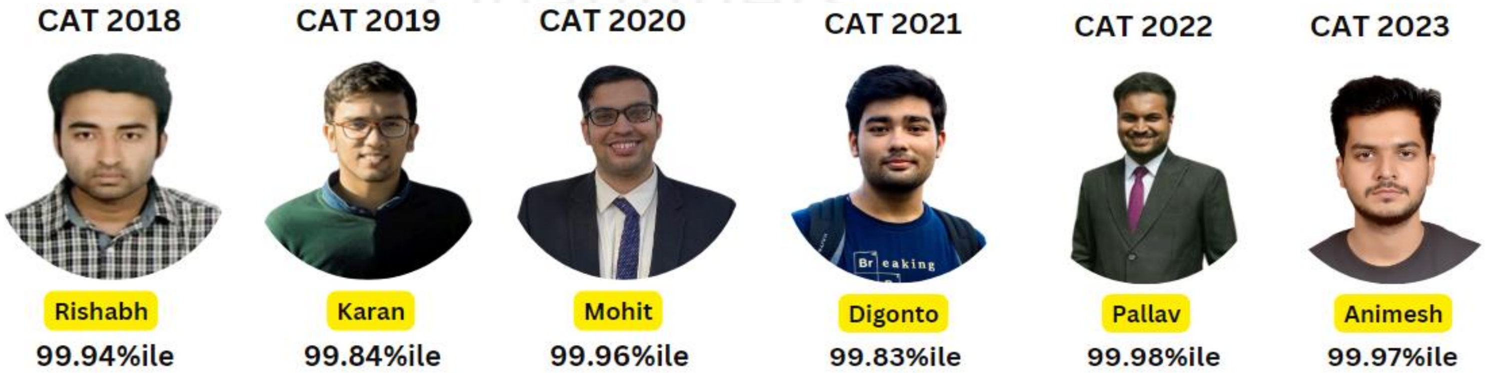
- If one side and perpendicular height is given

$$area\ of\ triangle = \frac{1}{2} \times base \times height = \frac{1}{2}ah1 = \frac{1}{2}bh2 = \frac{1}{2}ch3$$

$$a \times h1 = b \times h2 = c \times h3 = constant$$



$h1 : h2 : h3 = 1/a : 1/b : 1/C$

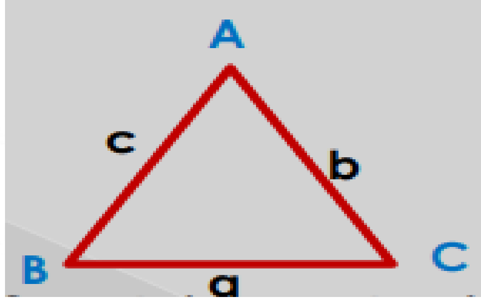


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

IMPORTANT CONCEPT OF SIDES OF TRIANGLE

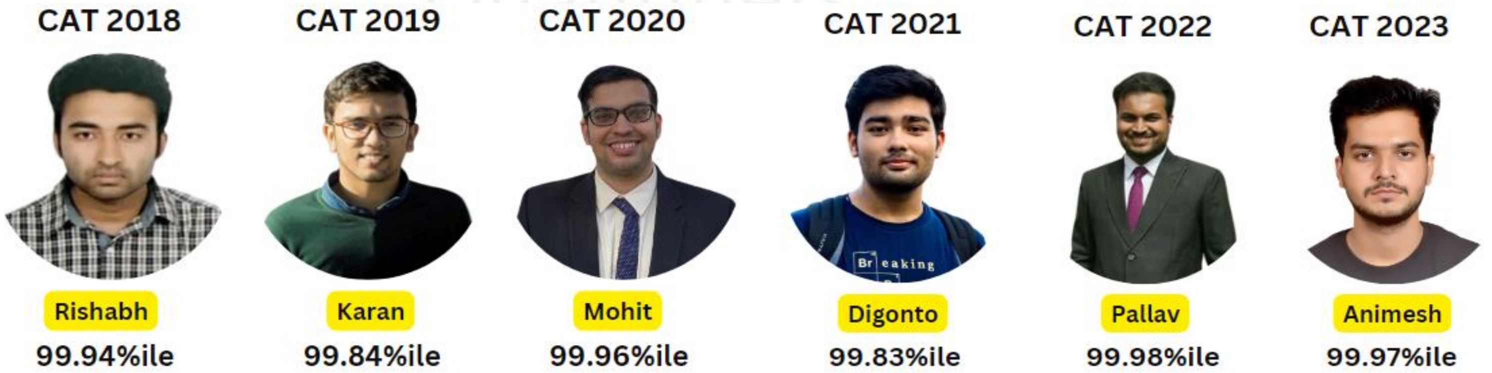
- The sum of two sides of a triangle is always greater than third sides of triangle.

$$a + b > c$$
$$b + c > a$$
$$c + a > b$$



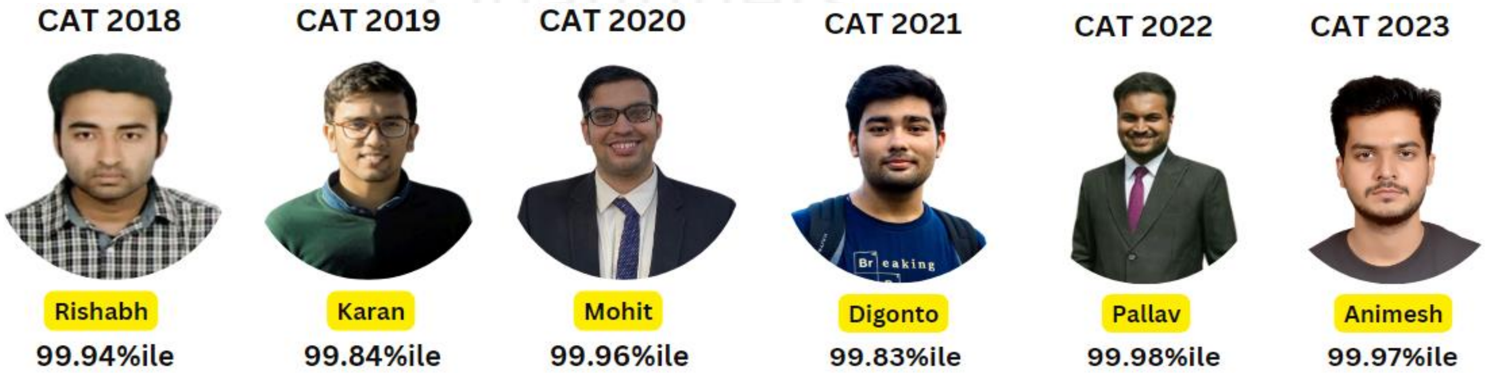
- The difference of any two sides of a triangle is less than third side of triangle.

$$|c - a| < b$$
$$|b - a| < c$$
$$|c - b| < a$$



CONCEPT OF RIGHT ANGLE TRIANGLE

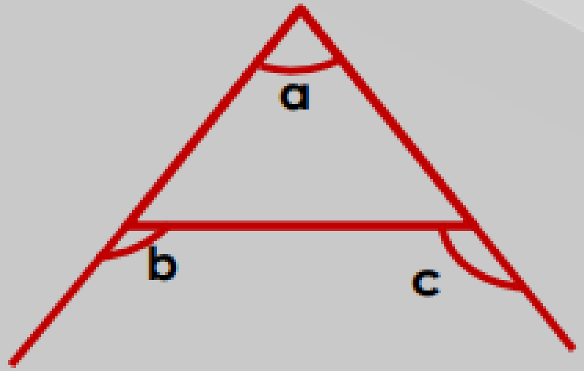
- $a^2 - b^2$, $2ab$ & $a^2 + b^2$ are sides of right angle triangle.
- $a^2 - b^2$, $2ab$ & $a^2 + b^2$ are also Pythogorean triplet.
- Some example of triplet.
- (3, 4, 5) (5, 12, 13) (7, 24, 25) (8, 15, 17)
- (9, 40, 41) (11, 60, 21) (12, 35, 37) (13, 84, 85) (16, 63, 65) (20, 21, 29)



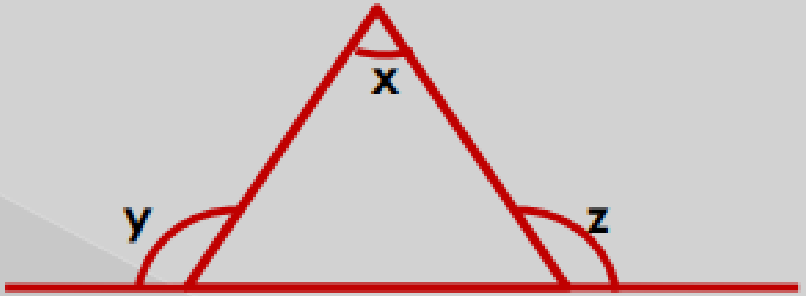
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

CONCEPT OF TRIANGLE

$$\angle b + \angle c = 180° + \angle a$$



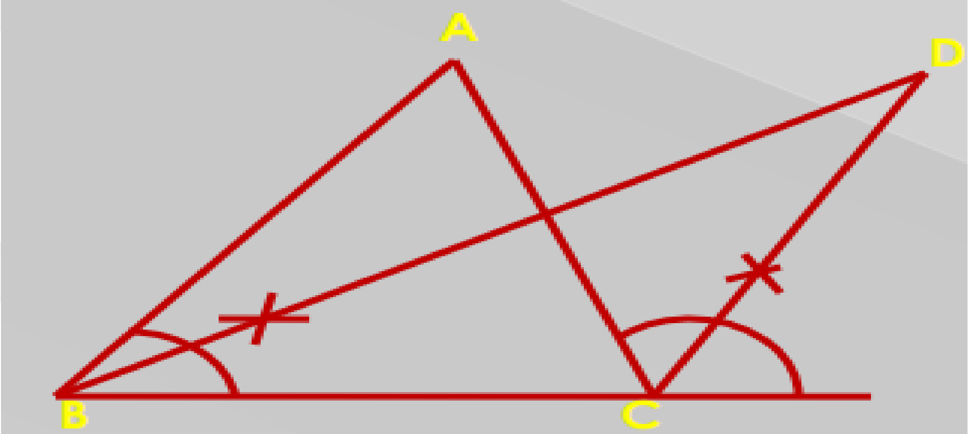
$$\angle x = \angle(x + y) - 180°$$



CAT 2018	CAT 2019	CAT 2020	CAT 2021	CAT 2022	CAT 2023
Rishabh	Karan	Mohit	Digonto	Pallav	Animesh
99.94%ile	99.84%ile	99.96%ile	99.83%ile	99.98%ile	99.97%ile

BISECTOR RULE OF TRIANGLE

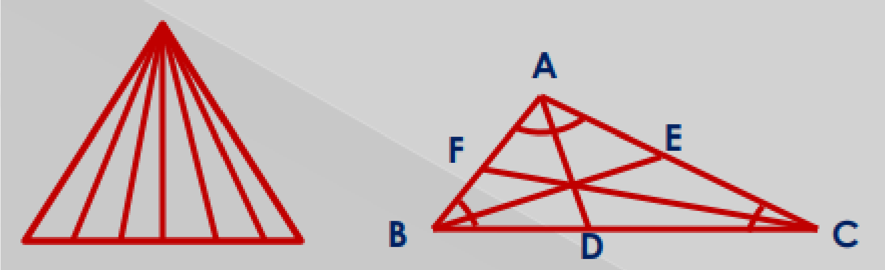
$$\angle D = 90^\circ - \frac{\angle C}{2} - \frac{\angle B}{2} = \frac{\angle A}{2}$$



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

CEVIANS AND ANGLE BISECTOR

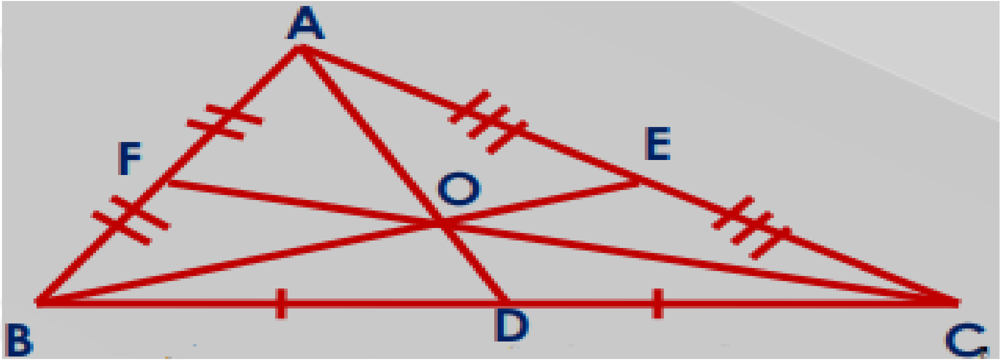
- The line vertex of a triangle to its opposite sides are called cevians.



- A line which bisect the angle in two equal part is called angle bisector (AD, BE, CF).
- Angle bisector is also a cevian.
- Where all three bisector met, that point is called in- center.

MEDIANS

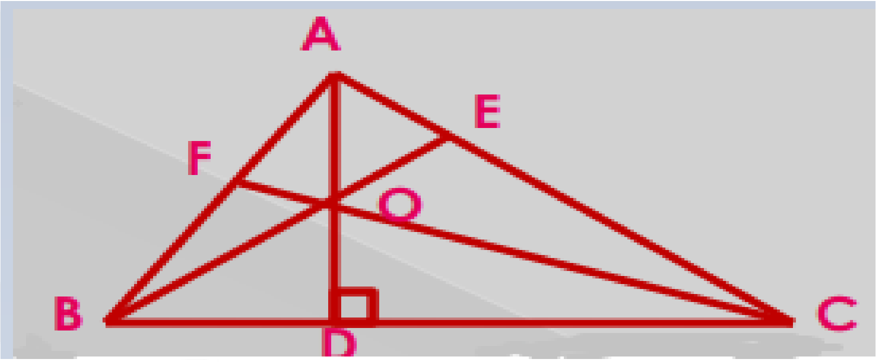
- If the midpoint of any side in a triangle is joined with the opposite vertex, then it will be called median.
- AD is the median of side BC, BE is the median of AC & CF is the median AB.



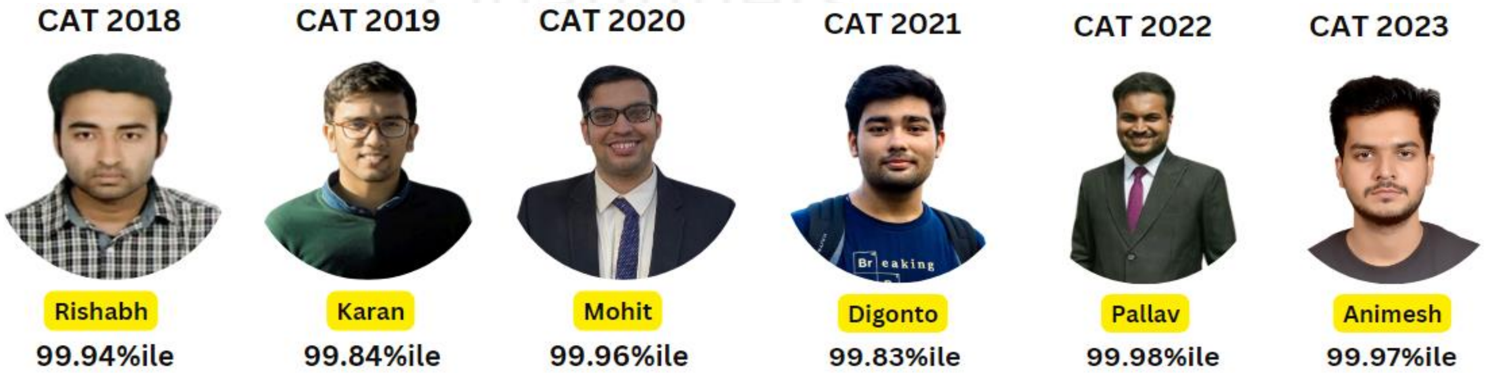
- Where three median met that point (o) is called centroid.

ALTITUDES

- The perpendicular drawn from the vertex of a triangle to its opposite side is called altitudes.
- AD, BE & CF are altitudes.

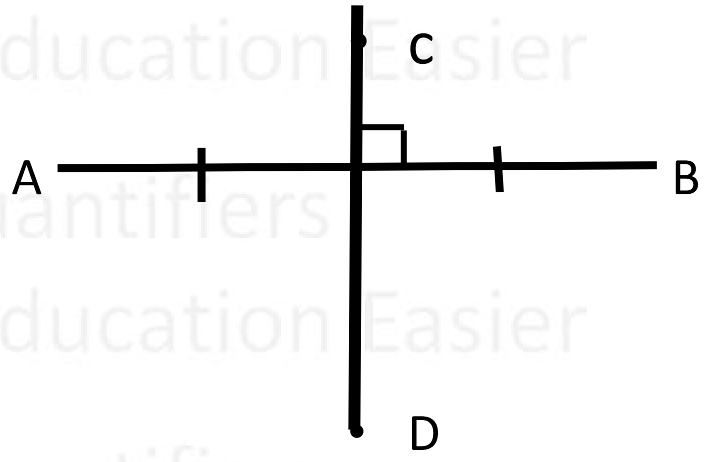


- Point where three altitudes meet is called ortho-center

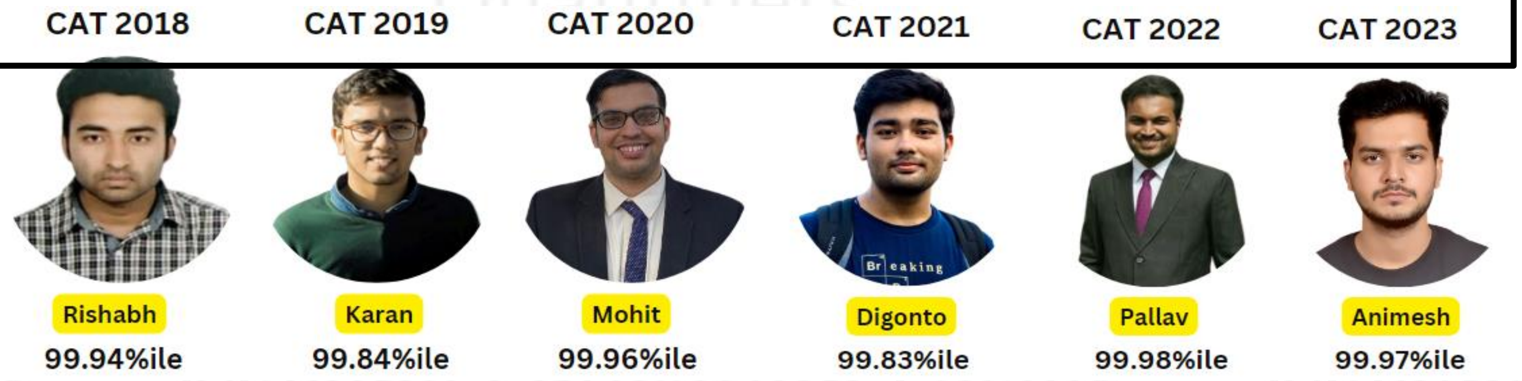


PERPENDICULAR BISECTOR

- A line which is perpendicular on another line and also bisect this other line is called perpendicular bisector.

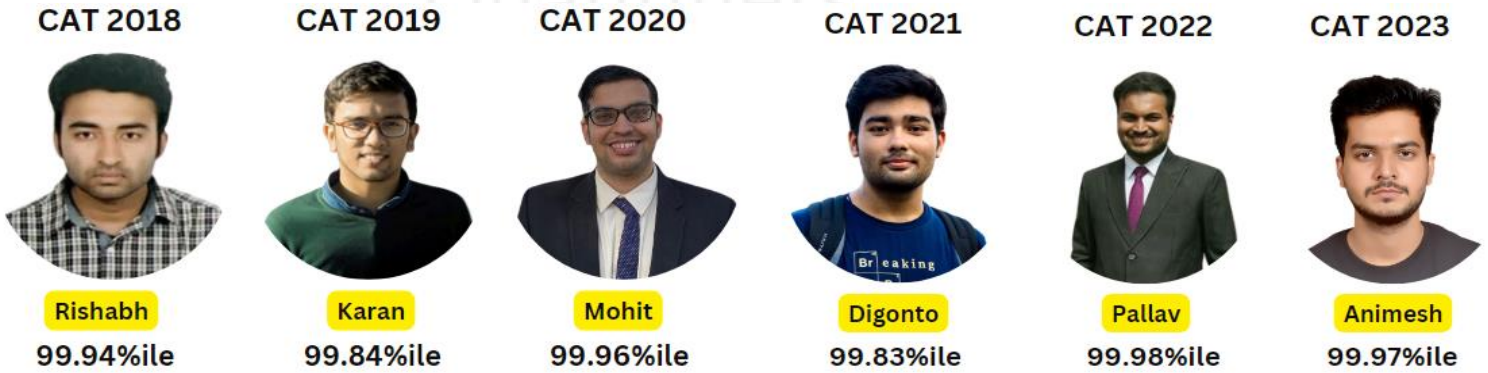


- CD is perpendicular bisector of AB.



IMPORTANT CONCEPT

- In equilateral and isosceles triangle in-center, ortho-center and centroid meet at one point.
- In scalene triangle in-center, ortho-center and centroid meet at different point.



QUANTIFIERS CAT ACADEMY

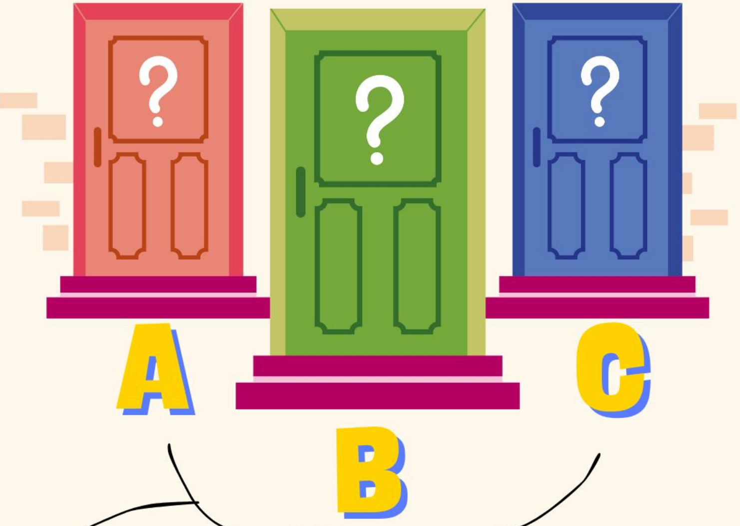
Sectional Tests

https://youtu.be/FtH4bZQmhMs

How to attempt sectional tests
CAT level tests

Join our WhatsApp FREE 24/7 Doubt Solving Group (if group is full, follow us on Instagram @quantifiers_cat for the new link)
Click here to join WhatsApp group
https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA

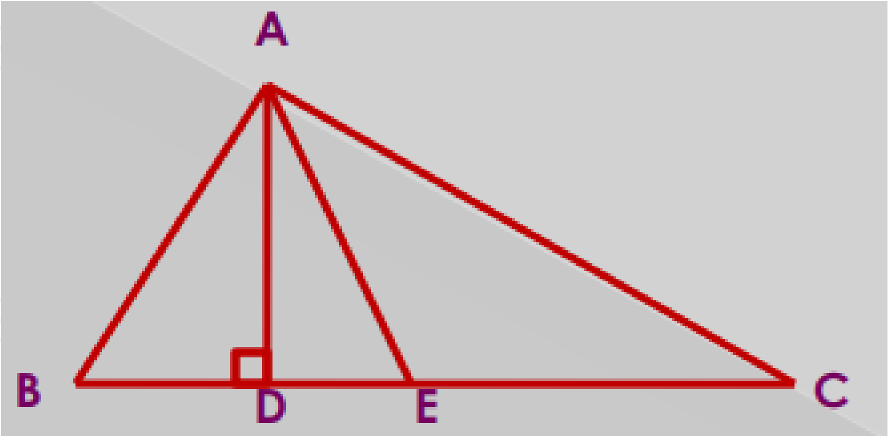
99.99%ile DILR Strategy



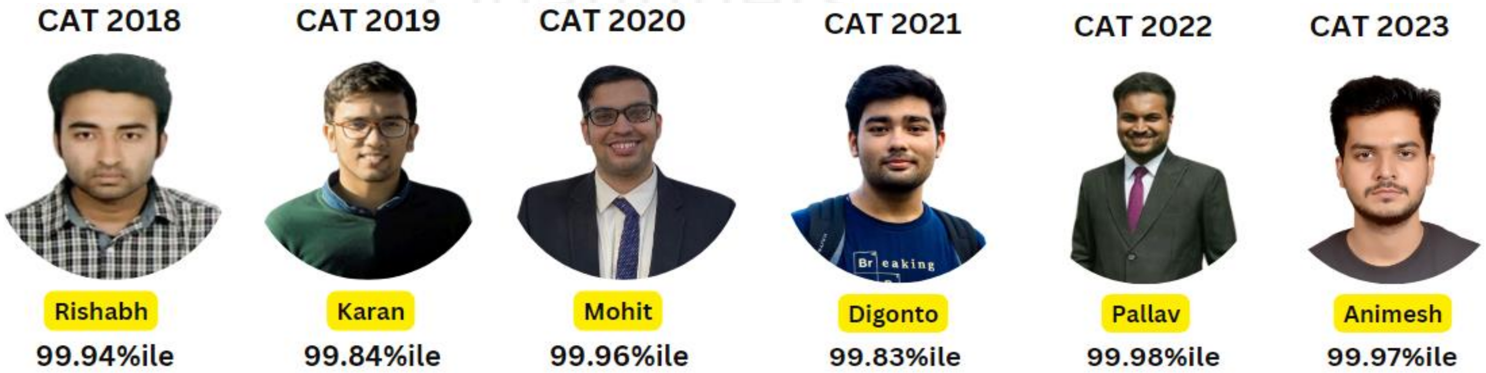
https://youtu.be/Kp6dphs-bvA

ANGLE BETWEEN ALTITUDE AND ANGLE BISECTOR

- When we draw altitude and angle bisector from same vertex then-
- AD is altitude and AE is angle bisector of angle A.



$$\angle DAE = \frac{\angle B - \angle C}{2}$$



Simran NMIMS

**Simran made it to
NMIMS Mumbai with
Quantifiers CAT Academy
so can you!!!!**

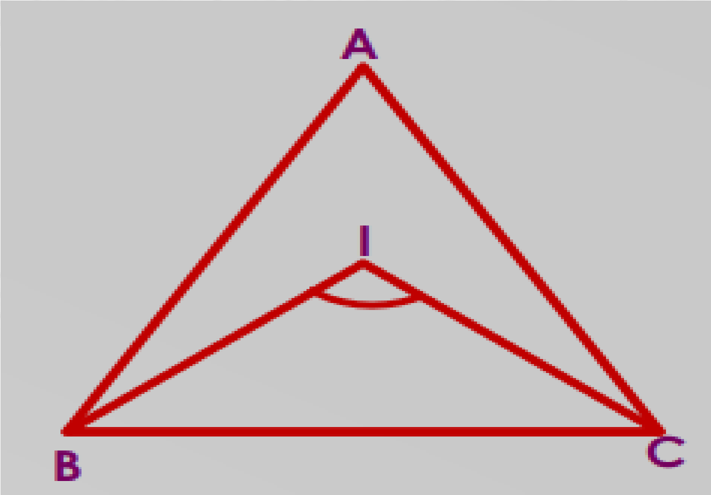
Aru XLRI

**Aru made it to
XLRI with
Quantifiers CAT Academy
so can you!!!!**

Join our WhatsApp FREE 24/7 Doubt Solving Group (if group is full, follow us on Instagram @quantifiers_cat for the new link)
https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA

IMPORTANT RESULT OF BISECTOR

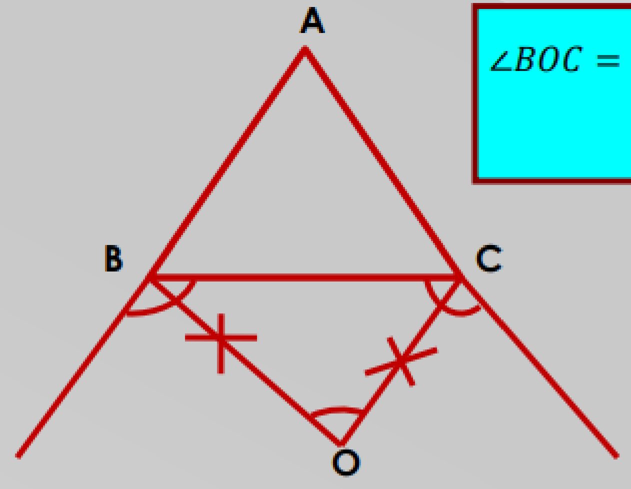
- When we draw two bisector of angle they meet at in-center (I).
- BI & CI are angle bisector



$$\angle BIC = 90^\circ + \frac{\angle A}{2}$$

BISECTOR OF EXTERIOR ANGLE

- If we draw external bisector angle then it meet at point O.

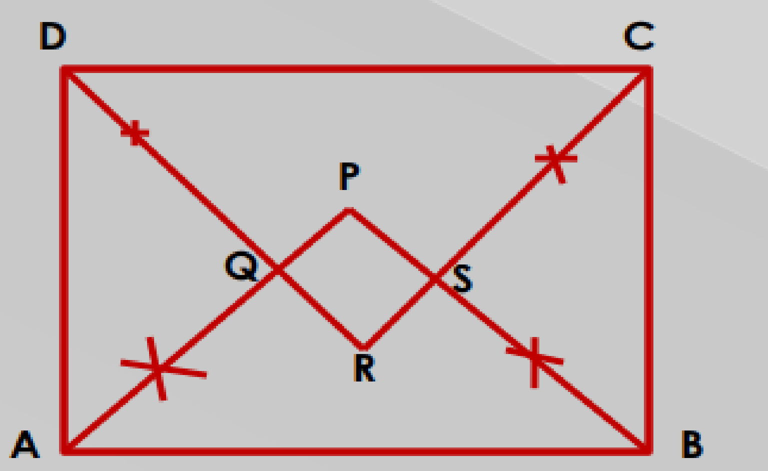


$$\angle BOC = 90^\circ - \frac{\angle A}{2}$$

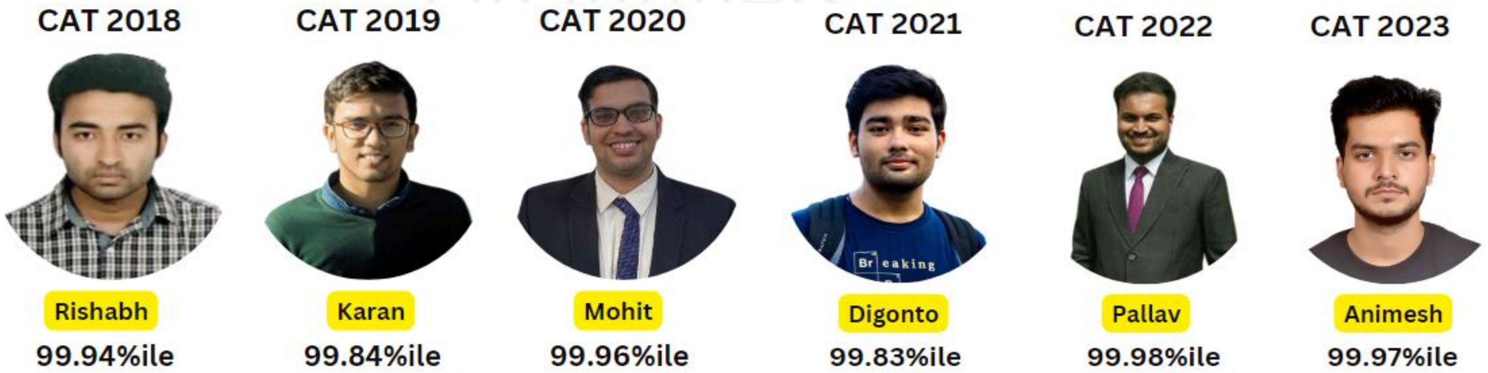
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

QUADRILATERAL

- Quadrilateral formed by angle bisector of another quadrilateral will always be a cyclic quadrilateral.

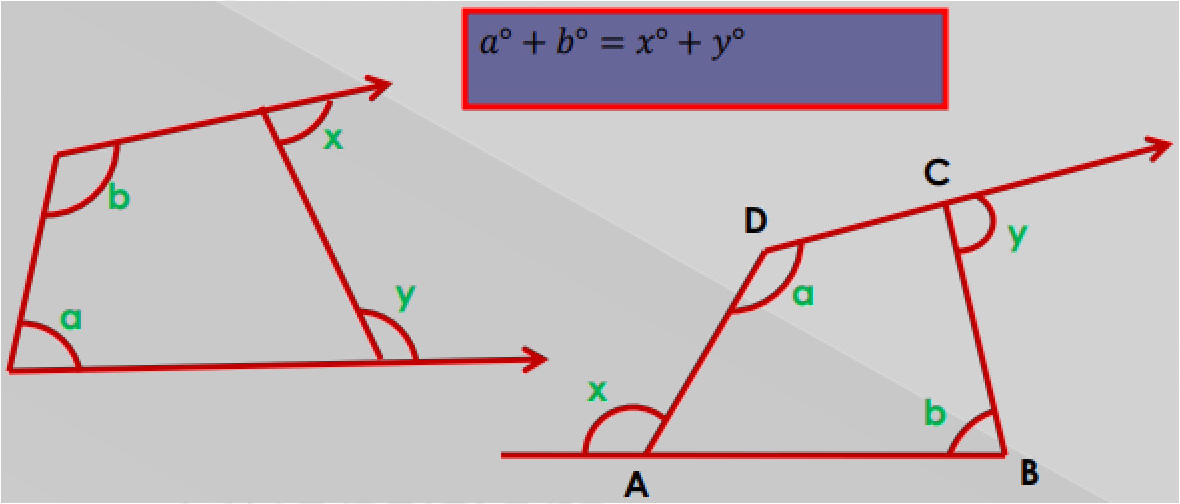


$$\angle APB = \frac{\angle C + \angle D}{2}$$
$$\angle CRD = \frac{\angle A + \angle B}{2}$$
$$\angle P + \angle R = 180°$$
$$\angle Q + \angle S = 180°$$



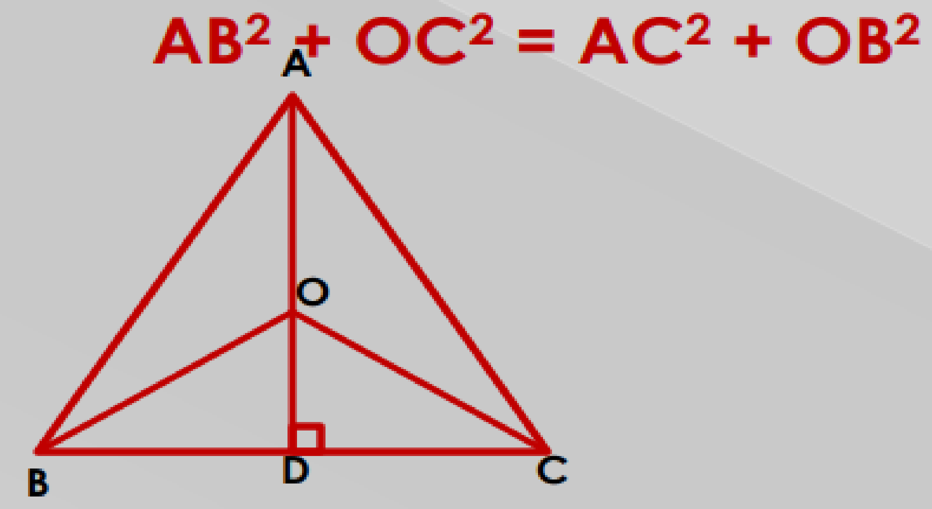
CONCEPT OF QUADRILATERAL

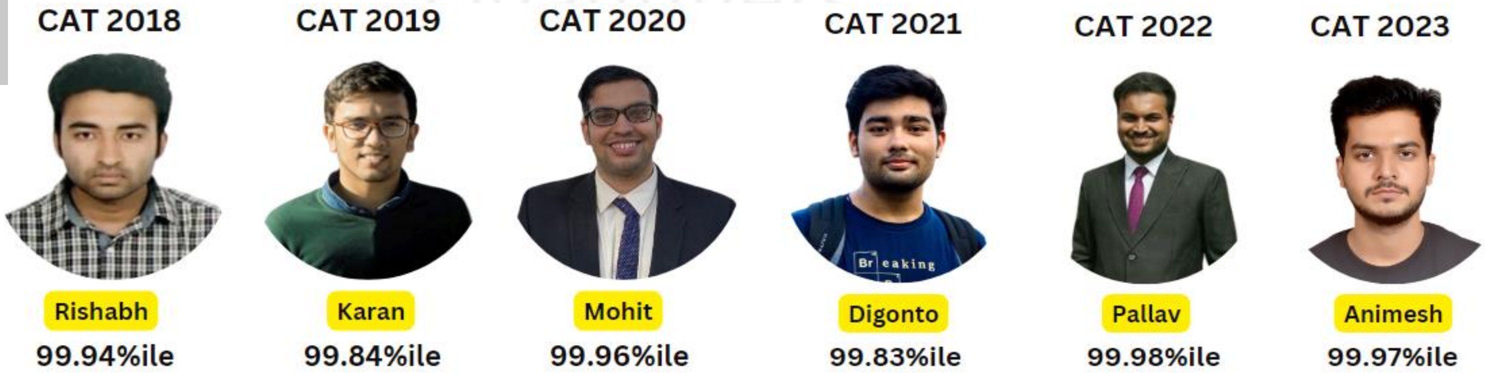
- In any quadrilateral-



IMPORTANT CONCEPT

- AD is perpendicular to side BC and O is any point on AD, then-





QUANTIFIERS CAT ACADEMY

Smriti - IIM C

Smriti made it to IIM Calcutta with Quantifiers CAT Academy So can you!!!!

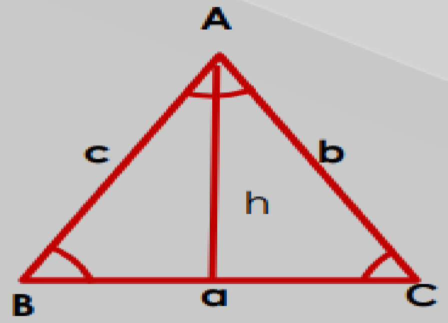
QUANTIFIERS CAT ACADEMY

Smriti - XLRI

Smriti made it to XLRI with Quantifiers CAT Academy So can you!!!!

Join our WhatsApp FREE 24/7 Doubt Solving Group (if group is full, follow us on Instagram @quantifiers_cat for the new link) https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA

SINE RULE

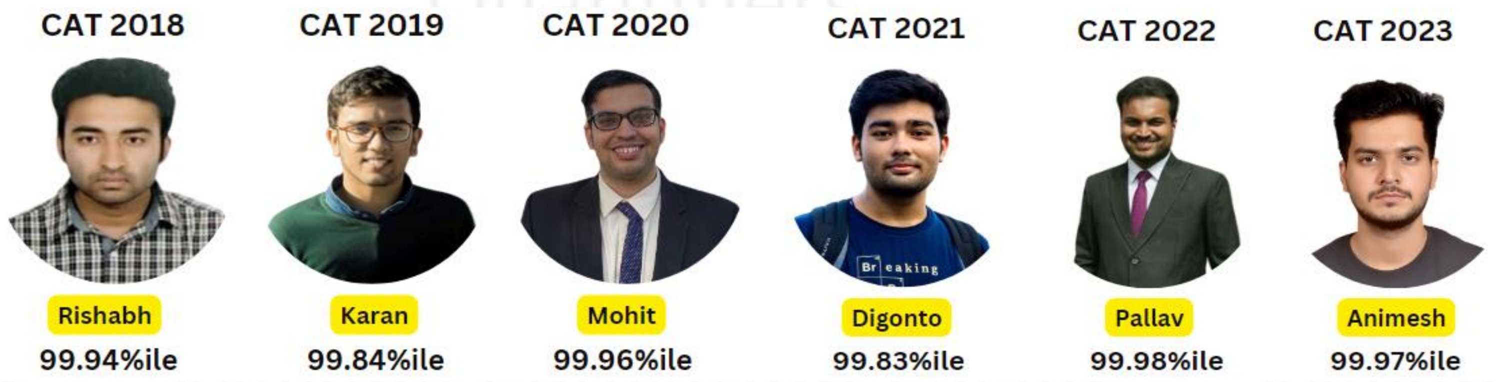


$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k(let)$$

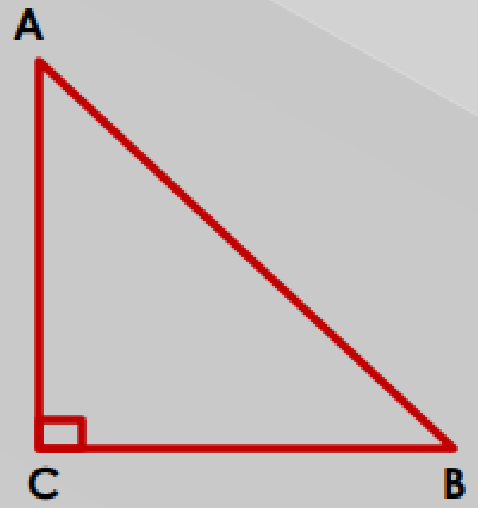
$$a : b : c = \sin A : \sin B : \sin C$$

$$h = c \sin B = b \sin C$$

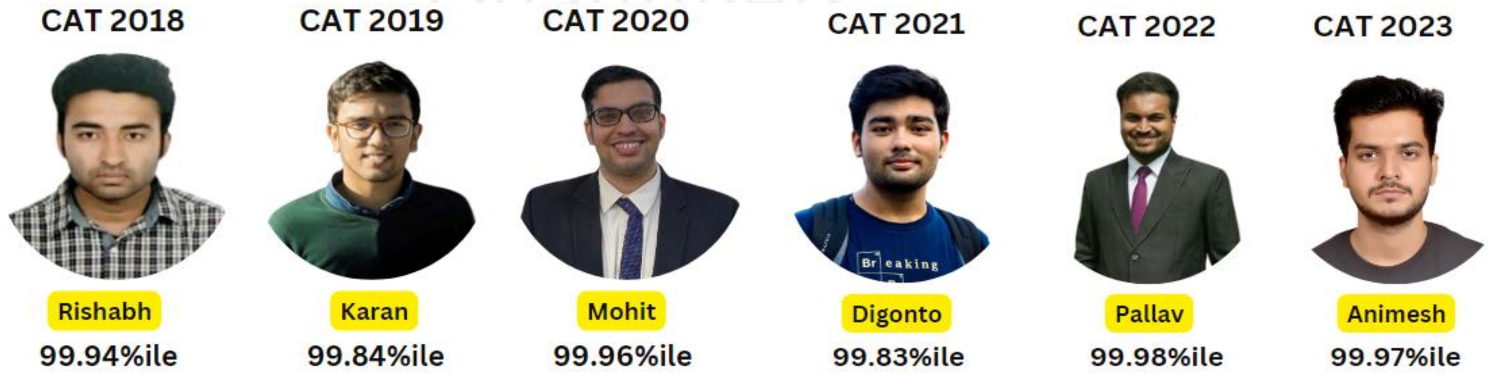
$$area of \Delta ABC = \frac{1}{2} ac \sin B = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A$$



ISOSCELES RIGHT ANGLE TRIANGLE

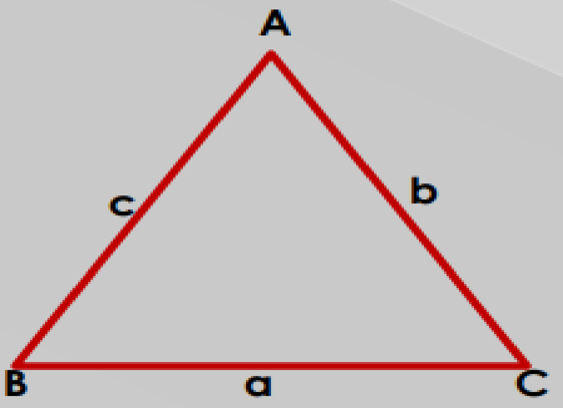


$$Angles = 45°, 45°, 90°$$
$$sides = 1:1:\sqrt{2}$$



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

COSINE RULE



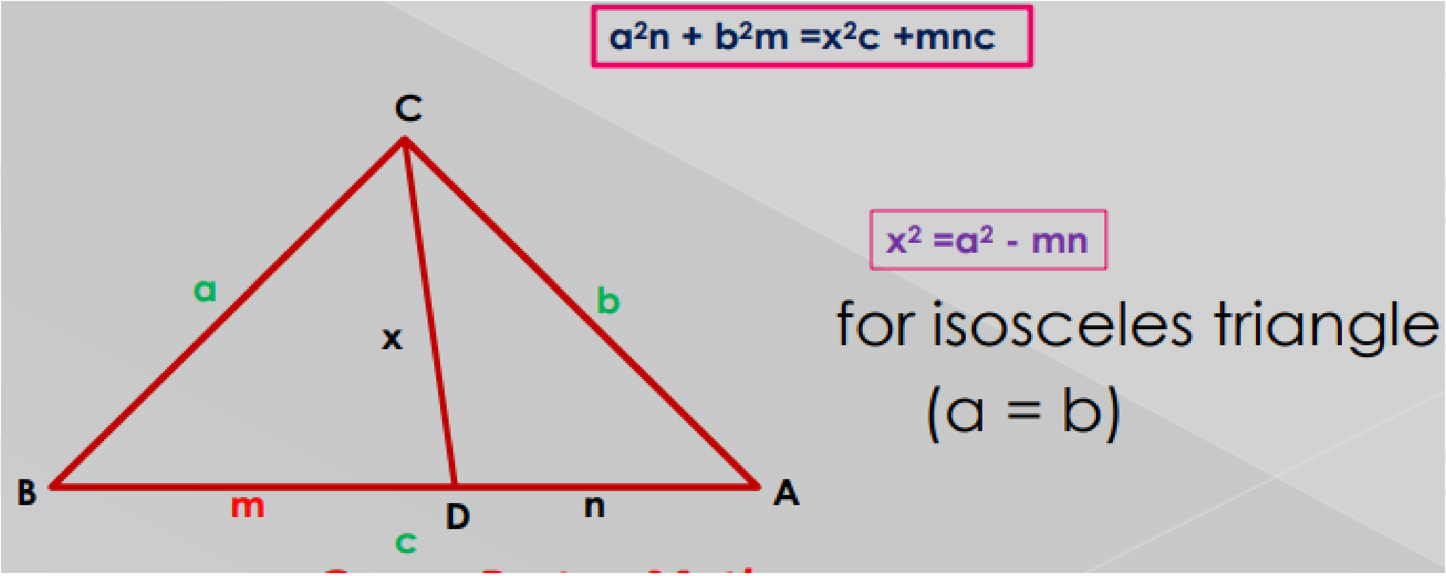
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

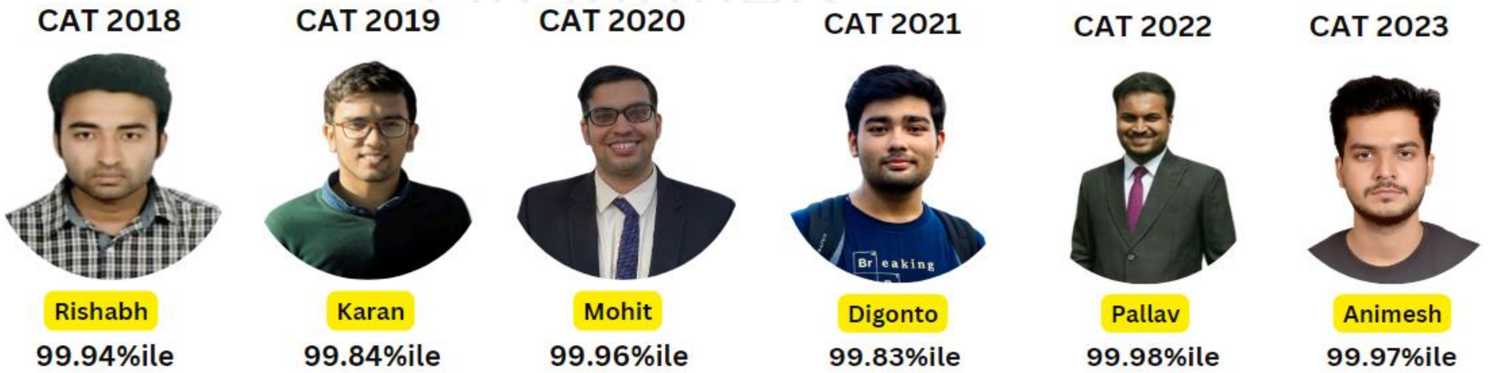
$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$a^2 = b^2 + c^2 - 2bc \cos A$
$b^2 = a^2 + c^2 - 2ac \cos B$
$c^2 = a^2 + b^2 - 2ab \cos C$

STEWART THEOREM

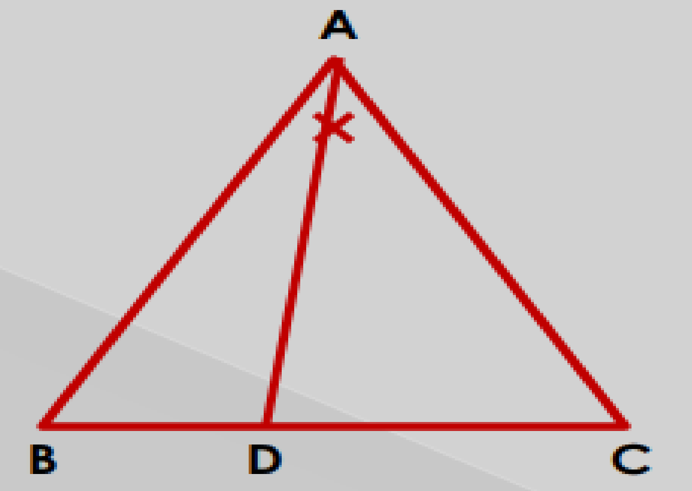




INTERIOR ANGLE BISECTOR THEOREM

ABC is a triangle and AD is the angle bisector.

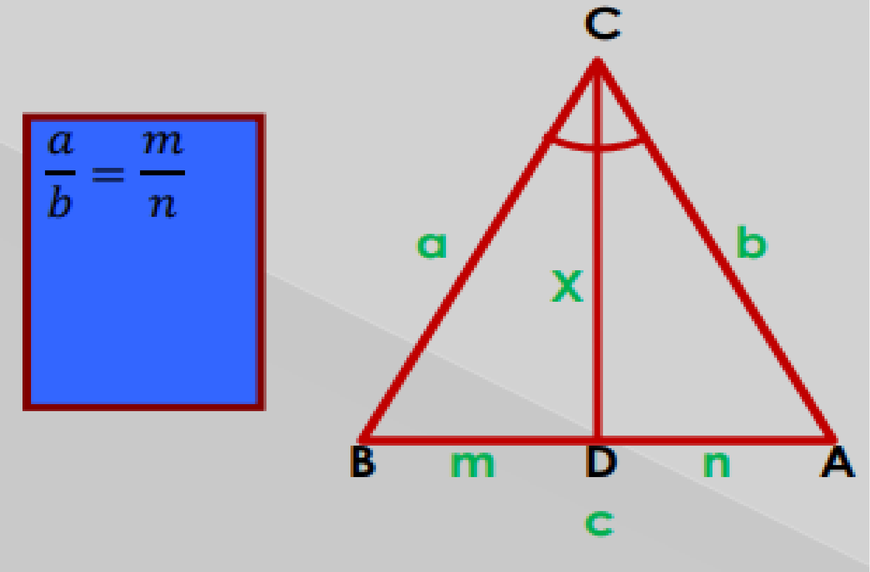
$$\frac{BD}{DC} = \frac{AB}{AC}$$

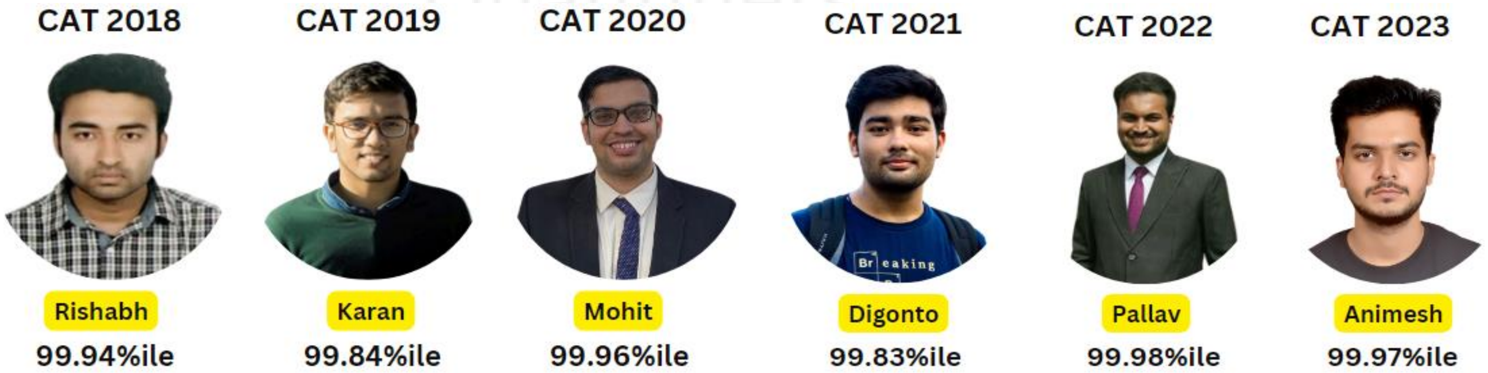


LENGTH OF ANGLE BISECTOR

CD is the angle bisector of angle ACB.

$X^2 = ab - mn$

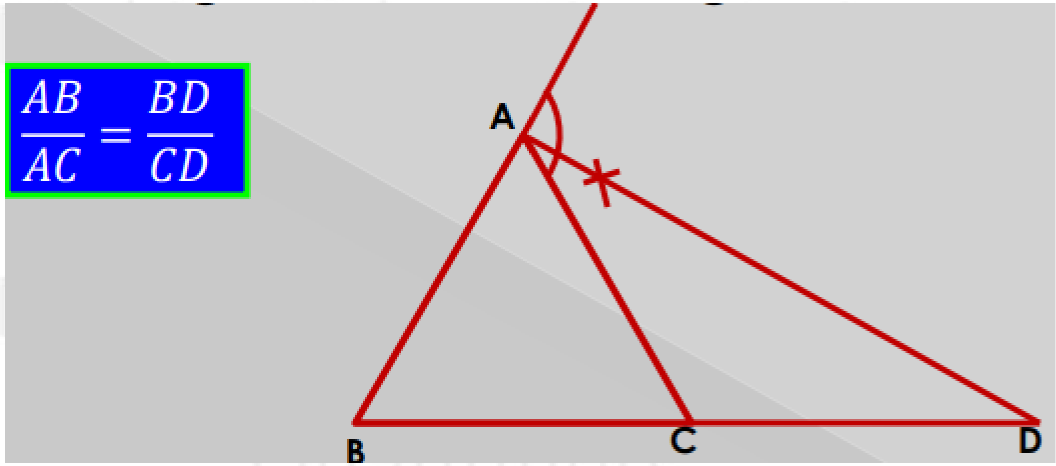




Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

EXTERIOR ANGLE BISECTOR THEOREM

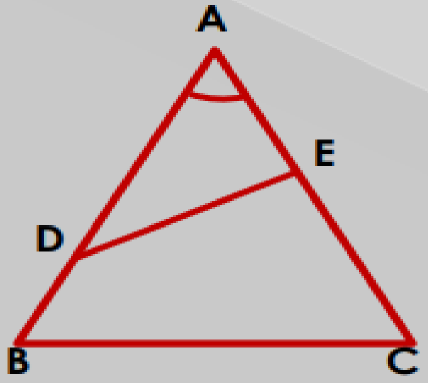
AD is the external angle bisector of angle BAC.



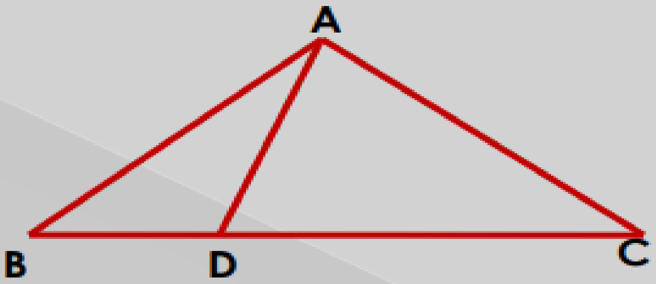
IMPORTANT CONCEPT

If perimeter of a triangle is fixed then area of will be maximum when triangle is equilateral.

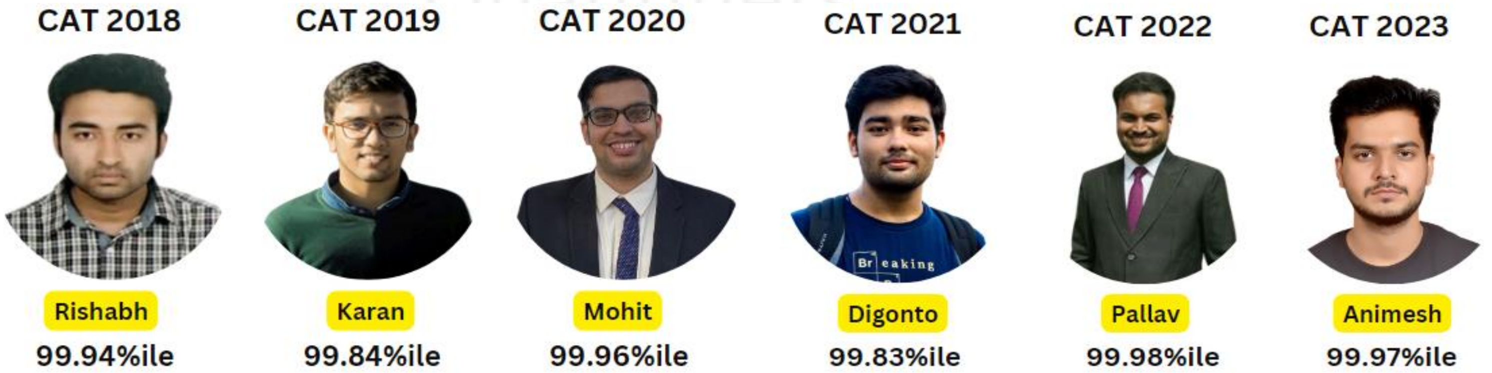
CONCEPT RELATED TO TRIANGLE



$$\frac{ar\Delta ADE}{ar\Delta ABC} = \frac{AD \times AE}{AB \times AC}$$



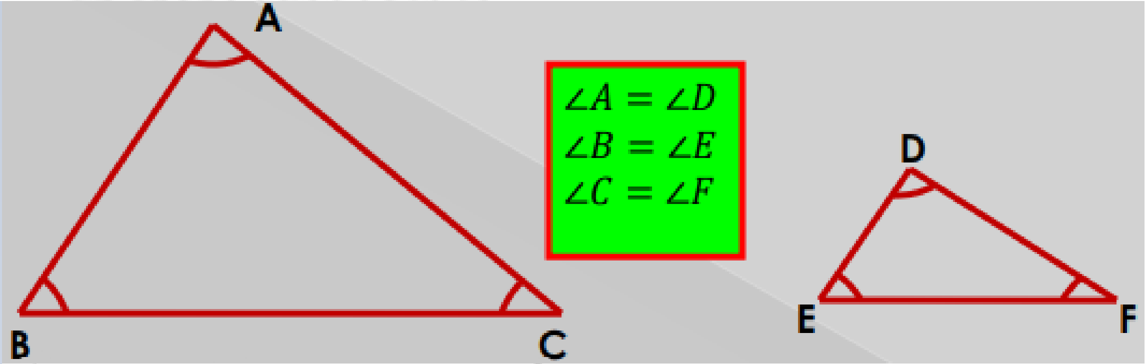
$$\frac{ar\Delta ADB}{ar\Delta ADC} = \frac{BD}{DC}$$



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

SIMILARITY OF TRIANGLE

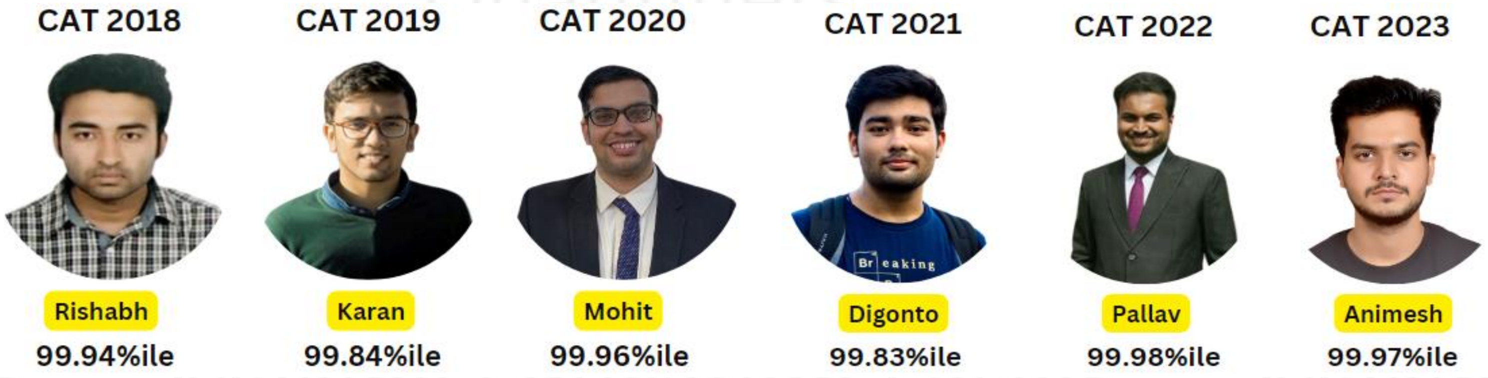
- Size may be different but shape is same.



$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta DEF} = \left(\frac{BC}{EF}\right)^2 = \left(\frac{AC}{DF}\right)^2 = \left(\frac{AB}{DE}\right)^2 = (\text{ratio of Corresponding length})^2$$

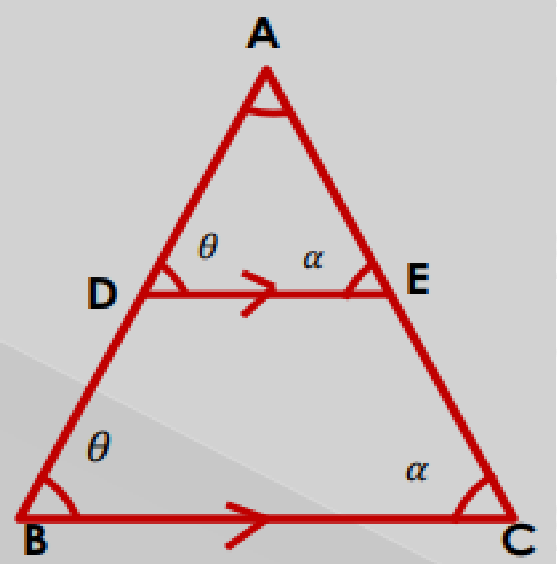
- Corresponding angles are equal.
- Corresponding sides are opposite sides of corresponding angles.

$$\frac{BC}{EF} = \frac{AC}{DF} = \frac{AB}{DE} = \frac{height1}{height2} = \frac{angle bisector1}{angle bisector2} = \frac{median1}{median2} = \frac{perimeter \Delta ABC}{perimeter \Delta DEF}$$



THALES THEOREM

- D & E are two points on AB and AC such that DE||BC.



$$\Delta ADE \approx \Delta ABC$$

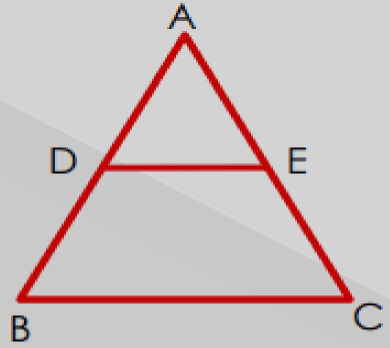
$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

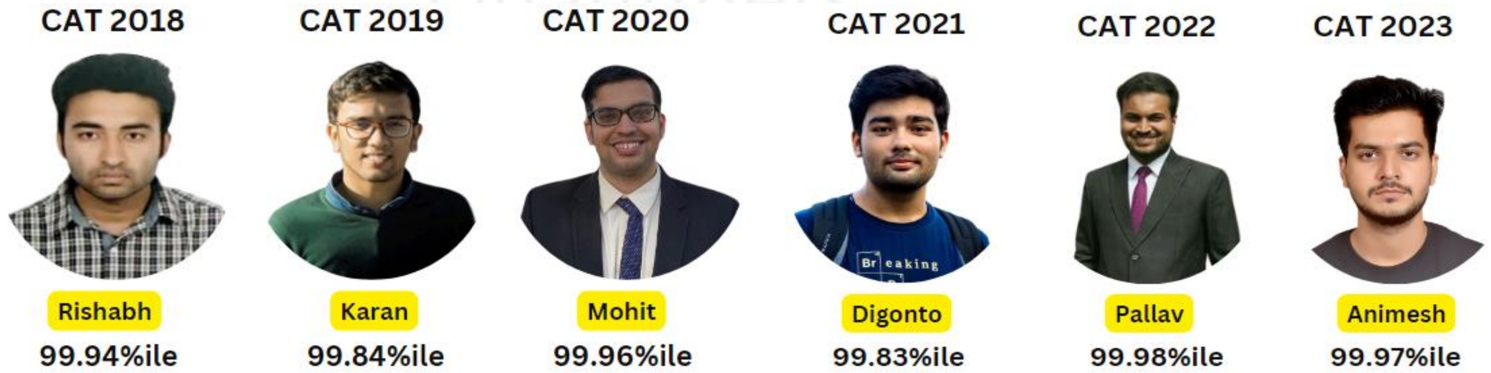
$$\frac{AD}{BD} = \frac{AE}{EC}$$

$$\frac{AD}{AB} = \frac{AE}{AC}$$

REVERSE OF THALES THEOREM

D & E are two points on side AB & AC respectively such that $\frac{AD}{DB}=\frac{AE}{EC}$, then DE || BC.





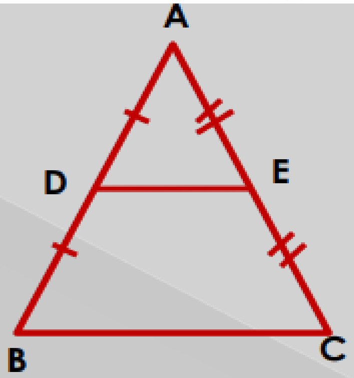
MID - POINT THEOREM

- D & E are midpoints of side AB & AC respectively

$$\frac{AD}{DB} = \frac{AE}{EC} = 1 \Rightarrow DE||BC \& \Delta ADE \approx \Delta ABC$$

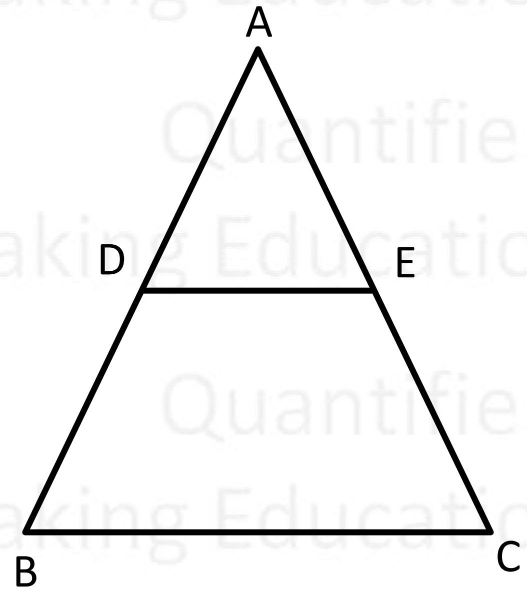
$$DE = \frac{BC}{2}$$

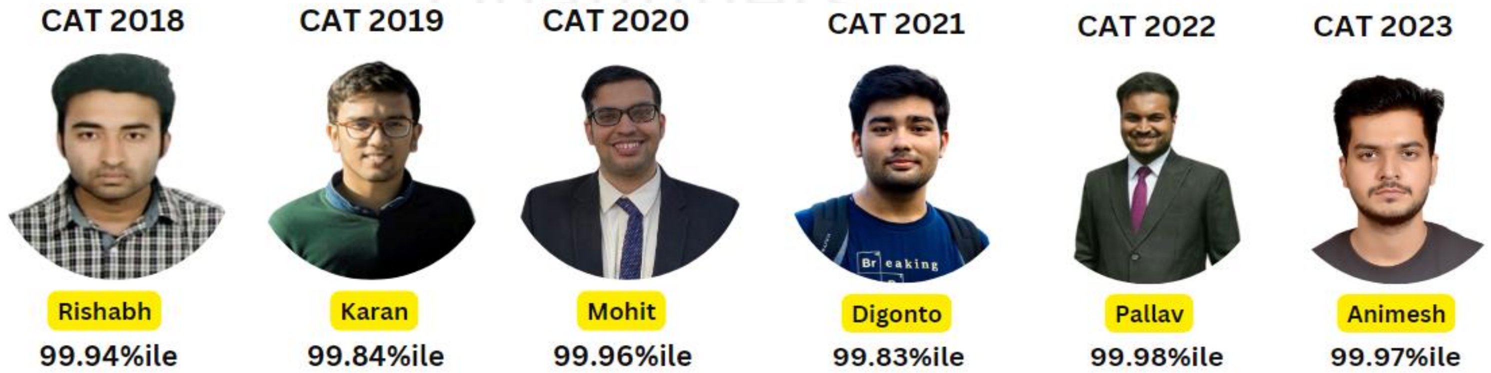
$$\frac{ar\Delta ADE}{ar\Delta ABC} = \frac{1}{4} \& \frac{ar\Delta ADE}{arDECB} = \frac{1}{3}$$



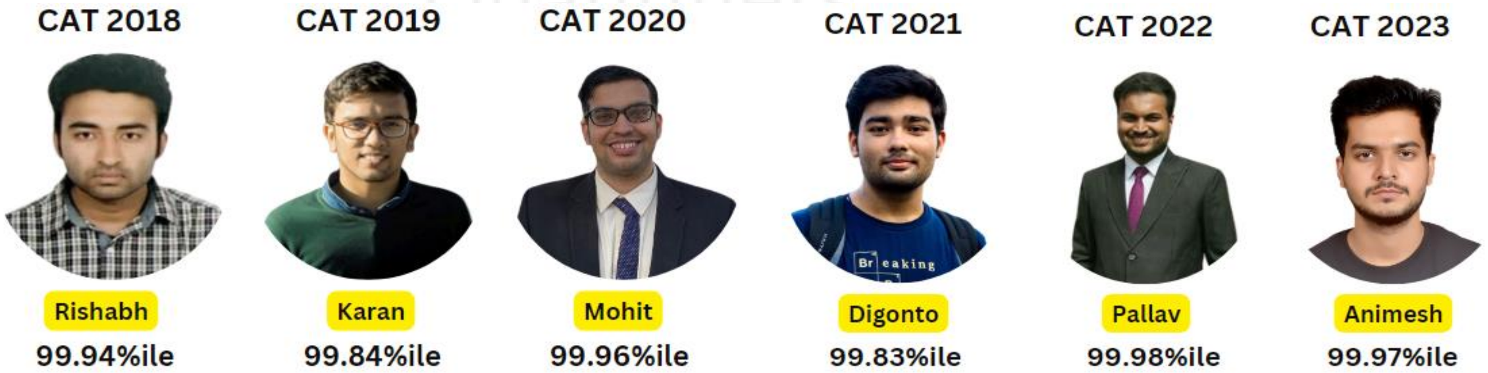
REVERSE OF MID - POINT THEOREM

- D is mid point of side AB and A line DE is drawn parallel to BC, then
- E will be the mid point of side AC. (AE = AC)





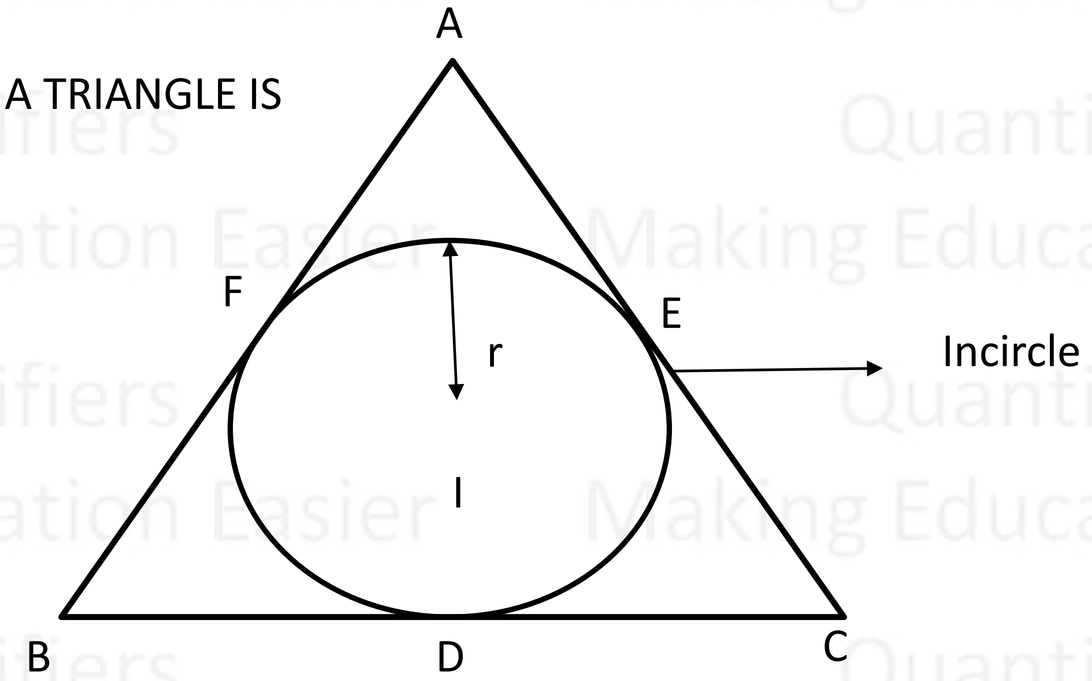
INCENTER & CIRCUMCENTER



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

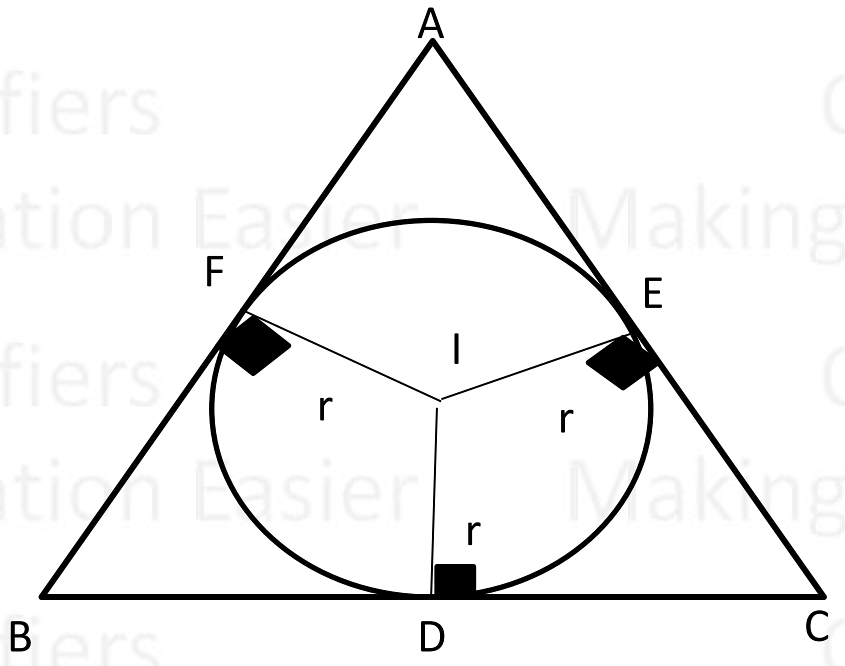
INCENTER

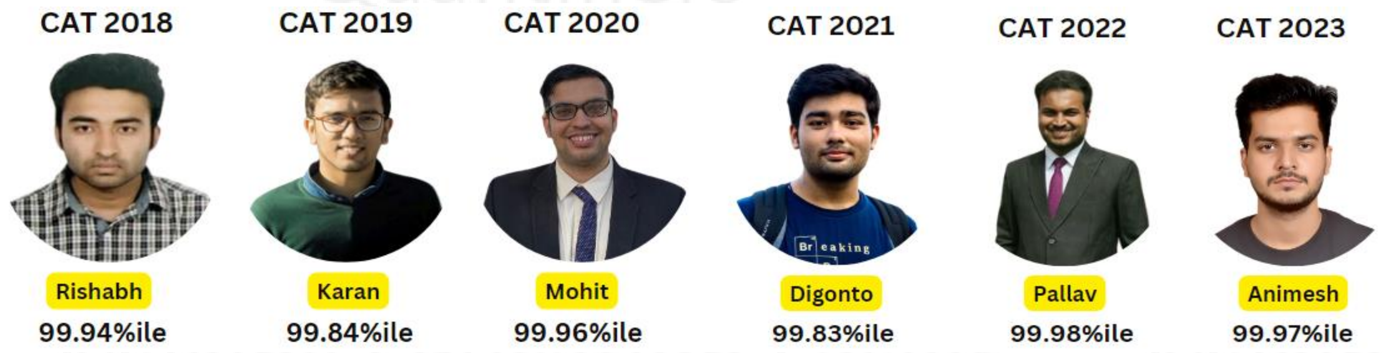
CENTER OF INCIRCLE OF A TRIANGLE IS CALLED INCENTER



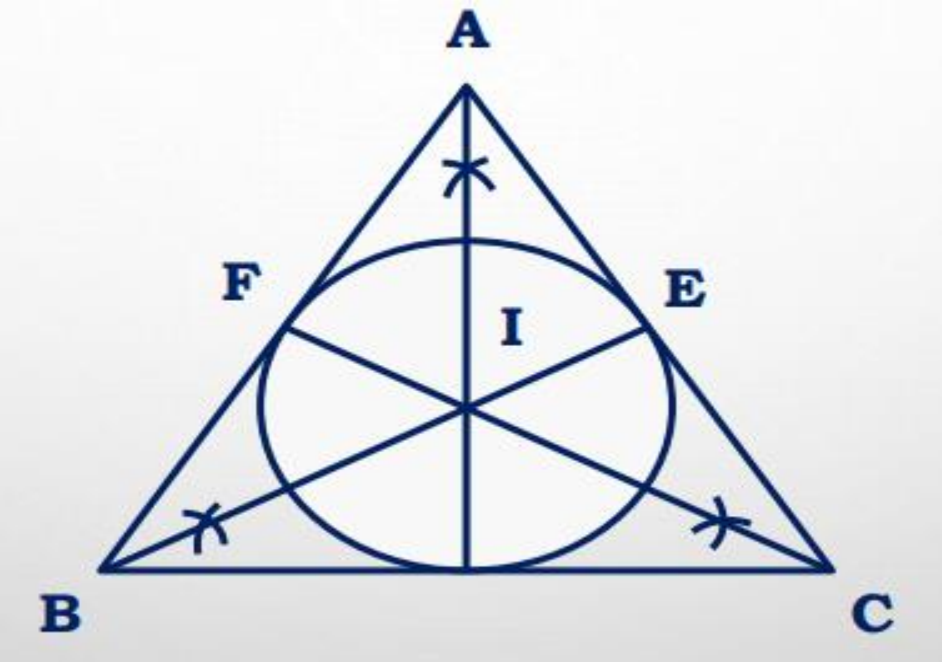
I = Incenter
r = Inradius

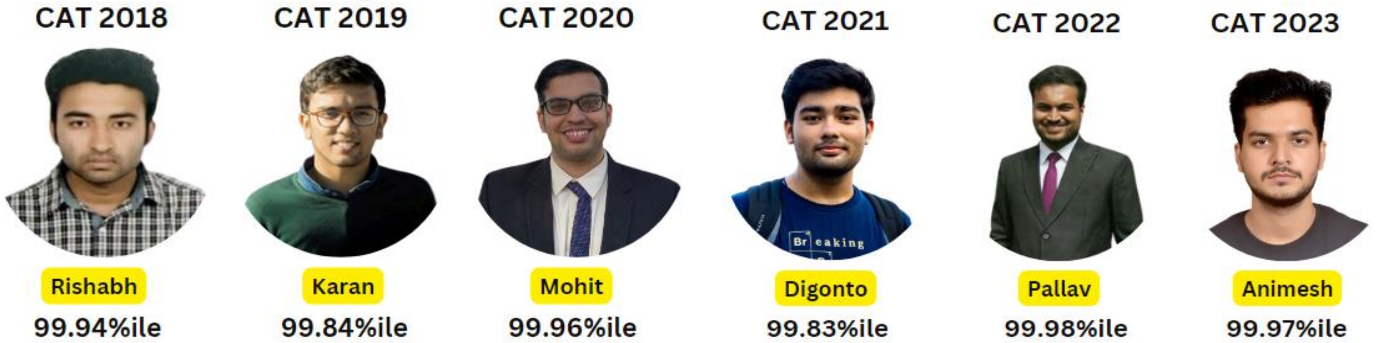
INCENTER IS THE INTERECTION OF ANGLE- BISECTORS OF A TRIANGLE



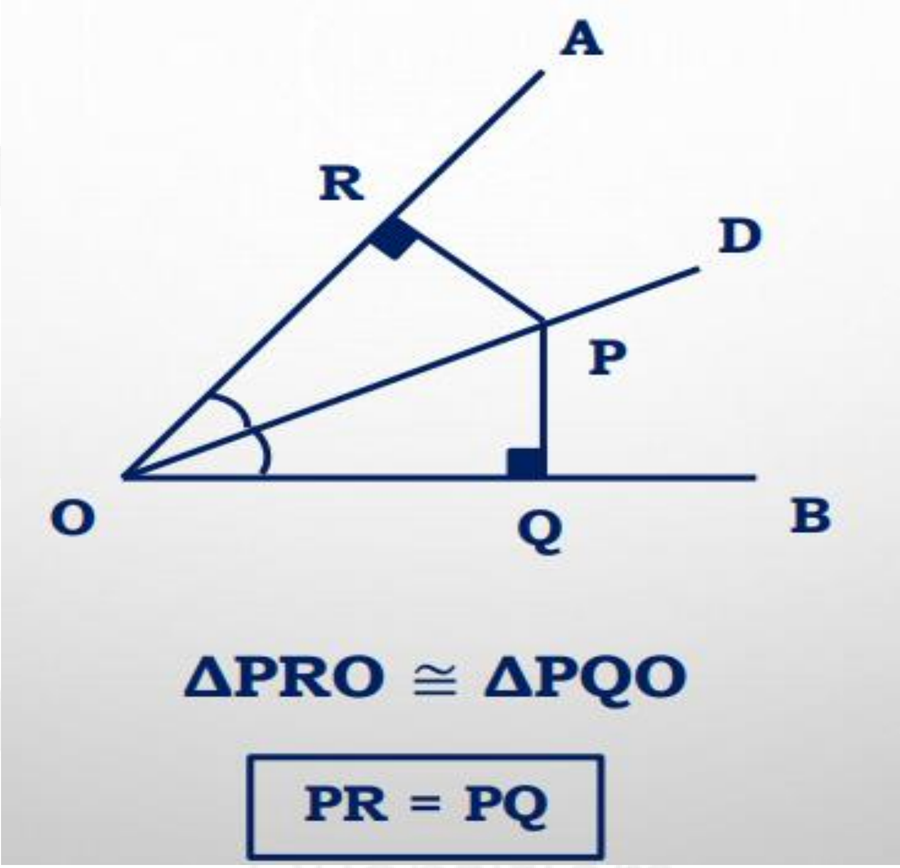


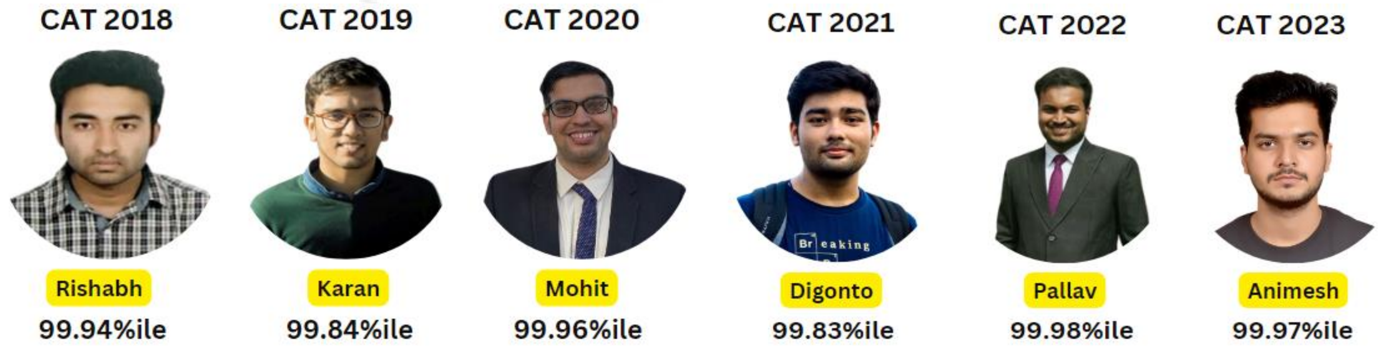
INCENTER IS EQUIDISTANT FROM ALL THE SIDES OF THE TRIANGLE



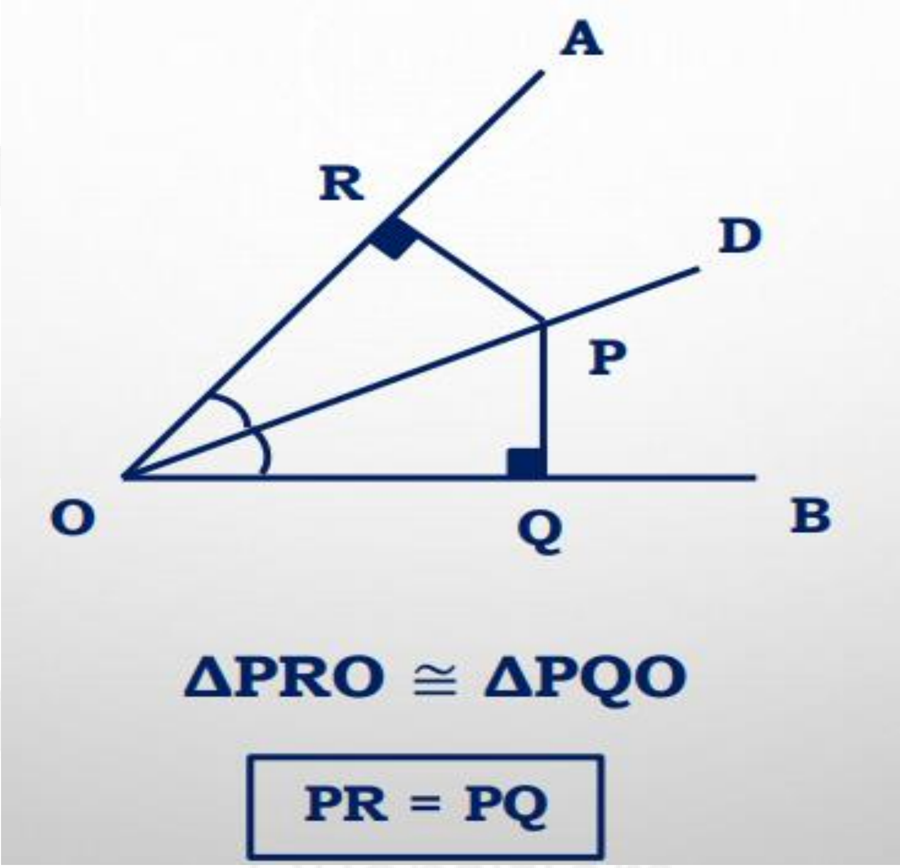


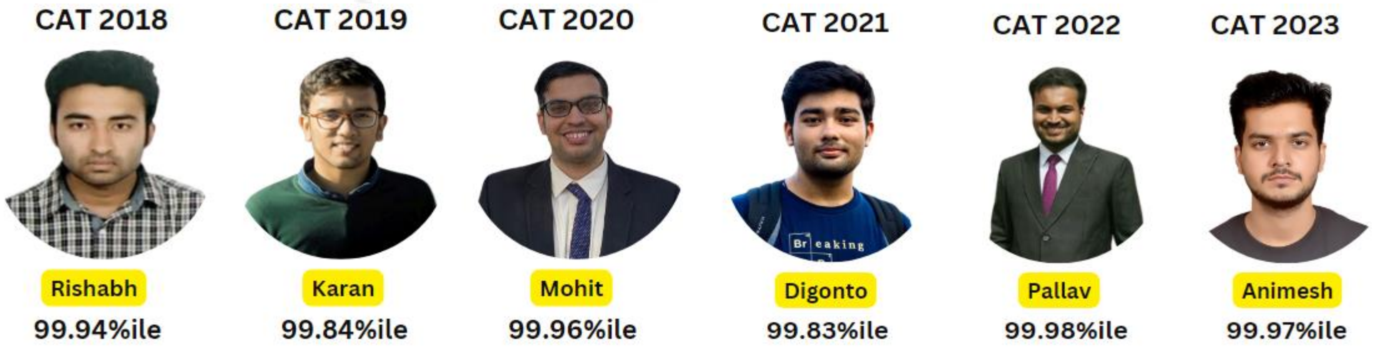
OD IS ANGLE BISECTOR OF ΔAOB





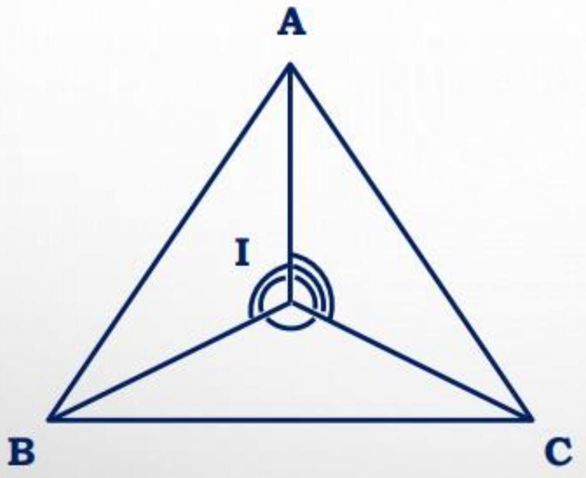
OD IS ANGLE BISECTOR OF ΔAOB





Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

OD IS ANGLE BISECTOR OF ΔAOB



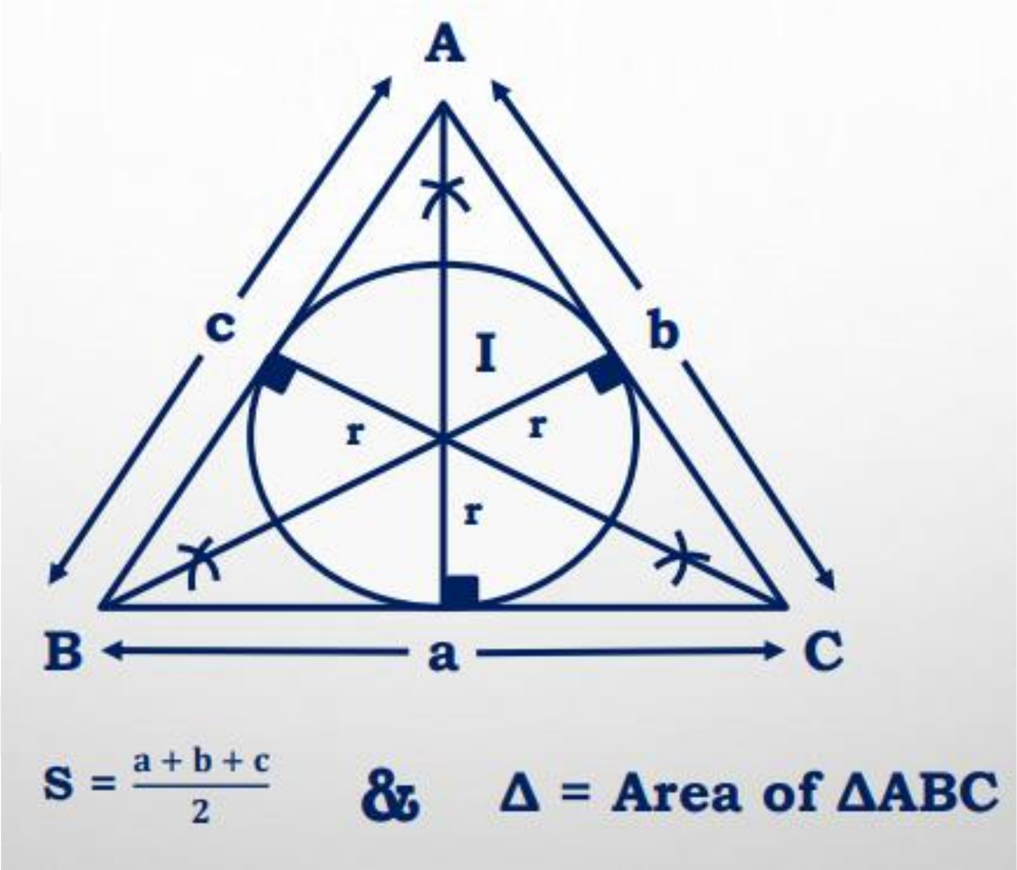
$$\angle BIC = 90^\circ + \frac{\angle A}{2}$$

$$\angle CIA = 90^\circ + \frac{\angle B}{2}$$

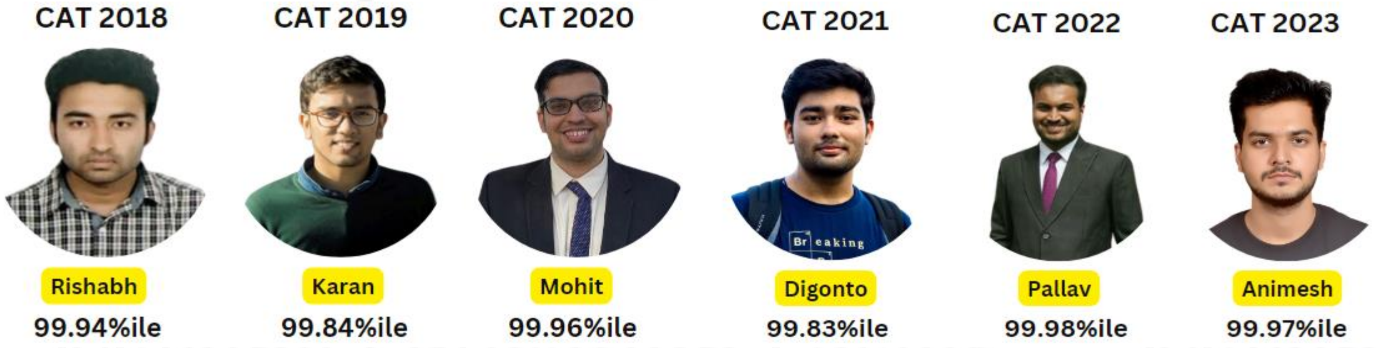
$$\angle AIB = 90^\circ + \frac{\angle C}{2}$$

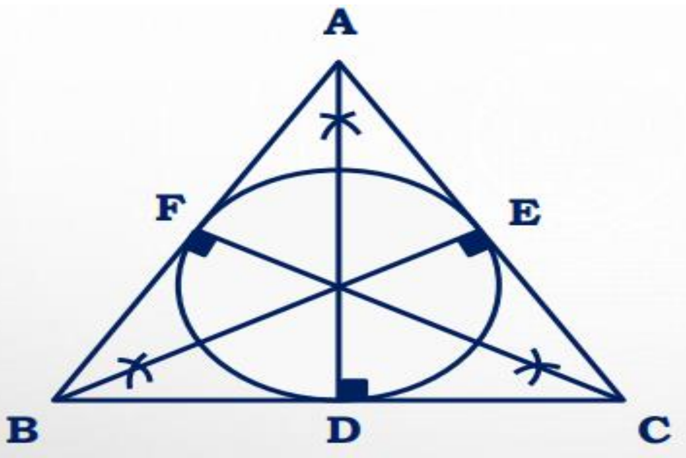
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

INRADIUS OF A TRIANGLE (R)



r= area/s



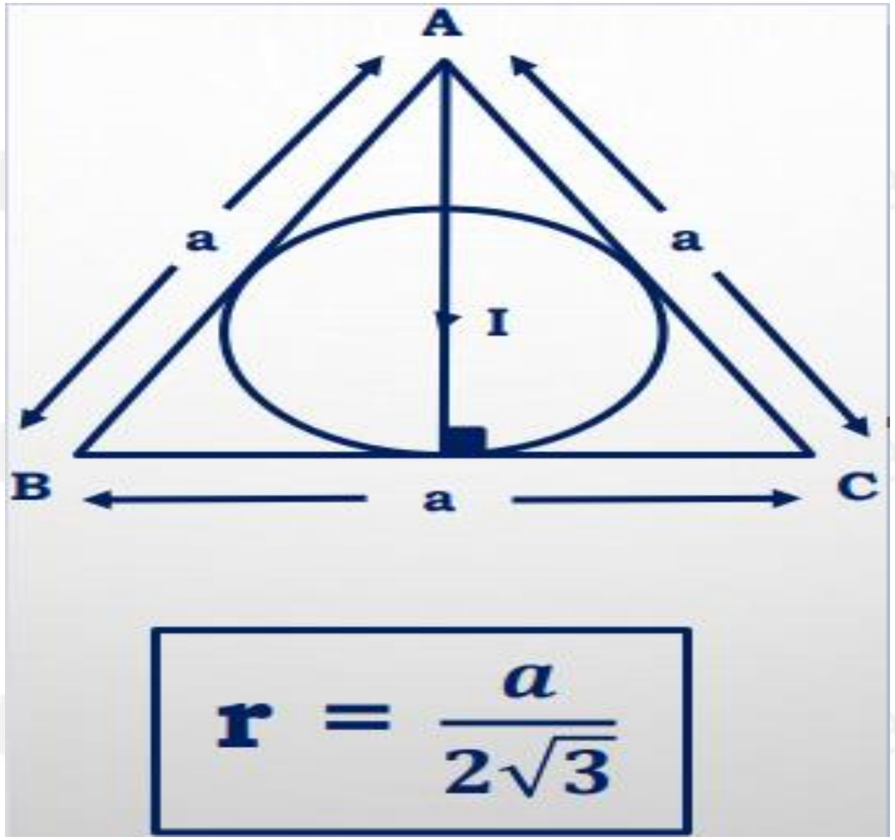


$$\left.\begin{aligned} AD &= h_1 \\ BE &= h_2 \\ CF &= h_3 \end{aligned}\right\} \text{Altitudes}$$

r = Inradius

$$\boxed{\frac{1}{r} = \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}}$$

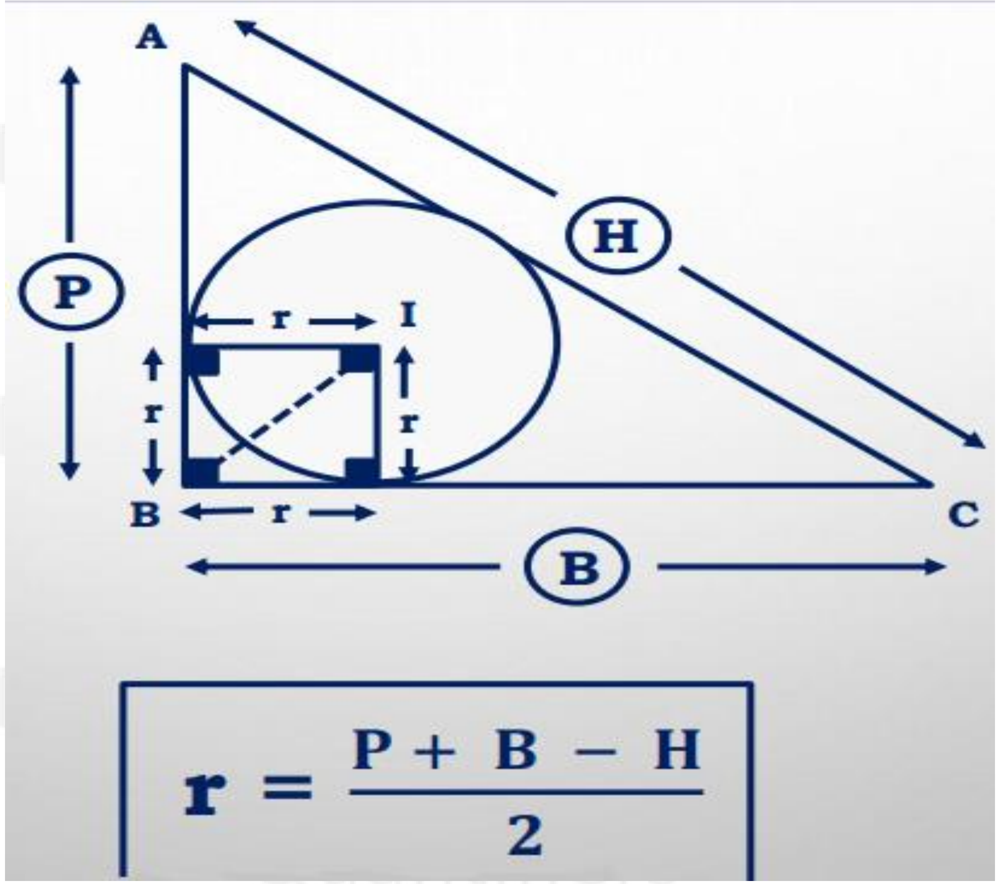
INRADIUS OF EQUILATERAL TRIANGLE

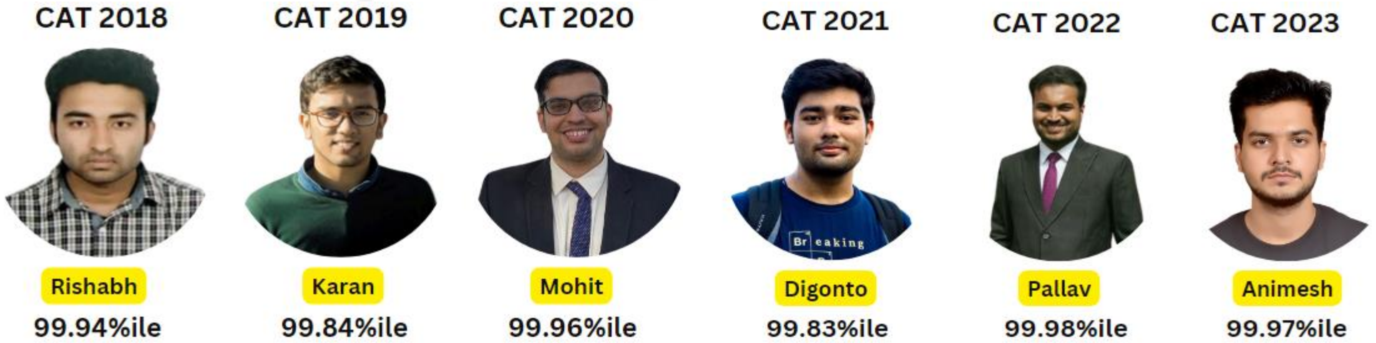


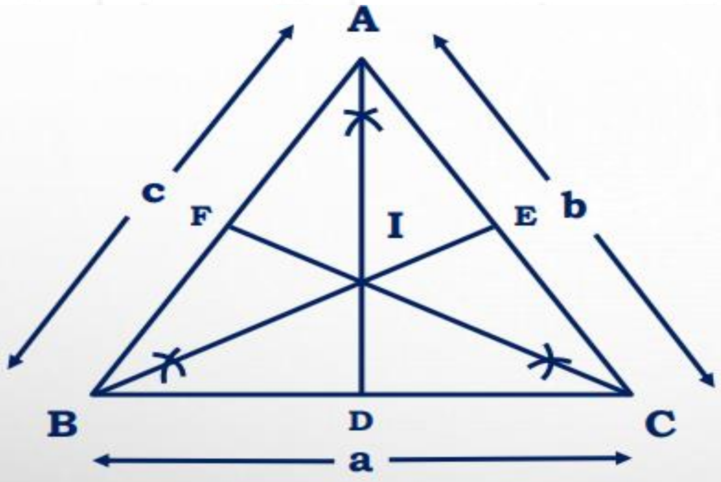
CAT 2018 | CAT 2019 | CAT 2020 | CAT 2021 | CAT 2022 | CAT 2023

Rishabh 99.94%ile | Karan 99.84%ile | Mohit 99.96%ile | Digonto 99.83%ile | Pallav 99.98%ile | Animesh 99.97%ile

INRADIUS OF RIGHT-ANGLED TRIANGLE







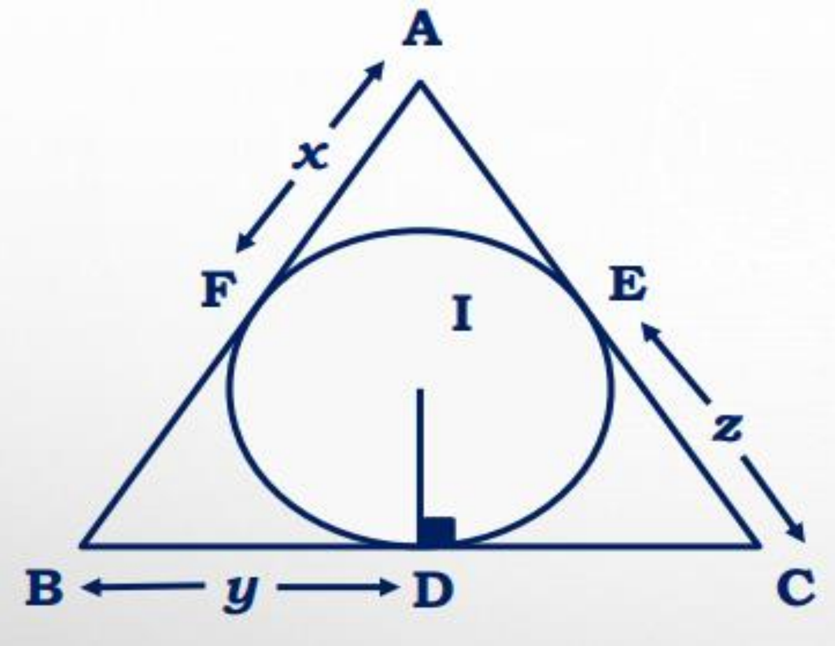
I = INCENTER

AD, BE and CF are angle-bisectors

$$\frac{AI}{ID} = \frac{c+b}{a}$$

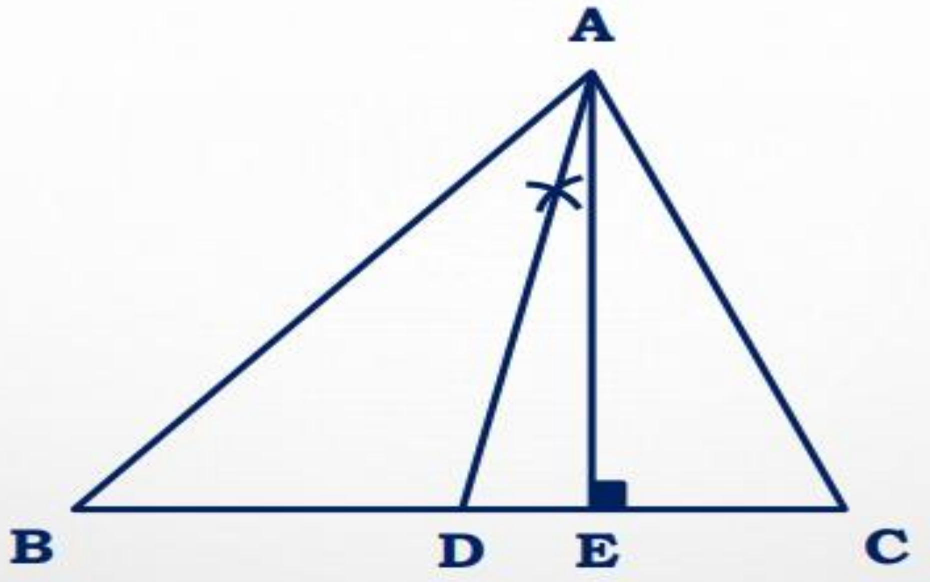
$$\frac{BI}{IE} = \frac{a+c}{b}$$

$$\frac{CI}{IF} = \frac{a+b}{c}$$

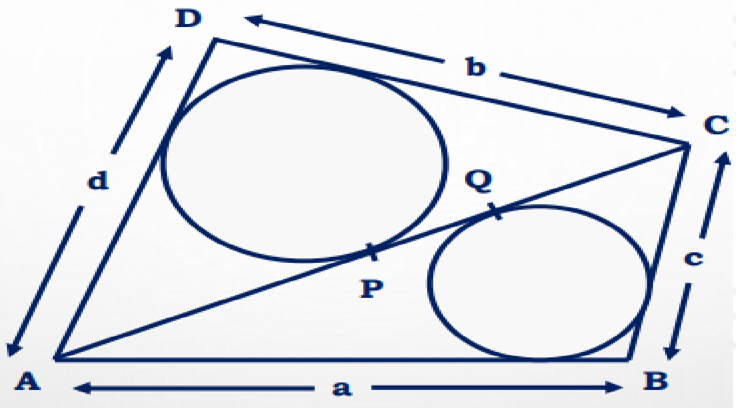


$$s = x + y + z$$

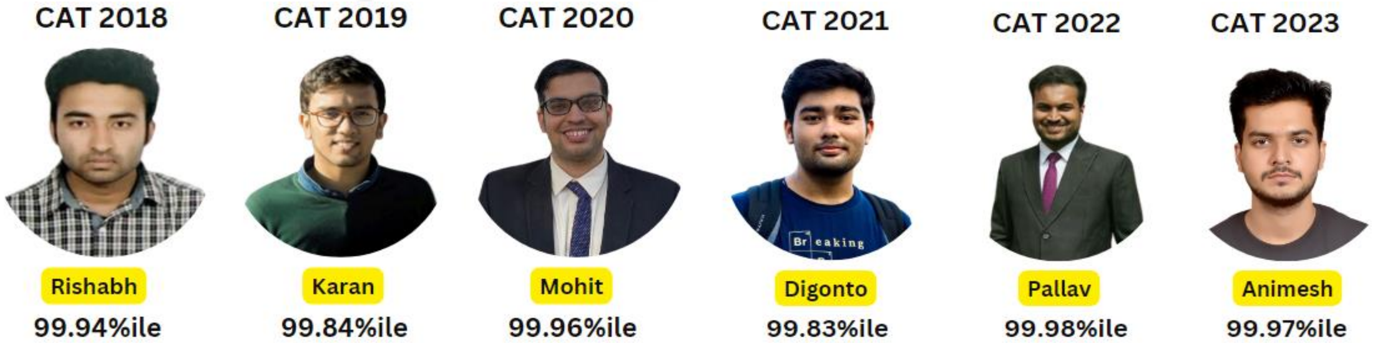
$$r = \sqrt{\frac{xyz}{x+y+z}}$$



$$\angle ADE = \frac{\angle C - \angle B}{2}$$

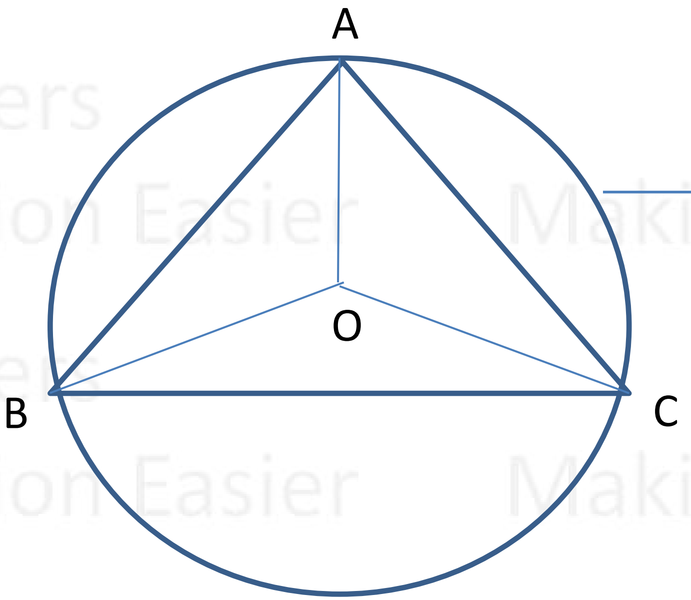


$$PQ = \frac{(a+b)\sim(c+d)}{2}$$



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

CIRCUMCENTER (परिकेन्द्र)

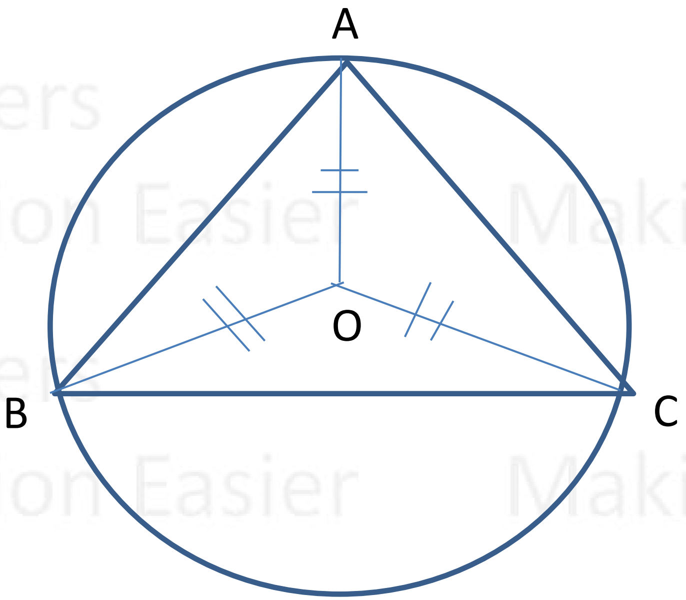


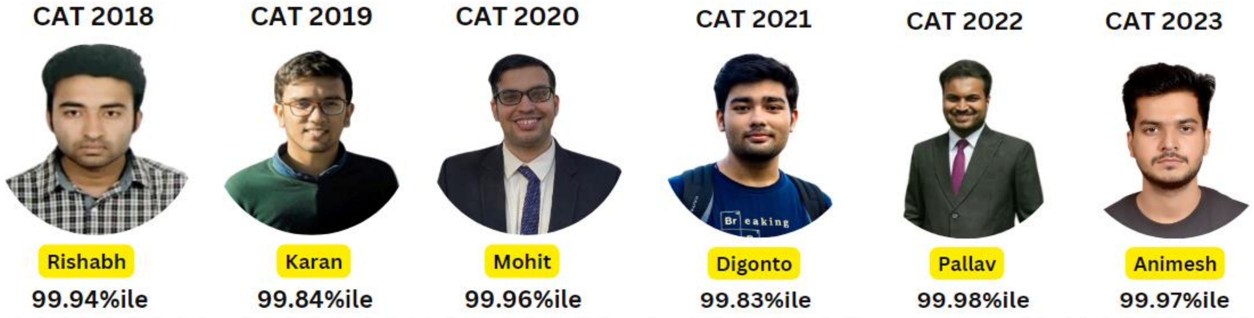
Circumcircle
(परिवृत्)

O is the Circumcenter
(परिकेन्द्र)

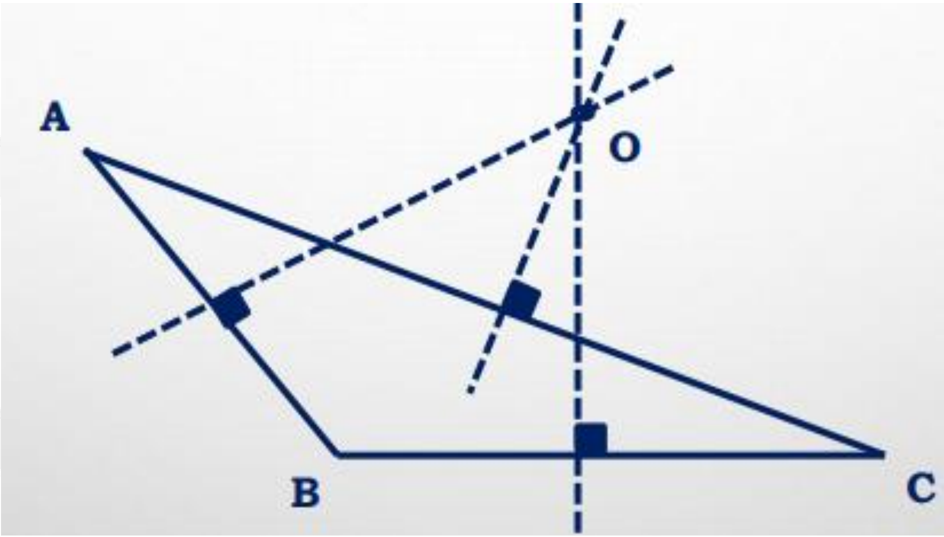
OA = OB = OC= R → Circumradius
(परित्रिज्या)

Circumcenter is equidistant from all the three vertices of triangle

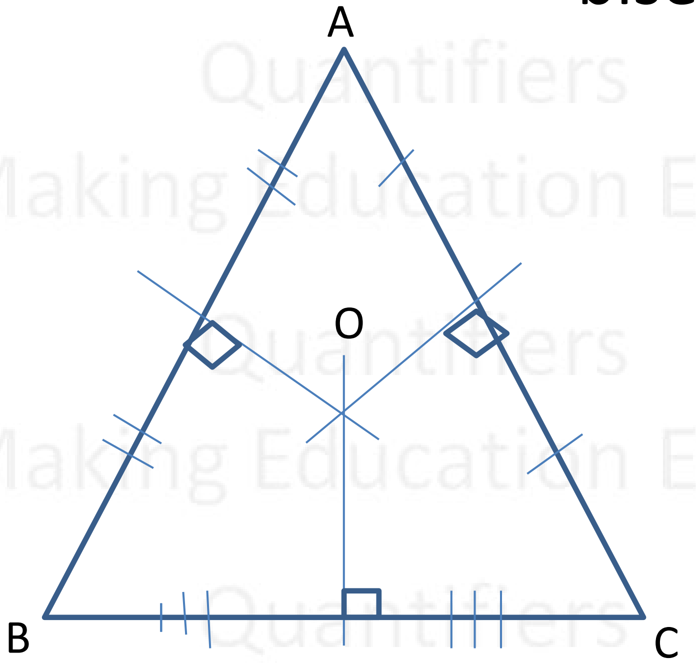


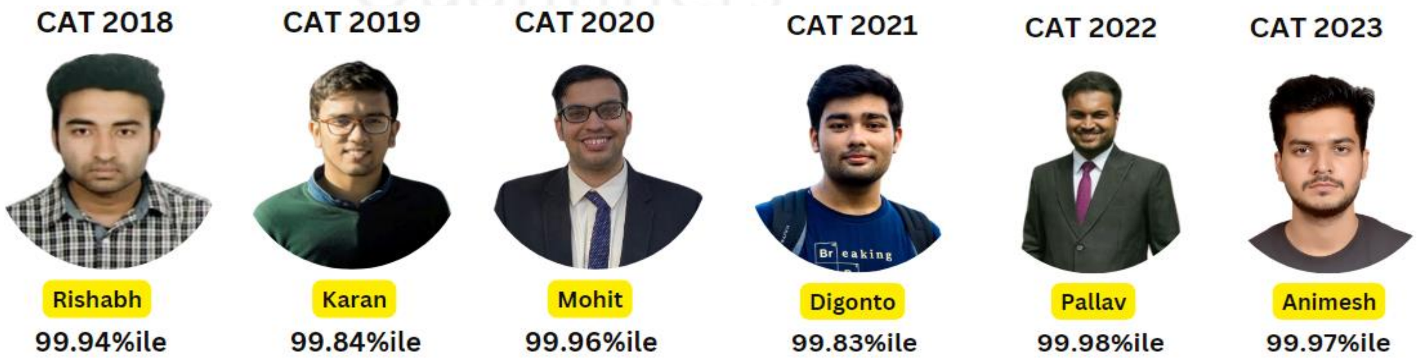


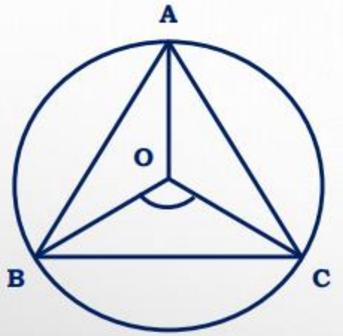
Circumcenter may lie outside or on the triangle



Intersection point of all the perpendicular bisectors of sides of triangle is called circumcenter



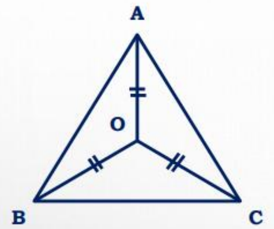




$\angle BOC = 2\angle BAC$

$\angle AOC = 2\angle ABC$

$\angle AOB = 2\angle ACB$

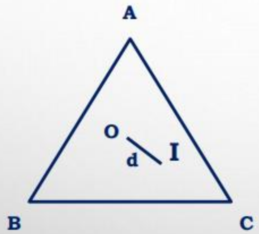


$$R = \frac{a}{2sinA} = \frac{b}{2sinB} = \frac{c}{2sinC}$$

Δ = Area of triangle ABC

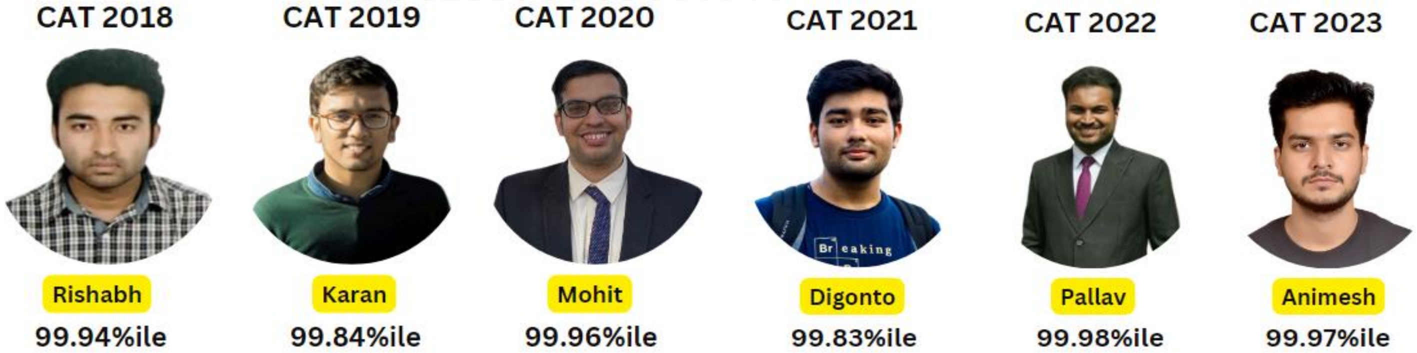
$$R = \frac{abc}{4\Delta}$$

DISTANCE BETWEEN CIRCUMCENTER AND INCENTER



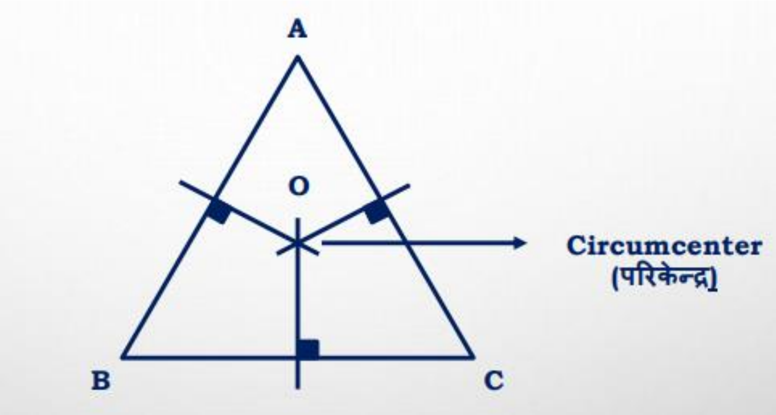
$$OI = d = \sqrt{R^2 - 2Rr}$$

$$\frac{R}{r} \geq 2$$

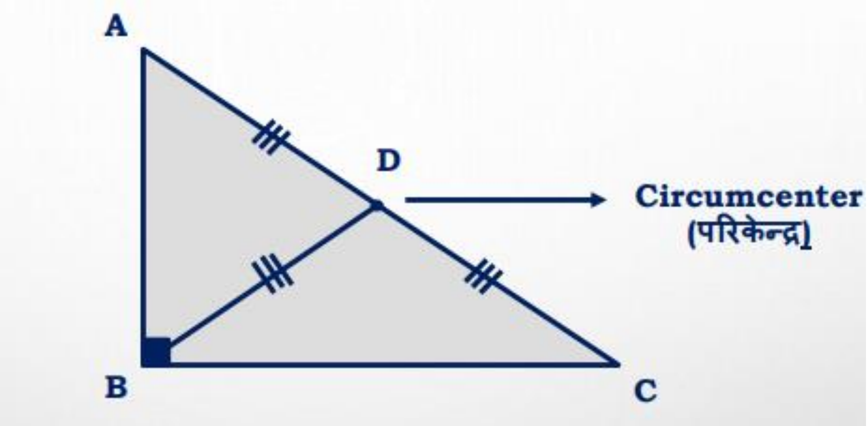


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

POSITION OF CIRCUMCENTER

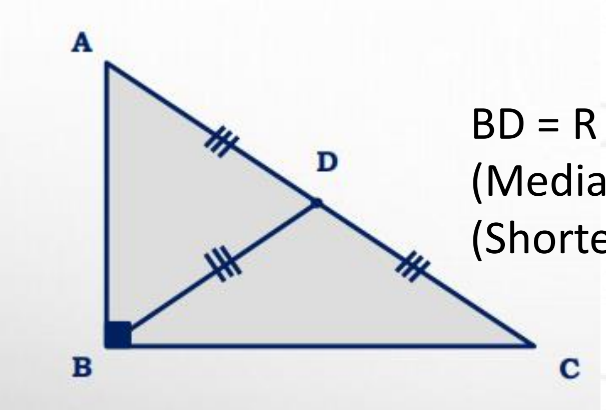


In acute angled triangle,
circumcenter lies inside the triangle



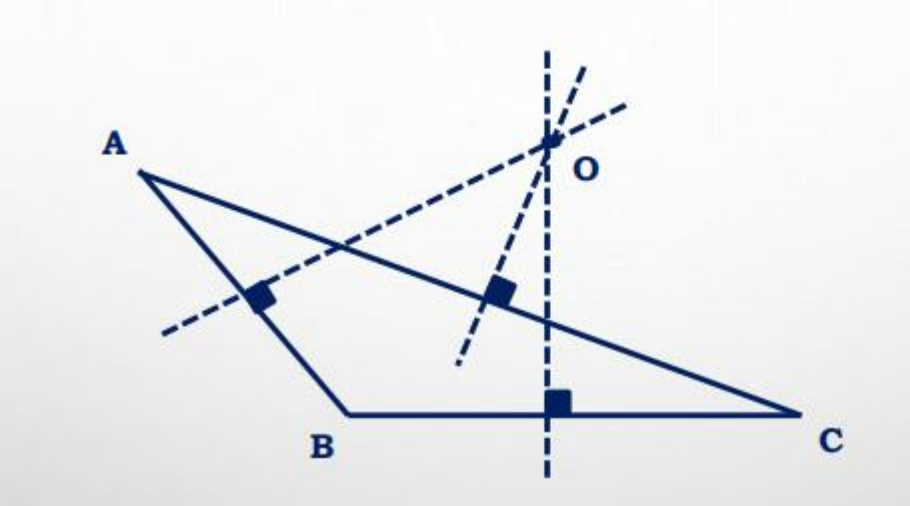
OA = OB = OC = R = $\boldsymbol{Hypotenuse}/2$

In right angled triangle, circumcenter lies on the mid-point of hypotenuse of the triangle



BD = R = $\boldsymbol{Hypotenuse}/2$
(Median of Hypotenuse)
(Shortest Median)

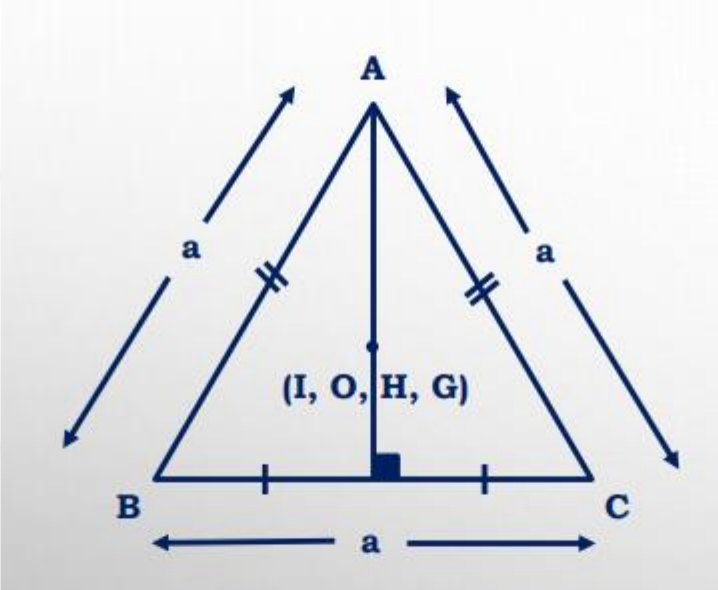
POSITION OF CIRCUMCENTER



In obtuse angled triangle, circumcenter lies outside the triangle

Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

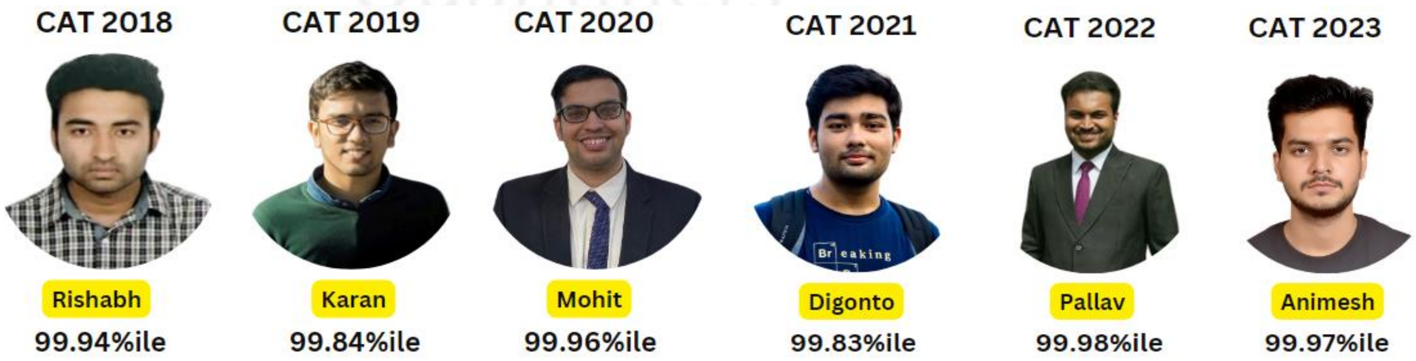
EQUILATERAL TRIANGLE



$$R = \frac{a}{\sqrt{3}} \qquad r = \frac{a}{2\sqrt{3}}$$

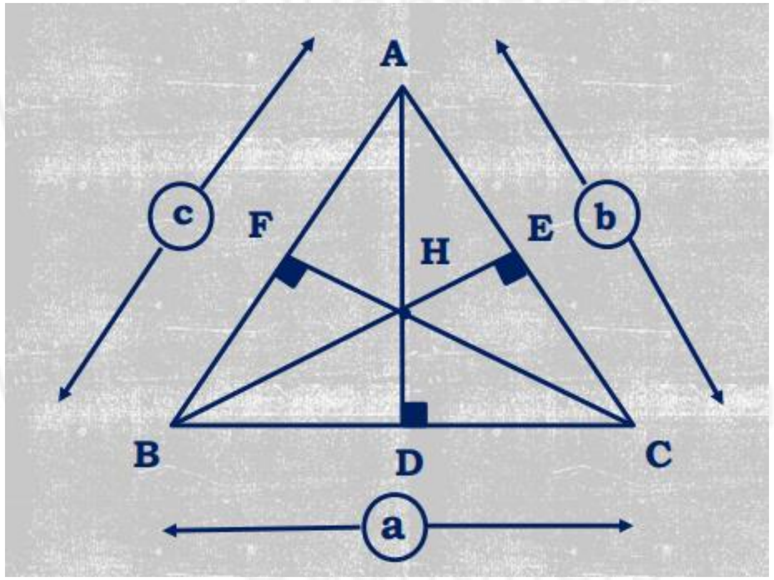
R : r = 2 : 1

Incenter, Circumcenter, orthocenter and centroid all lie on same point in equilateral triangle



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

ORTHOCENTER

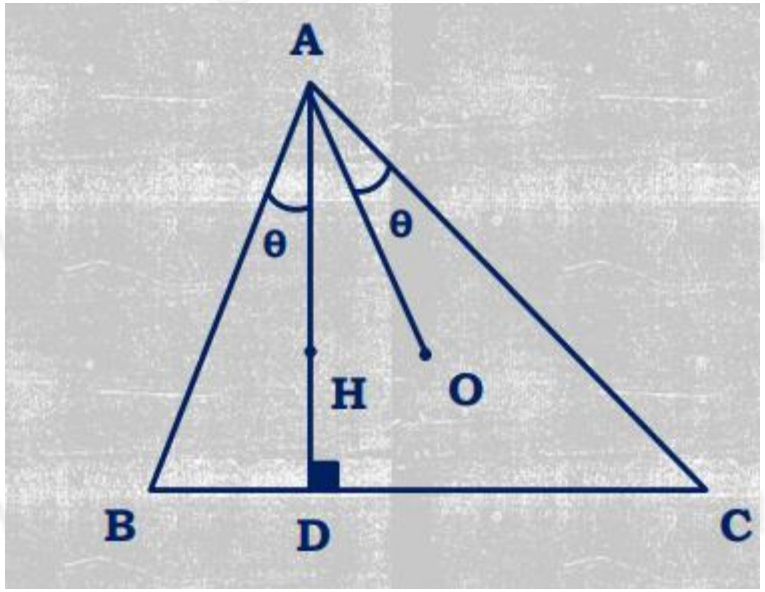


- Orthocenter is the intersection of all 3 altitudes of triangle
- Orthocenter may lie outside the triangle

H = Orthocenter (लंबकेन्द्र)

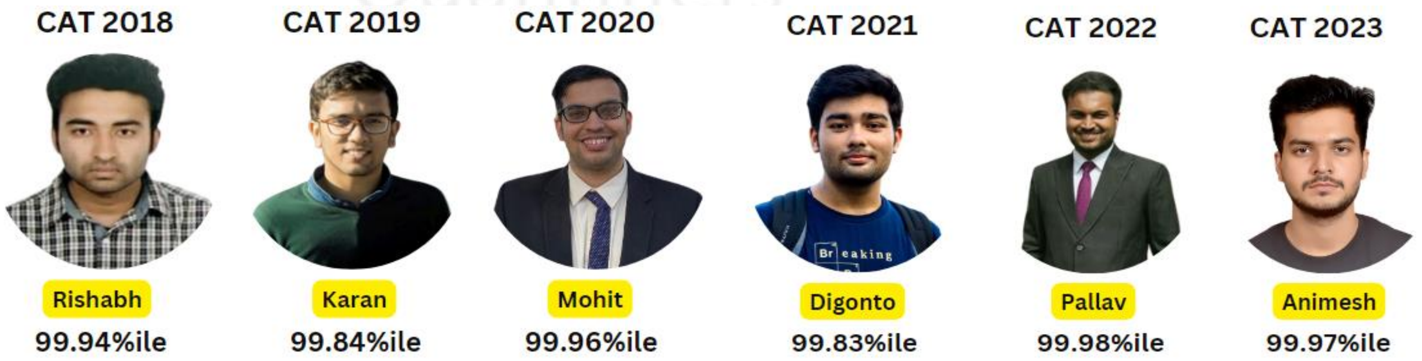
AD = h1, BE = h2 & CF = h3

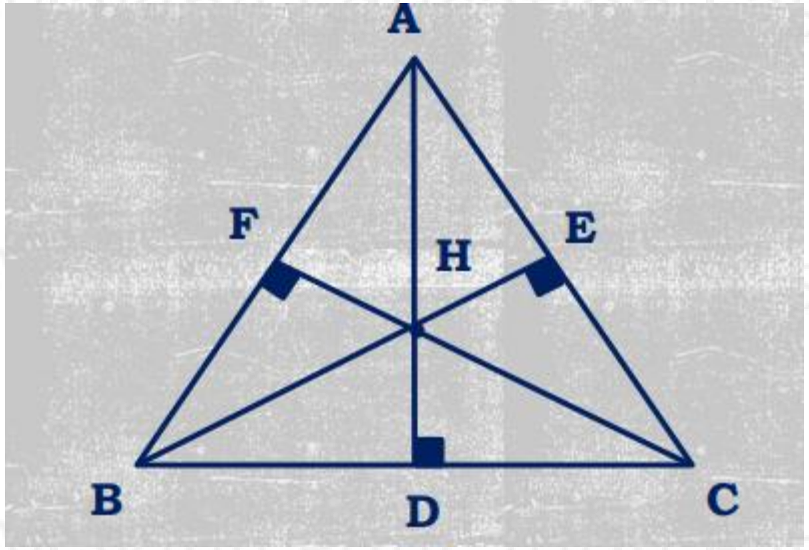
h1 : h2 : h3 = $\mathbf{1}/\mathbf{a}$: $\mathbf{1}/\mathbf{b}$: $\mathbf{1}$/c



H = Orthocenter (लंबकेन्द्र)
O = Circumcenter (परिकेन्द्र)

∠BAH = ∠OAC



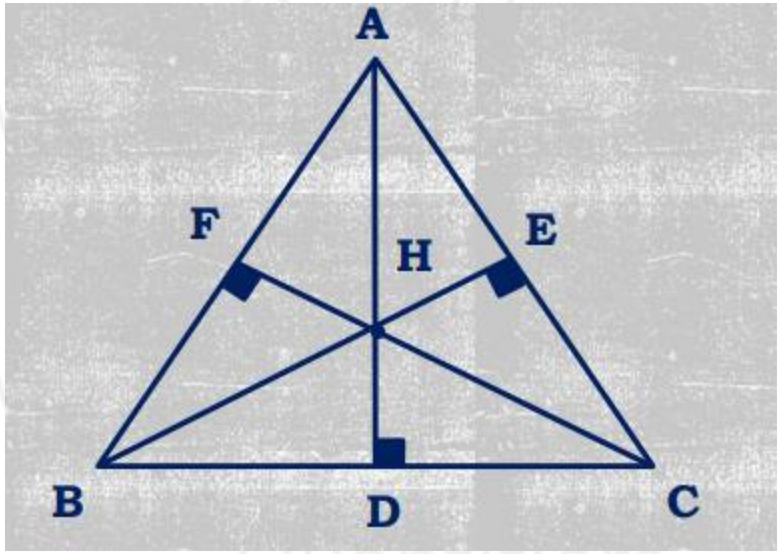


(AB + BC + CA) > (AD + BE + CF)

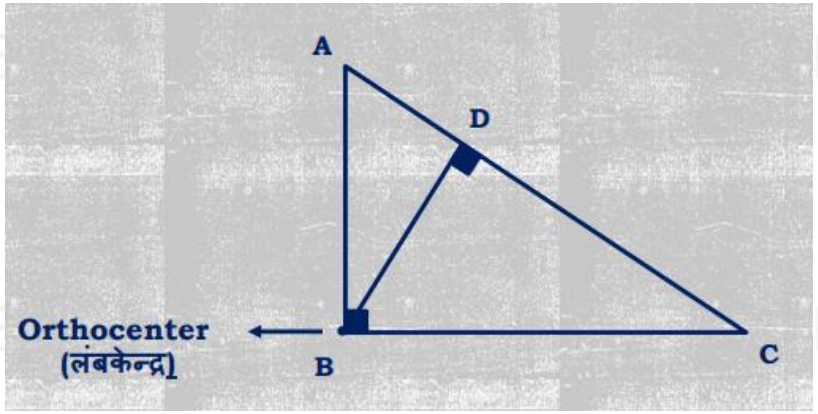
(AB + BC + CA) /(AD + BE + CF) > 1

Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

POSITION OF ORTHOCENTER

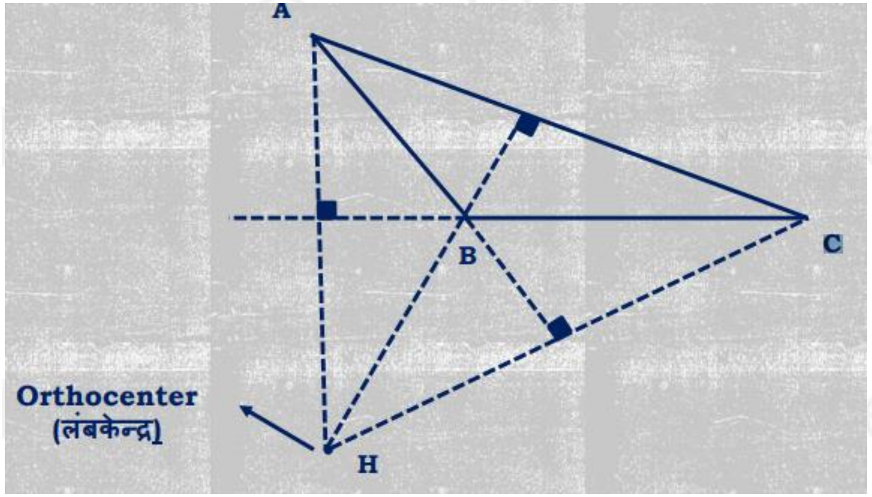


In acute angled triangle, orthocenter lies inside the triangle

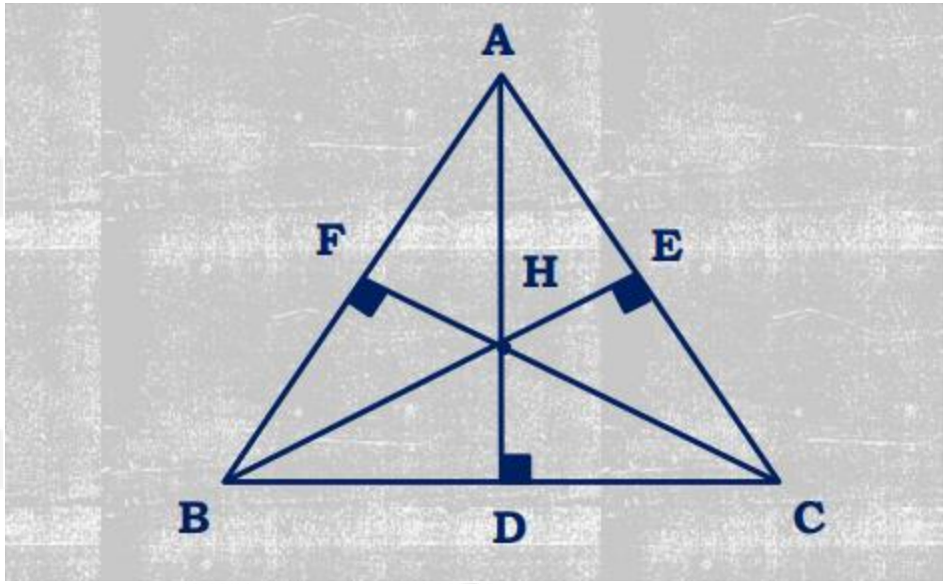


In right angled triangle, orthocenter lies at 90° vertex of the triangle

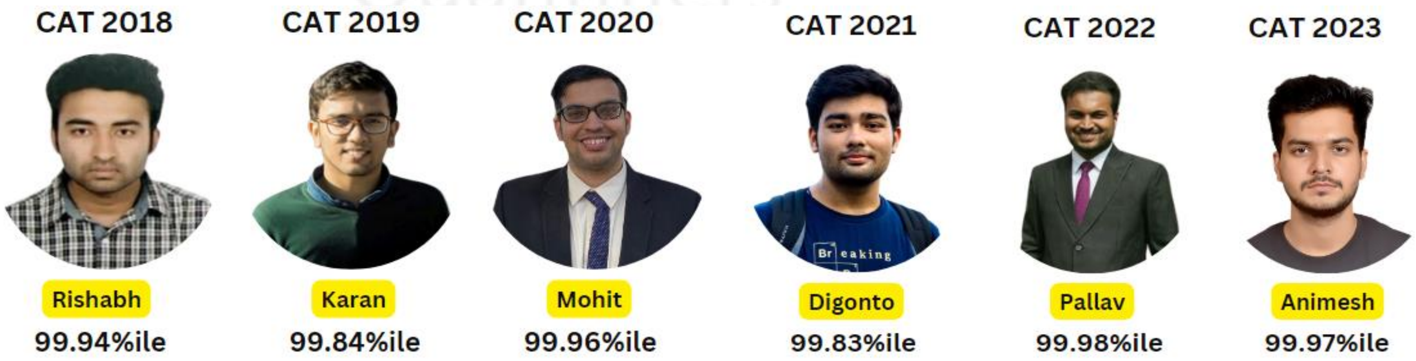
POSITION OF ORTHOCENTER



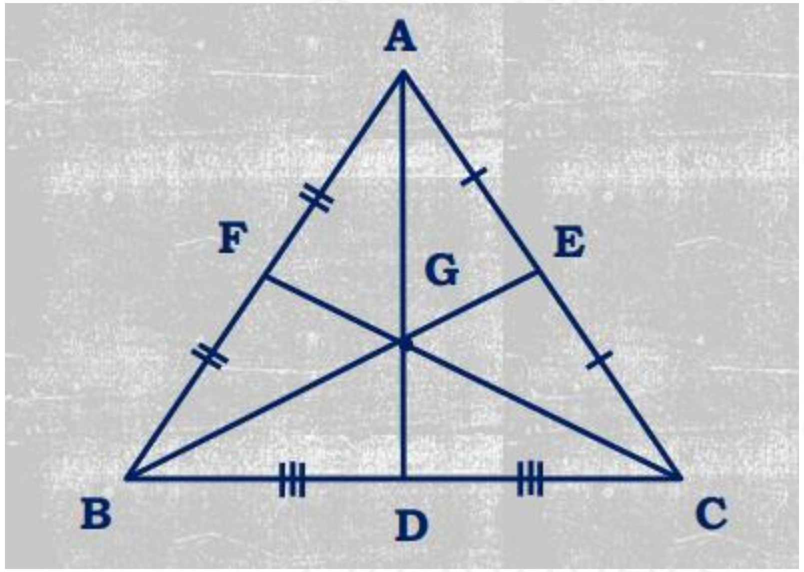
In obtuse angled triangle, orthocenter lies outside the triangle (near the largest angle)



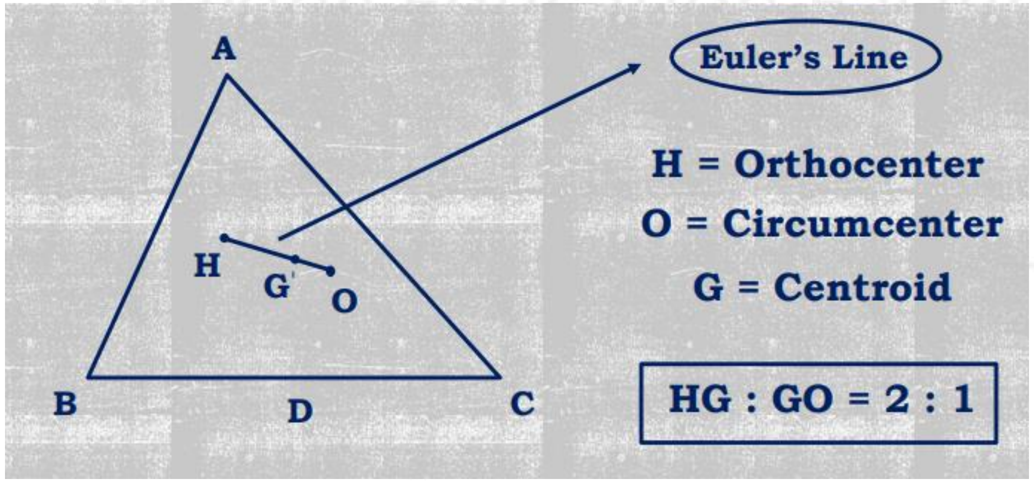
AH × HD = BH × HE = CH × HF



CENTROID

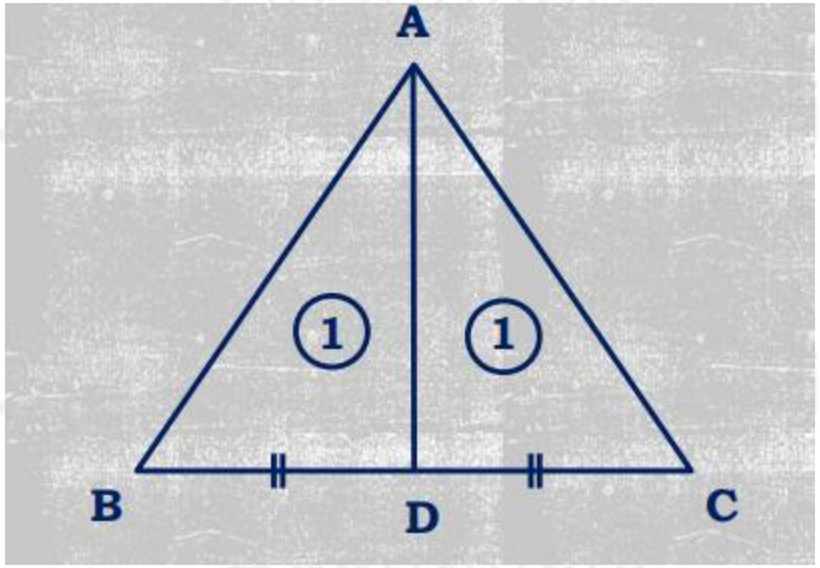


AG : GD = BG : GE = CG : GF = 2 : 1



H, G and O are always collinear (i.e. they lie in a straight line)

MEDIAN

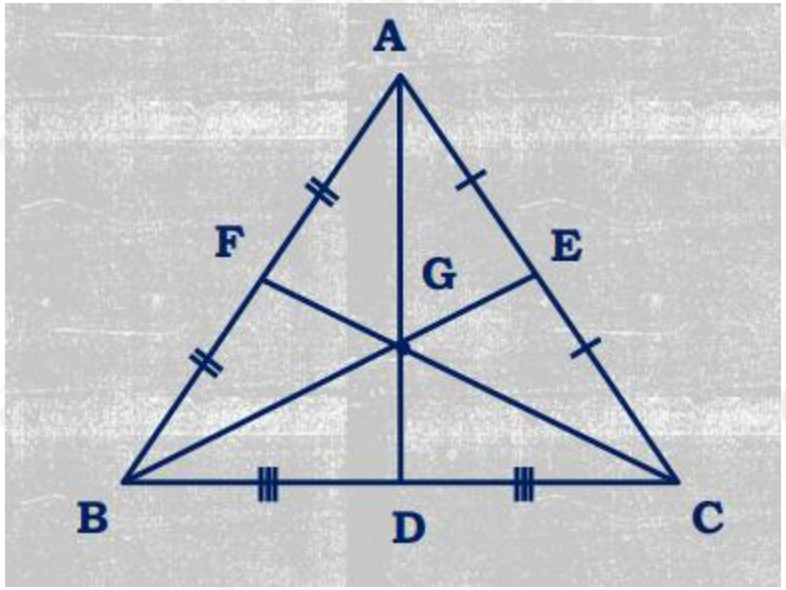


BD = DC

Area of ΔADB = Area of ΔADC

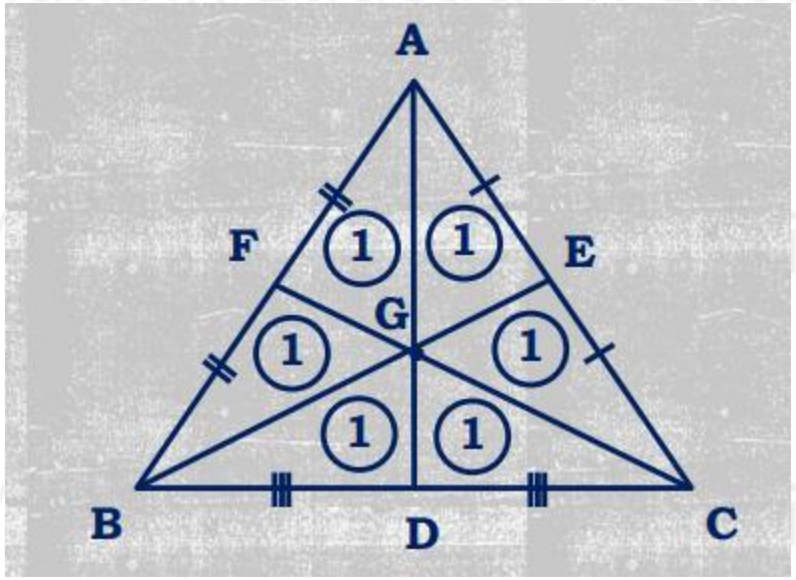
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

CENTROID

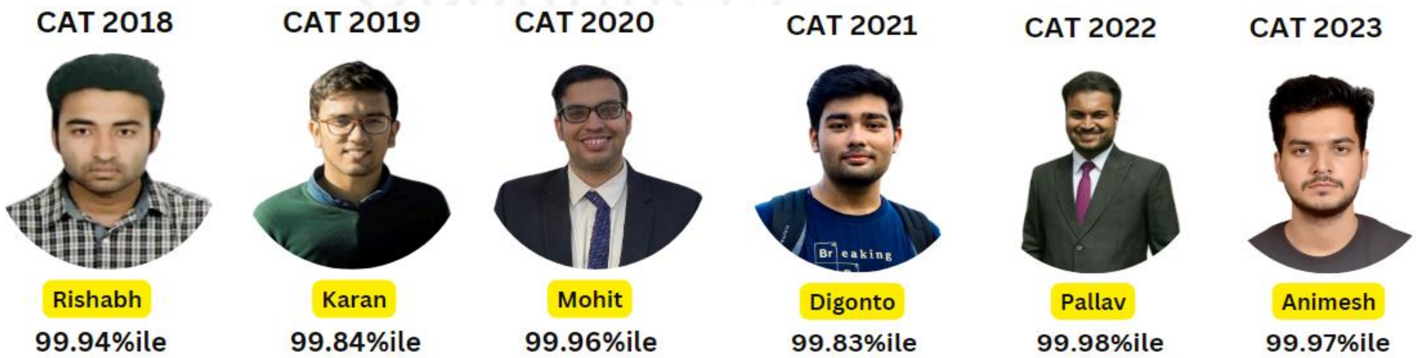


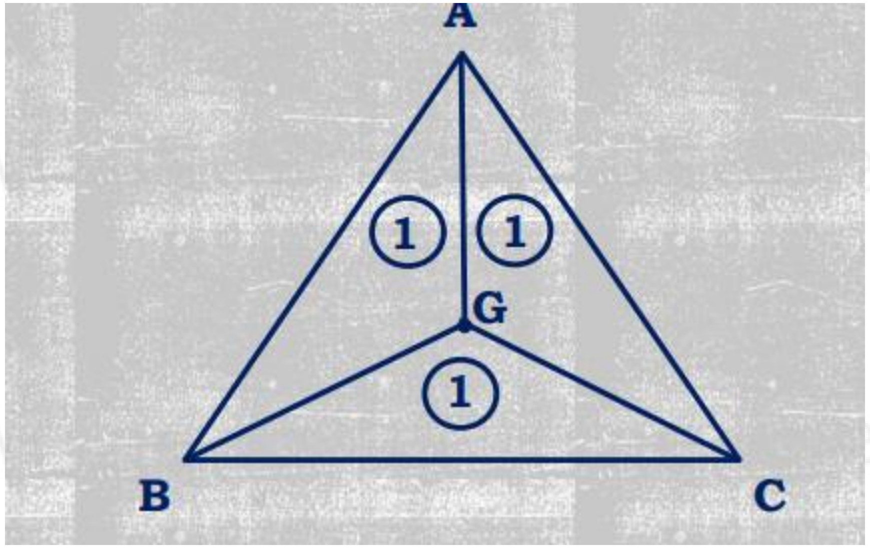
Centroid is the intersection point of all 3 medians of triangle ➢
Centroid always lies inside the triangle

G = Centroid

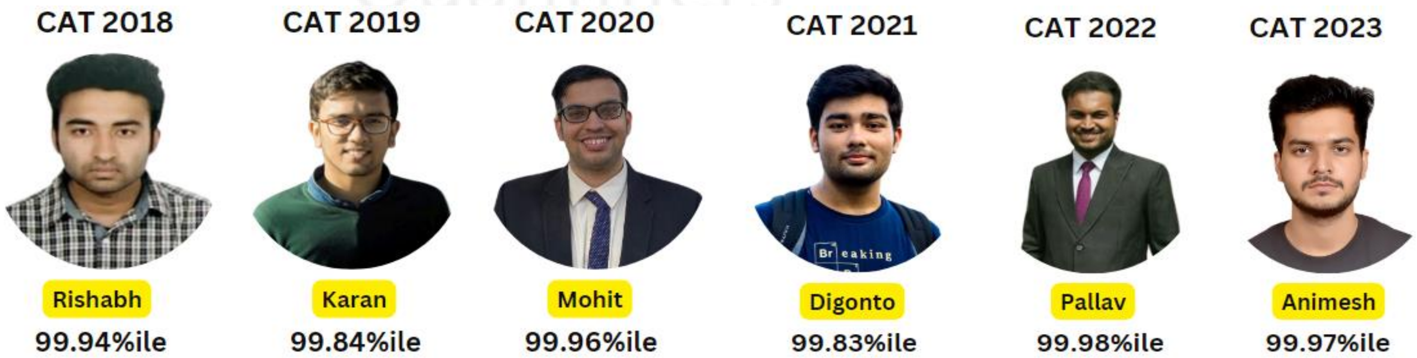


ΔBGD = ΔGDC = ΔGEC = ΔGAE = ΔGFA = ΔGFB = $\mathbf{1}/\mathbf{6}$ th Area of ΔABC

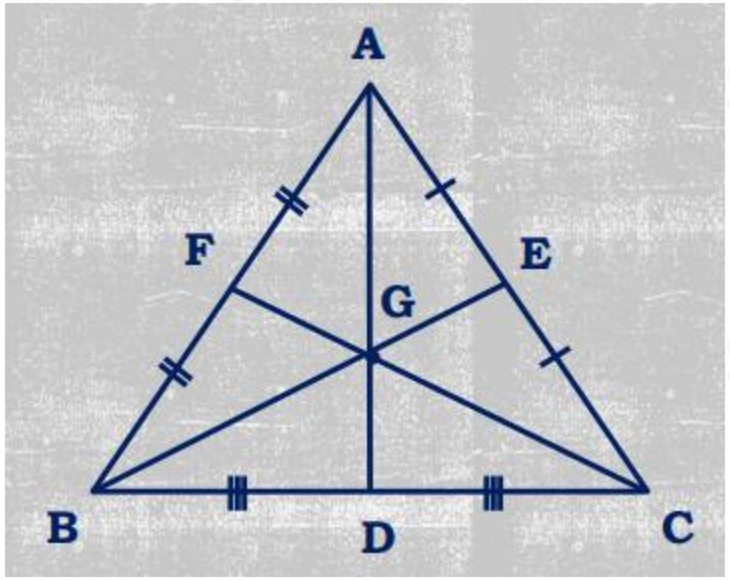




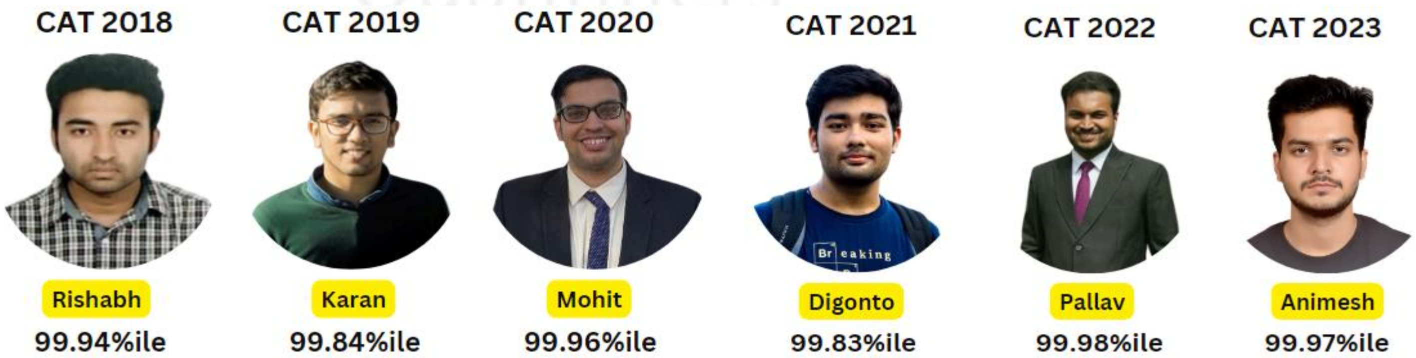
$\Delta AGB = \Delta BGC = \Delta AGC = \mathbf{1/3\ rd}$ Area of ΔABC



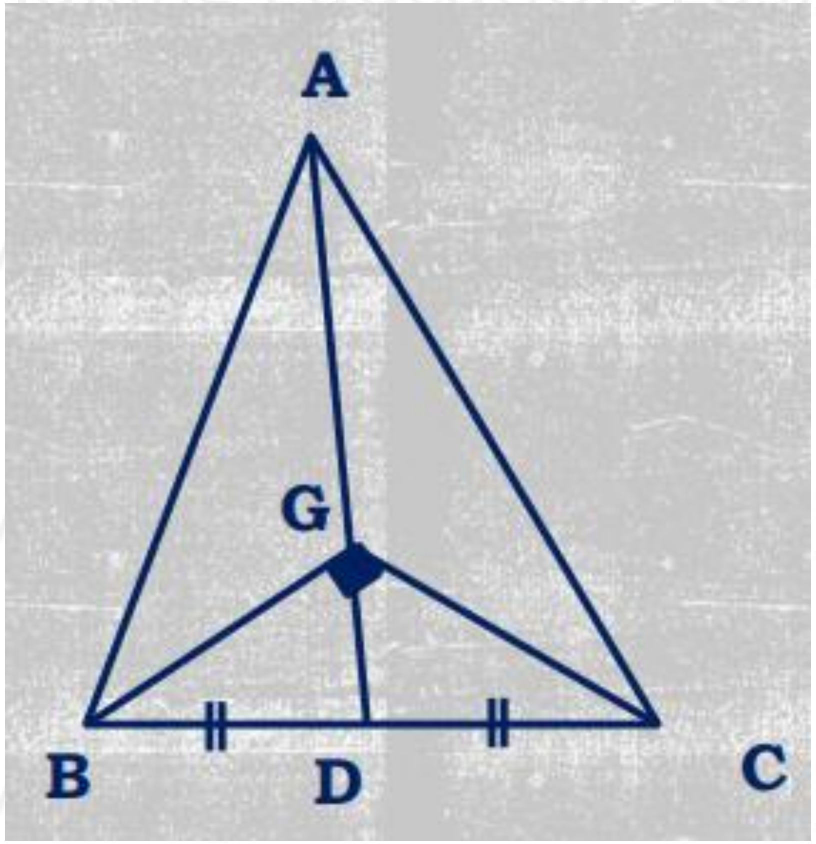
Centroid divides median in the ratio 2:1



AG : GD = 2 : 1
BG : GE = 2 : 1
CG : GF = 2 : 1

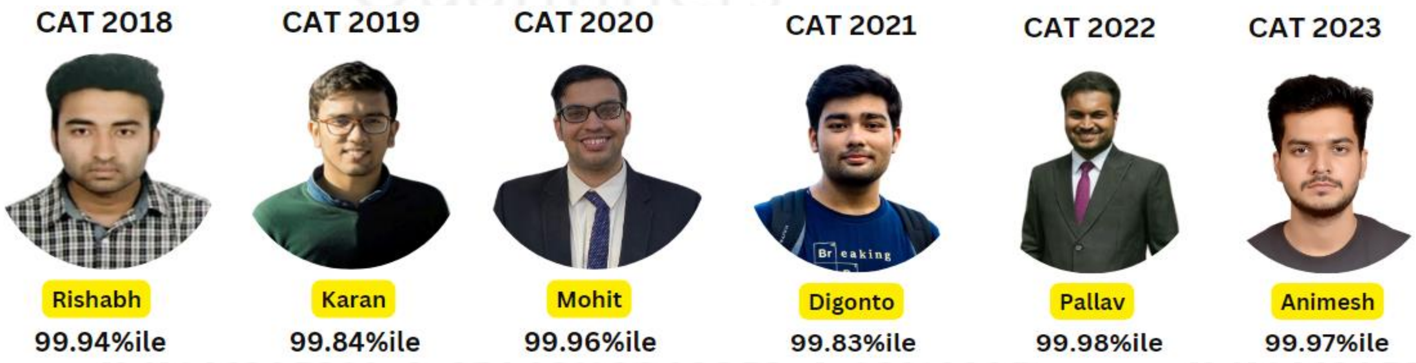


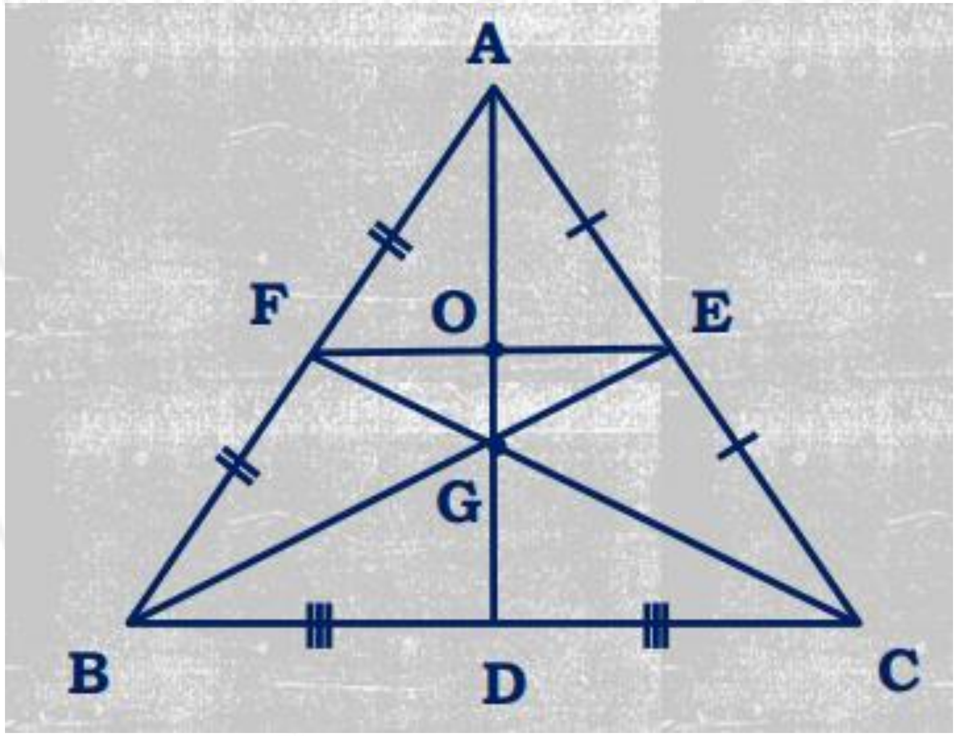
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408



BD = DC, AD = Median, G = Centroid

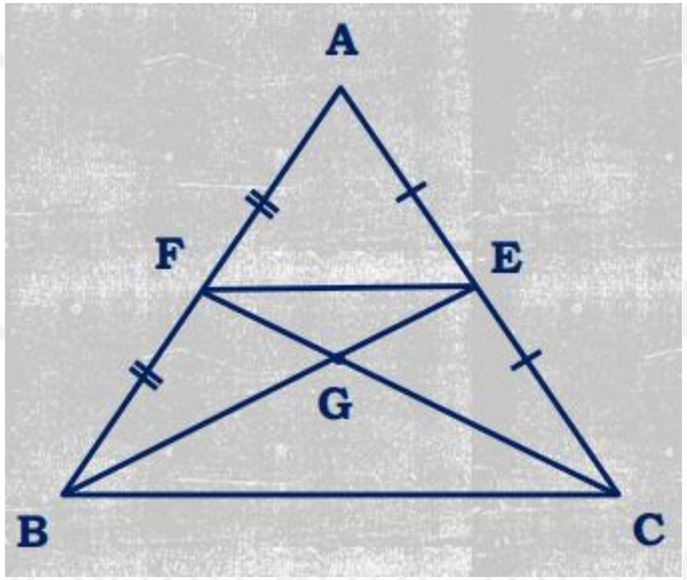
If AG = BC or AD = 1.5 BC
Then, ∠BGC = 90





EF ∥ BC and EF = $1/2$ BC

AO : OG : GD = 3 : 1 : 2

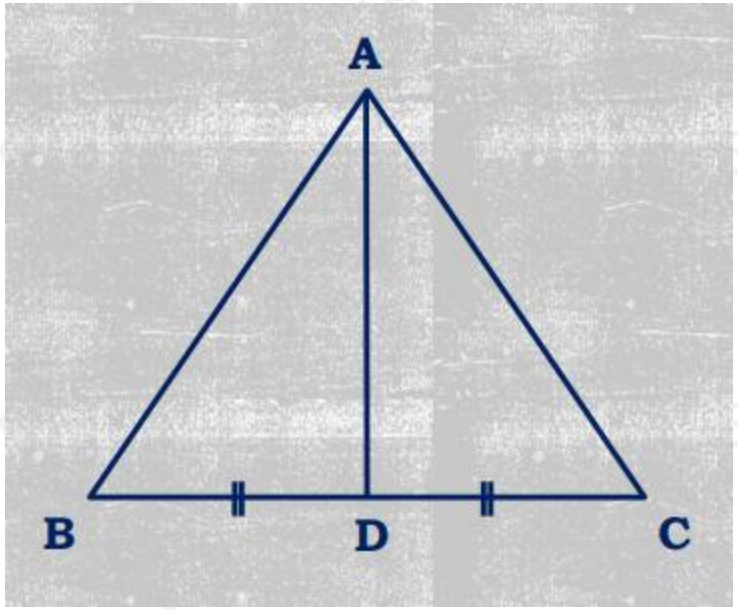


G = Centroid

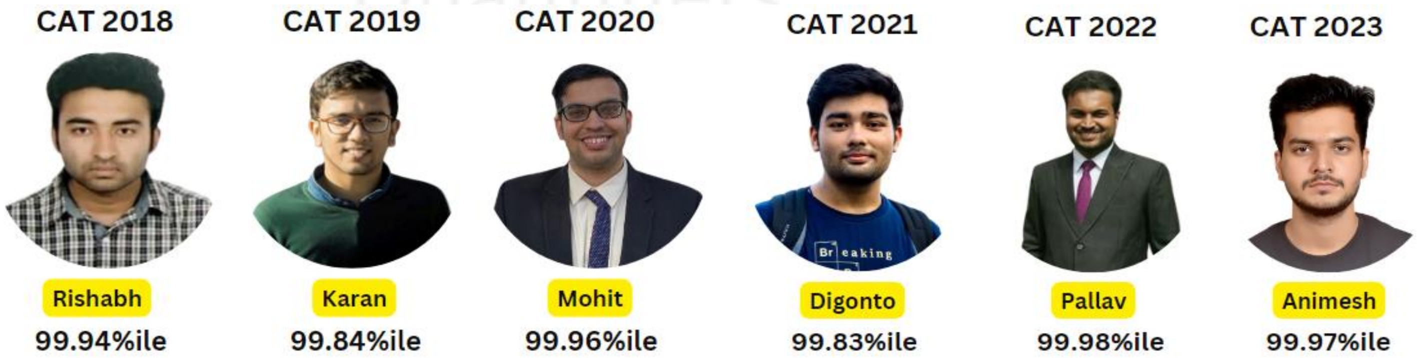
Area of ΔGEF = $1/12$ of area of ΔABC

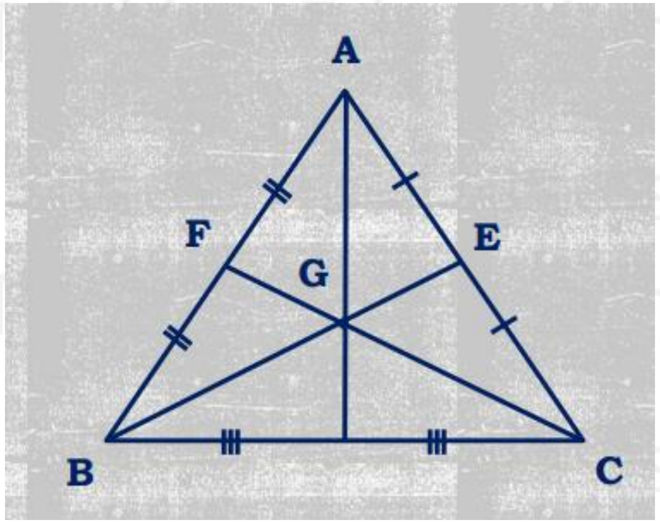
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

APOLLONEUS THEOREM (To find the length of median)

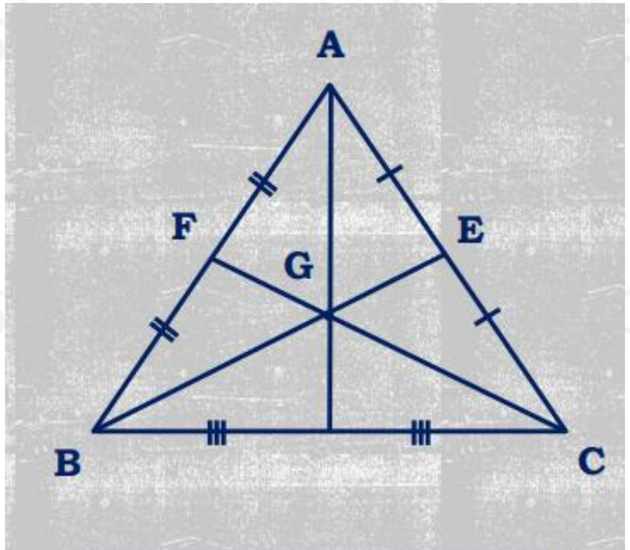


$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$



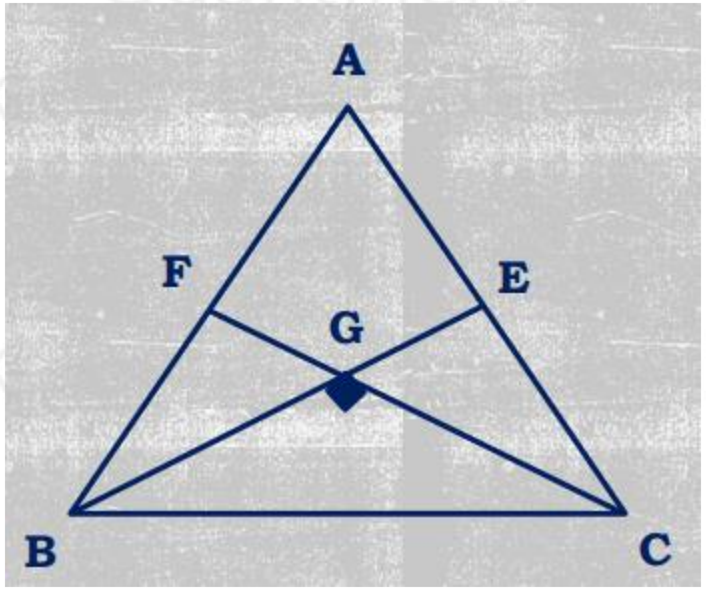


$$\frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{4}{3}$$



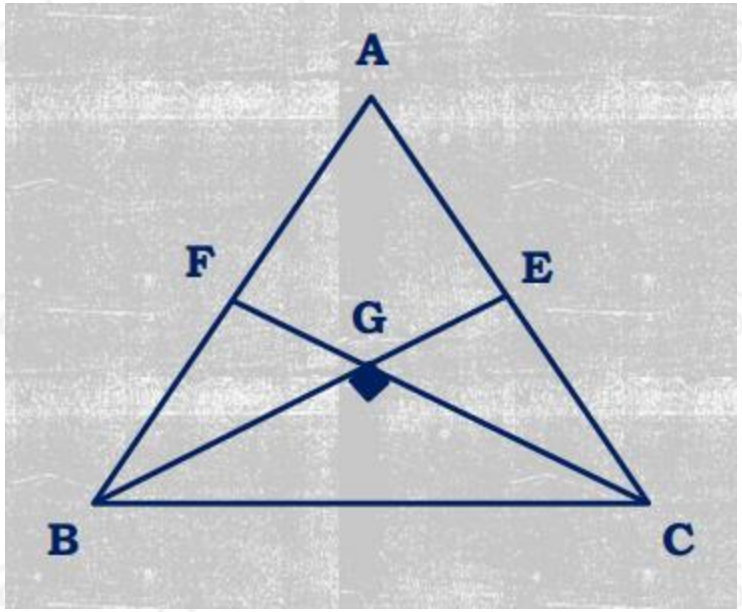
$$\boxed{1 < \frac{AB+BC+CA}{AD+BE+CF} < \frac{4}{3}}$$

When two medians intersect each other at 90°



AB^2 + AC^2 = 5 BC^2

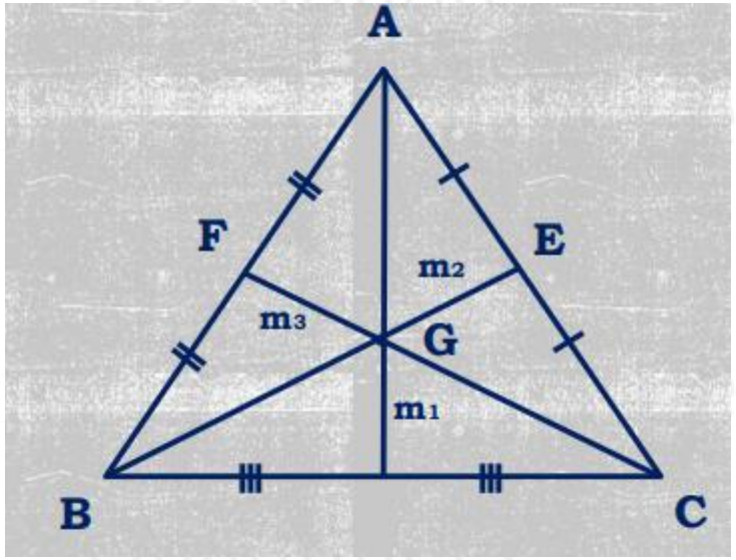
When two medians intersect each other at 90° Special Case – Isosceles Triangle



Equal sides
AB = BC

$$\frac{AB^2}{BC^2} = \frac{5}{2}$$

Area of triangle when length of all three medians are given



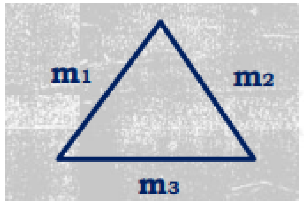
Medians

AD = m1

BE = m2

CF = m3

Sm = (m1 + m2 + m3) /2

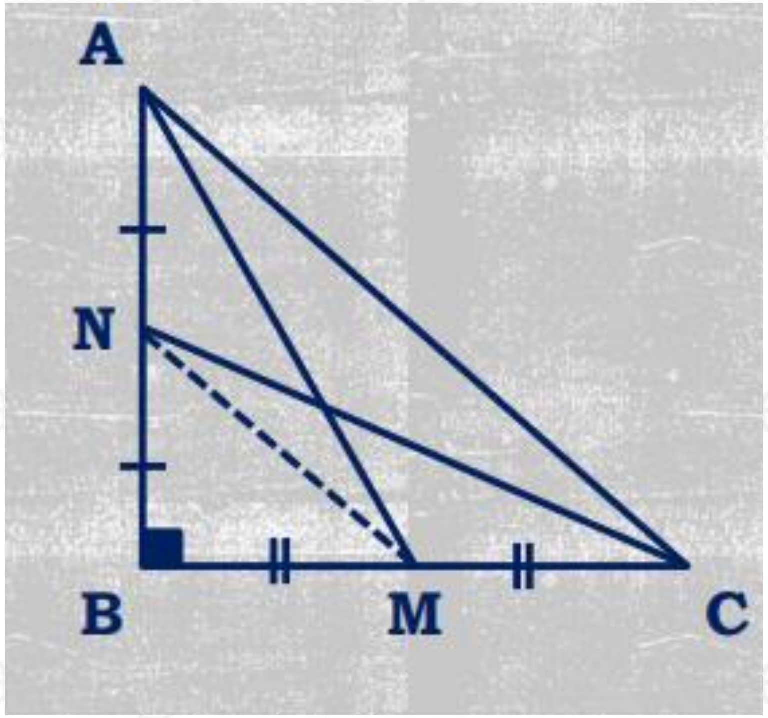


Area of ΔABC = $\frac{4}{3} \times \sqrt{S_m (S_m - m_1)(S_m - m_2)(S_m - m_3)}$

CAT 2018	CAT 2019	CAT 2020	CAT 2021	CAT 2022	CAT 2023

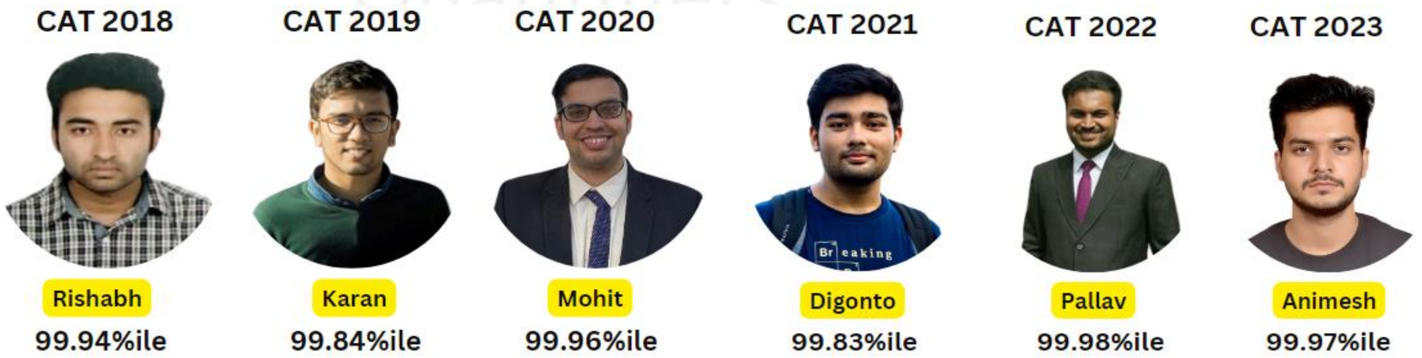
Rishabh	Karan	Mohit	Digonto	Pallav	Animesh
99.94%ile	99.84%ile	99.96%ile	99.83%ile	99.98%ile	99.97%ile

Medians in Right-angled triangle



$AM^2 + CN^2 = AC^2 + MN^2$

$4\,(AM^2 + CN^2) = 5\,AC^2$



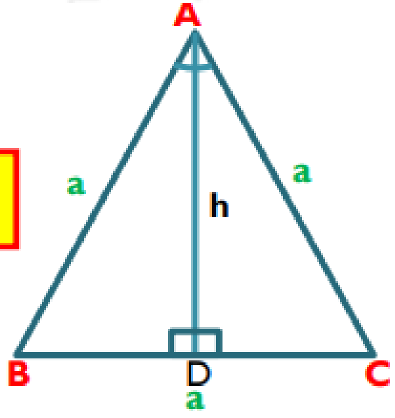
Introduction of Equilateral triangle

- All three sides of an equilateral triangle are equal. Each of its angles is 60 degrees.
- Median divides the triangle in to two equal congruent triangle.

$$\Delta ABD \cong \Delta ACD$$

$$Height = Median = angle\ bidector AD = h = \frac{\sqrt{3}}{2}a$$

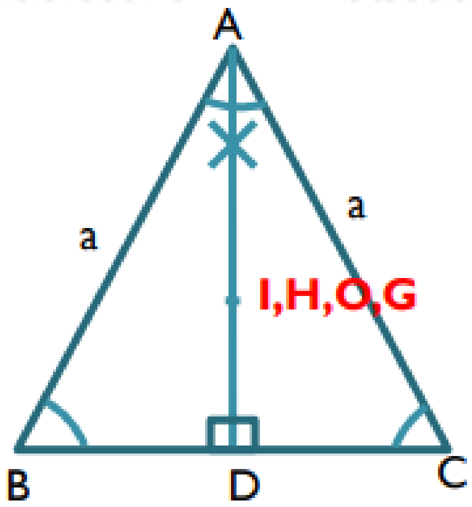
$$Area = \frac{\sqrt{3}}{4}(a \times a)$$



2018	CAT 2019	CAT 2020	CAT 2021	CAT 2022	CAT 2023
Rishabh	Karan	Mohit	Digonto	Pallav	Animesh
99.94%ile	99.84%ile	99.96%ile	99.83%ile	99.98%ile	99.97%ile

CENTRES IN EQUILATERAL TRIANGLE

- The altitude, median, angle bisector of an equilateral triangle are all the same line.
- Therefore its in-centre, orthocenter, circum-centre and centroid are at the same place

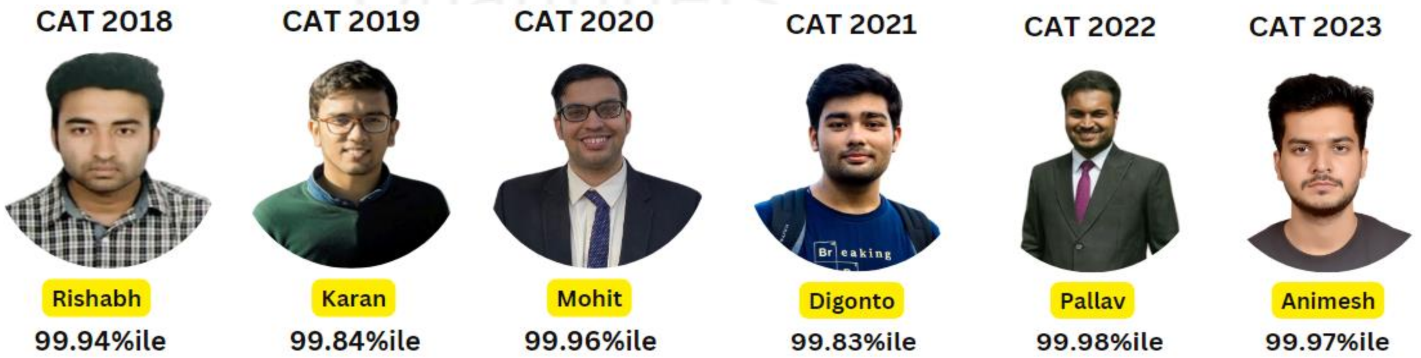


$$Inradius(r) = \frac{a}{2\sqrt{3}}$$

$$Circumradius(R) = \frac{a}{\sqrt{3}}$$

$$\frac{R}{r} = \frac{2}{1}$$

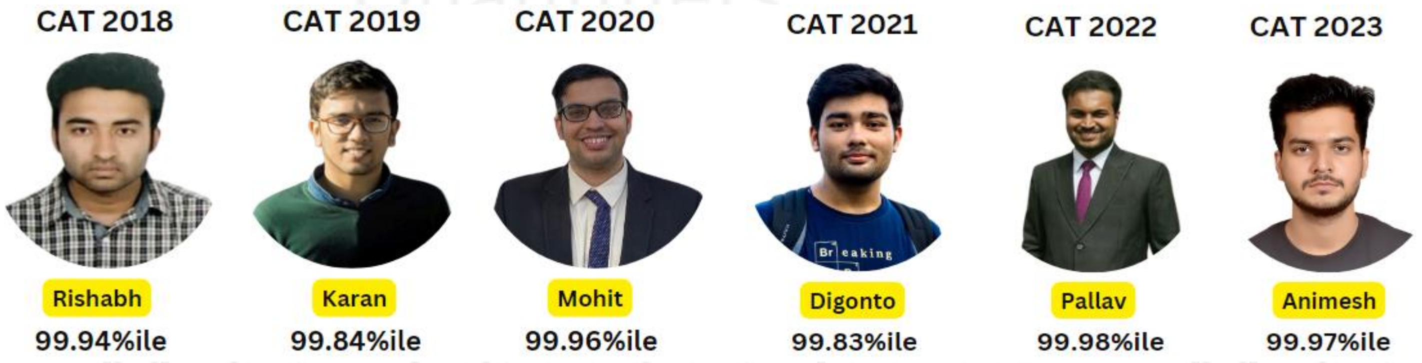
$$\frac{area\ of\ circumcircle}{area\ of\ incircle} = \frac{4}{1}$$



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

Chart for equilateral triangle

Side (a)	Height= $\frac{\sqrt{3}}{2}a$	Area= $\frac{\sqrt{3}}{4}(a\times a)$
2*k	$\sqrt{3}\times k$	$\sqrt{3}\times(k\times k)$
2 cm	$\sqrt{3}$ cm	$\sqrt{3}$ cm^2
14 cm	$7\sqrt{3}$ cm	$49\sqrt{3}$ cm^2
48 cm	$24\sqrt{3}$ cm	$576\sqrt{3}$ cm^2
15 cm	$7.5\sqrt{3}$ cm	$56.25\sqrt{3}$ cm^2

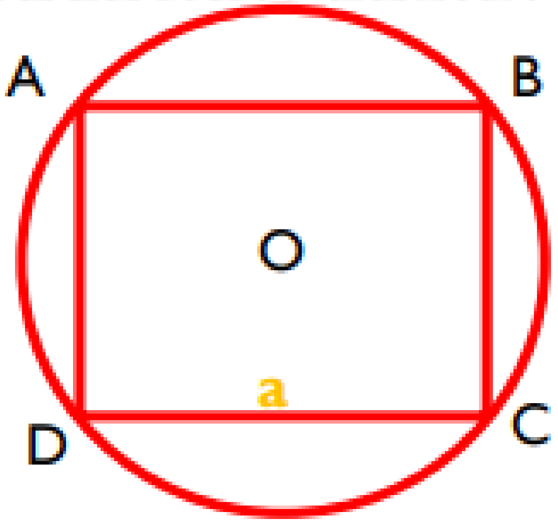


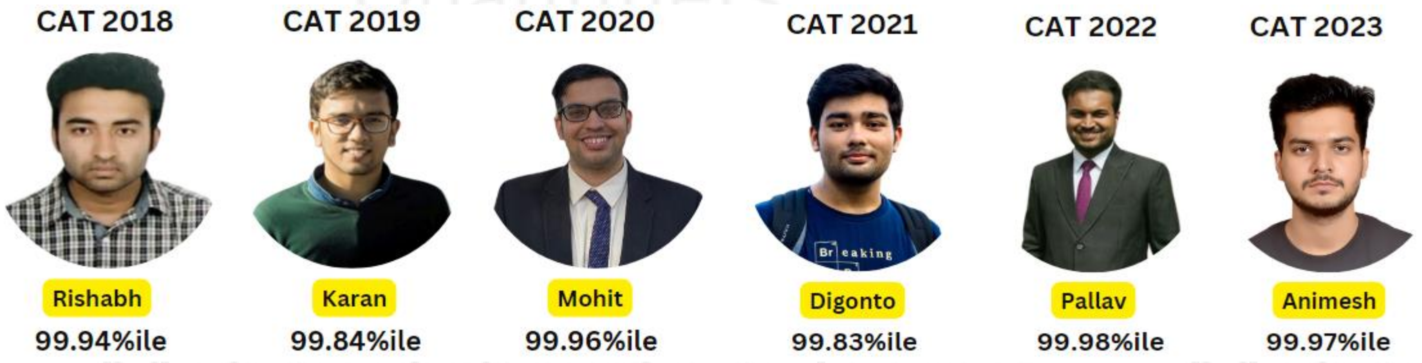
A square inscribed in a circle

A square ABCD is inscribed in a circle.
If side of square is a and diagonal is d, then

$$\textit{area of square} = (a \times a) = (d \times d)/2$$

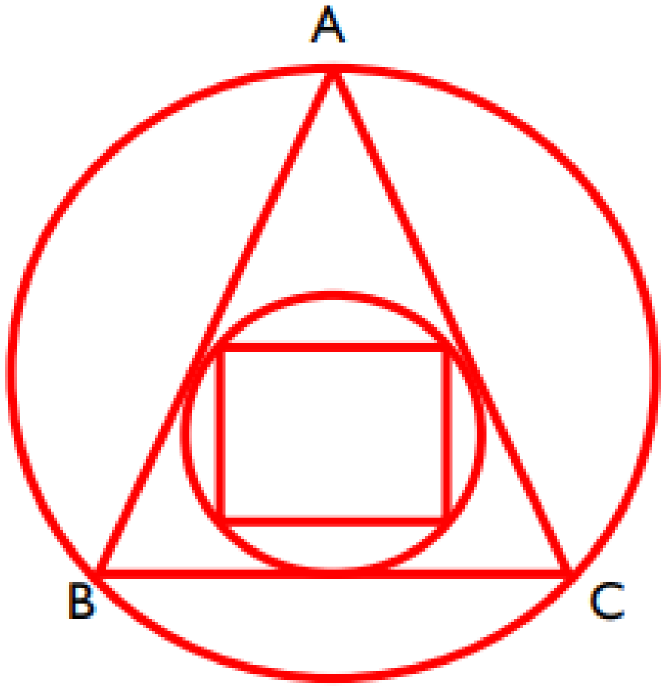
$$\textit{area of circle} : \textit{area of square} = \pi : 2 = 11 : 7$$





Area of circum-circle and in-circle of equilateral triangle

Triangle ABC is an equilateral triangle.



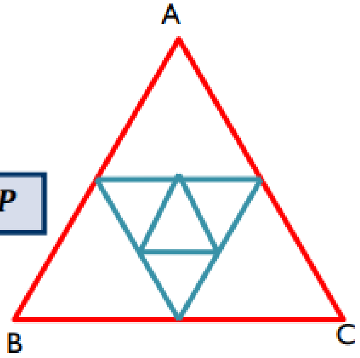
$area\ of\ circum\ circle\ of\ \Delta : area\ of\ \Delta : area\ of\ incircle\ of\ \Delta : area\ of\ square = 4\pi : 3\sqrt{3} : \pi : 2$

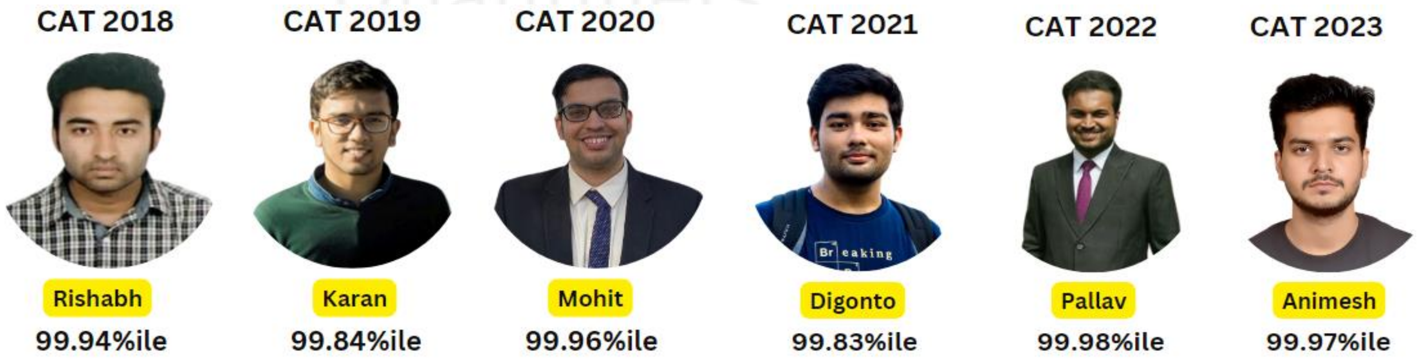
GEOMETRIC PROGRESSION FOR ANY TRIANGLE

- The area of triangle ABC is A and perimeter is P.
- When the midpoints of the three sides are joined, a medial triangle will be formed. And vice versa.

$$\boxed{\text{sum of area of all}\,\Delta = \frac{4}{3}A}$$

$$\boxed{\text{Perimeter of all triangle} = 2P}$$



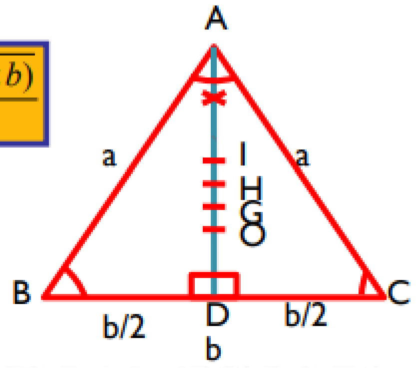


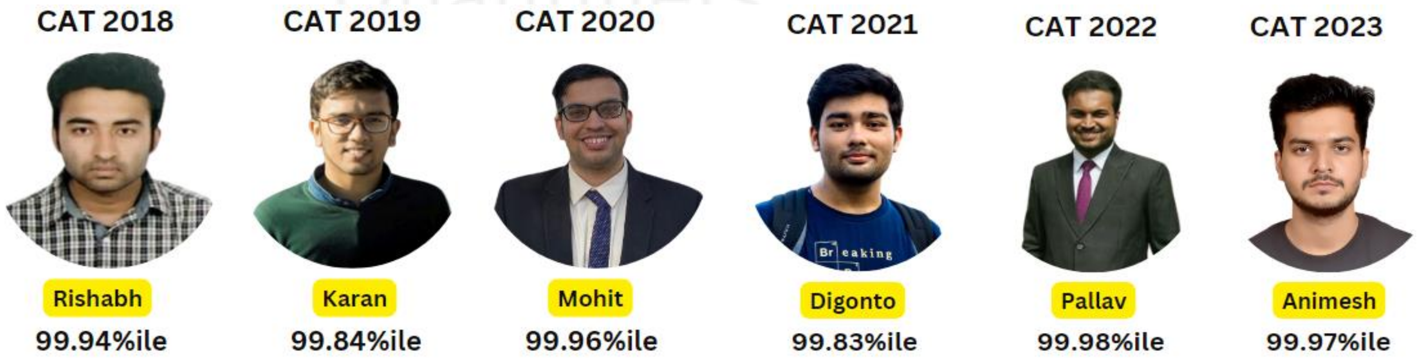
INTRODUCTION TO ISOSCELES TRIANGLE

- Two sides of an equilateral triangle are equal. Two angles of triangle are equal.
- Median divides the triangle in to two equal congruent triangle.
- The altitude, median, angle bisector of an equilateral triangle are all the same line (AD).
- Therefore its in-centre, orthocenter, circum-centre and centroid are at the same line.

$$Height = median = AD = \frac{\sqrt{4(a \times a - b \times b)}}{2}$$

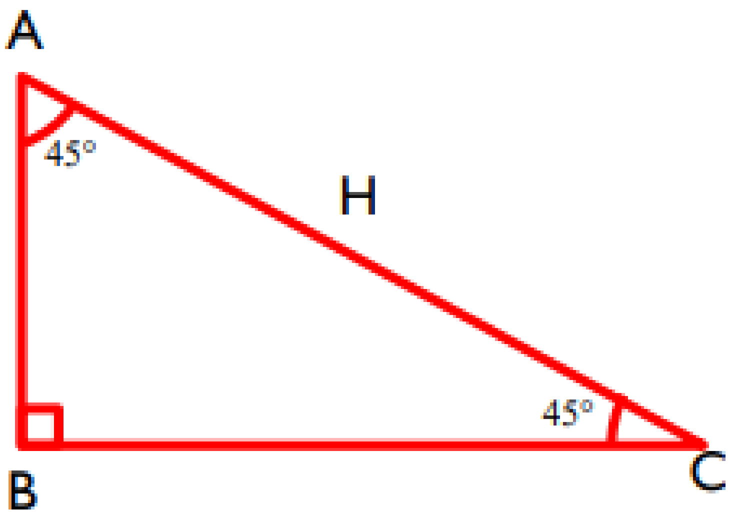
$$area = \frac{b}{4}\sqrt{4(a \times a) - (b \times b)}$$





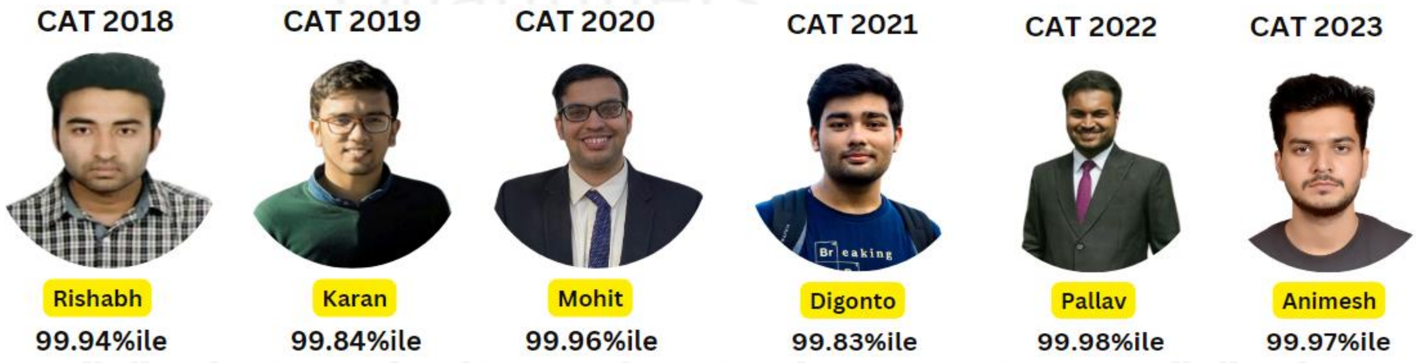
Isosceles right angle triangle

- ABC is a isosceles right angle triangle



$$\boldsymbol{Area = \frac{H \times H}{4}}$$

$$\boldsymbol{Perimeter = H(\sqrt{2} + 1)}$$



RIGHT ANGLE TRIANGLE

- One angle of right angle triangle is 90 degree.

AB^2 + BC^2 = AC^2

$$Area = \frac{1}{2} \times AB \times BC$$

Circumcentre of right angled triangle

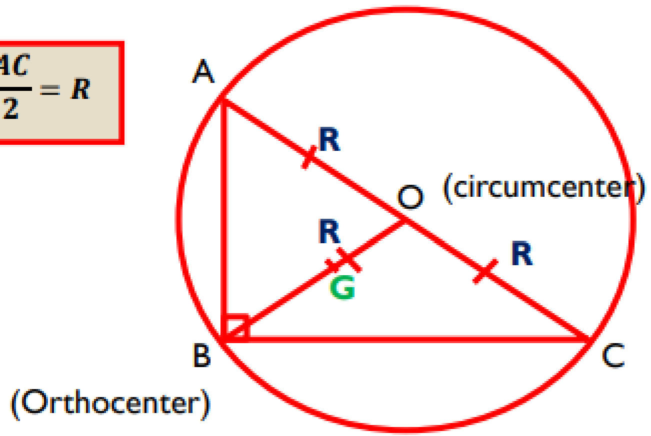
ABC is right angle triangle and AC is diameter of circle.
H is hypotenuse

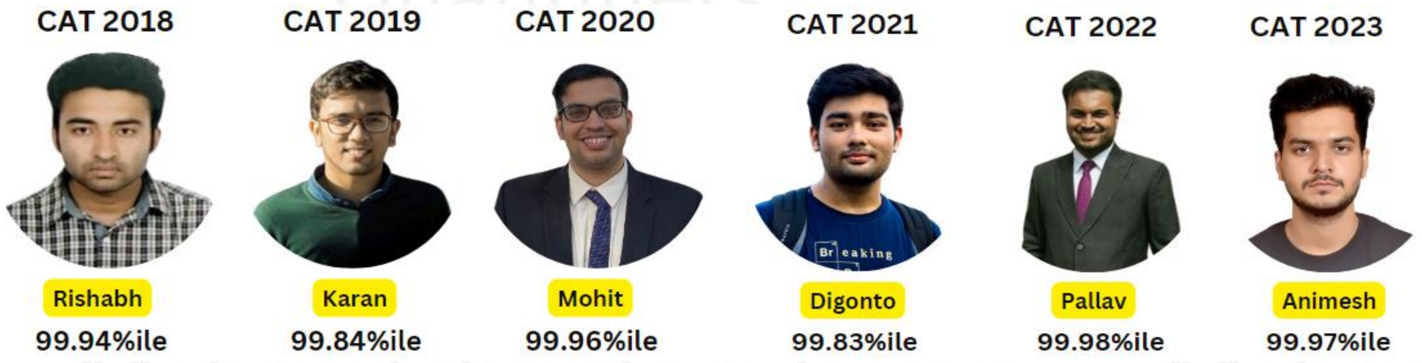
R = distance b/w orthocenter & circumcenter = median of hypotenuse = shortest median

$$BO = Circumradius = \frac{AC}{2} = R$$

$$BG = \frac{2}{3}R = \frac{H}{3}$$

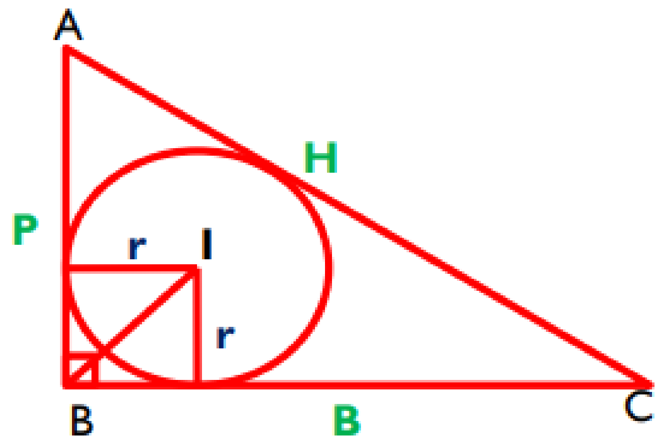
$$GO = \frac{R}{3} = \frac{H}{6}$$





In-center of right angle triangle

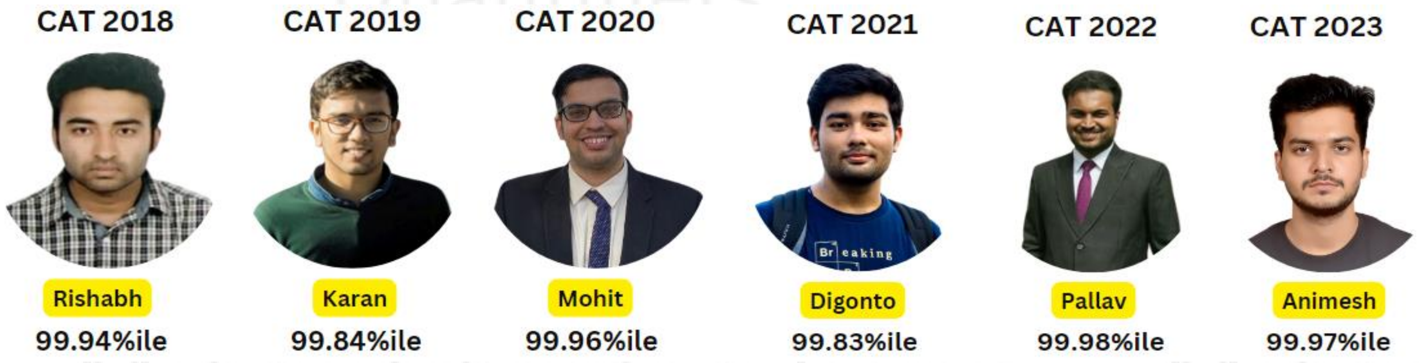
r is the in-radius of right angle triangle. I is incenter



$$R = \frac{H}{2}$$

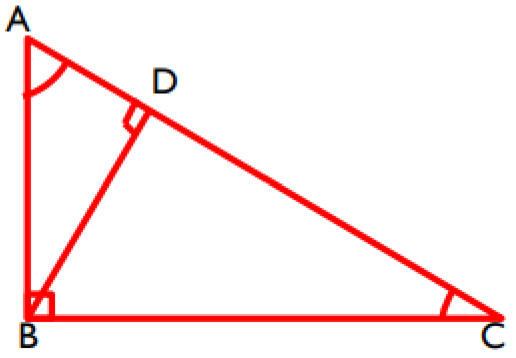
$$r = \frac{P + B - H}{2}$$

$$R + r = \frac{P + B}{2}$$



SIMILARLY concept

ABC is right angle triangle. BD is height on hypotenuse



$$\Delta ABC \approx \Delta ADB \approx \Delta BDC$$

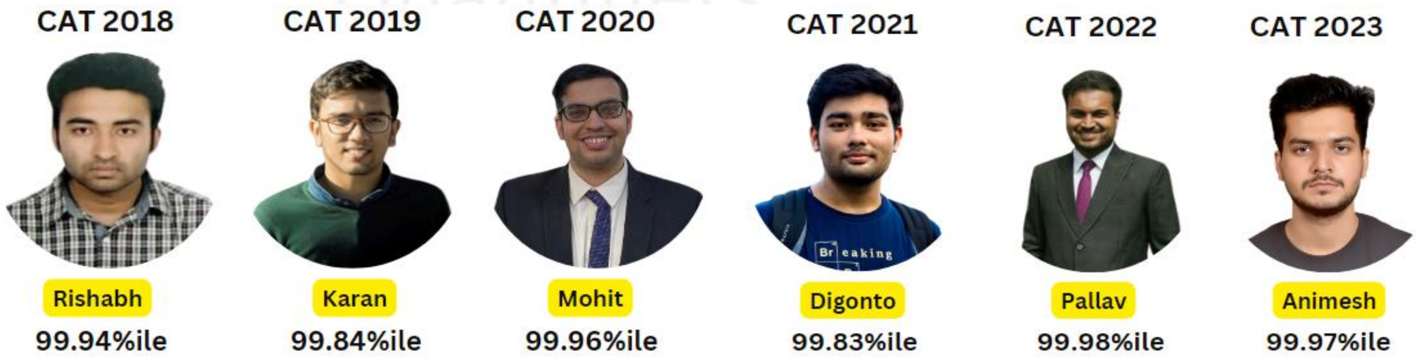
BD^2 = AB^2 – AD^2 = BC^2 –DC^2

$$\boxed{\frac{AB}{BC} = \sqrt{\frac{AD}{CD}}}$$

AB^2 =AD*AC

BC^2 = CD*CA

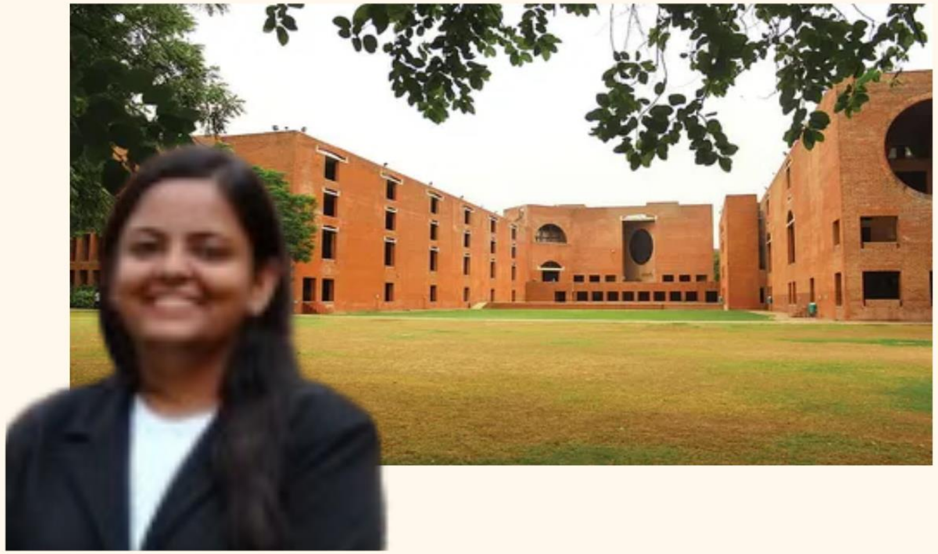
BD^2 = AD*CD



Alirpiya - MDI Gurgaon

Alirpiya made it to
MDI Gurgaon with
Quantifiers CAT Academy.
So can you

Anshika - IIM A



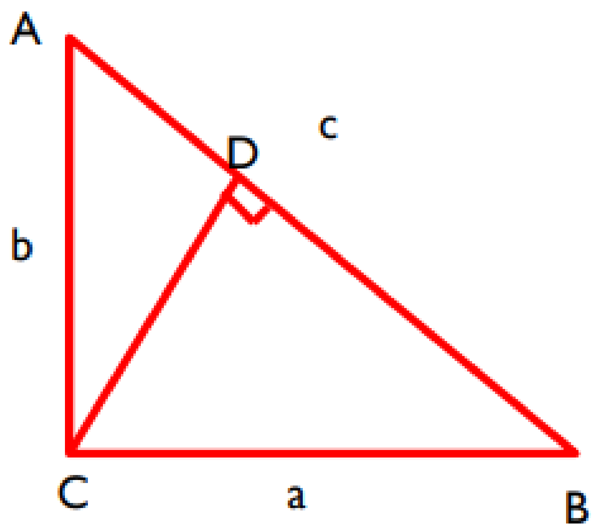
Anisha made it to
IIM Ahmedabad
Converted IIM A,B,C,K,L and top 20 colleges
with Quantifiers CAT Academy

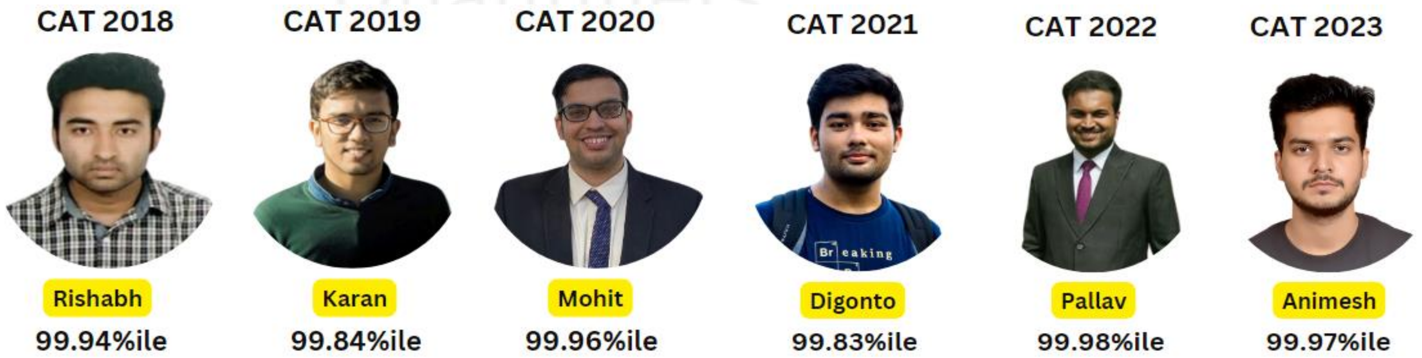
Height on hypotenuse

ACB is right angle triangle. A

$$p = ab/C$$

$$\boxed{1/p^2 = 1/a^2 + 1/b^2}$$



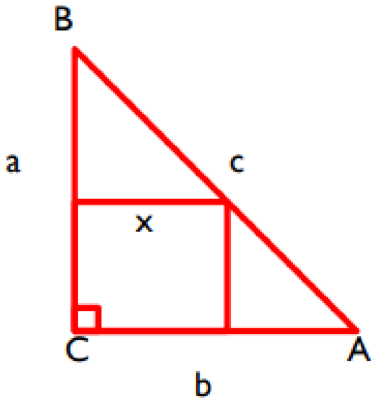


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

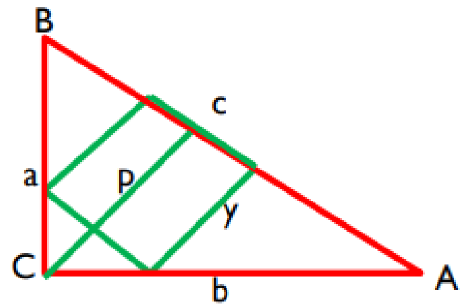
To find side of squares in triangle

- ABC is right angle triangle.

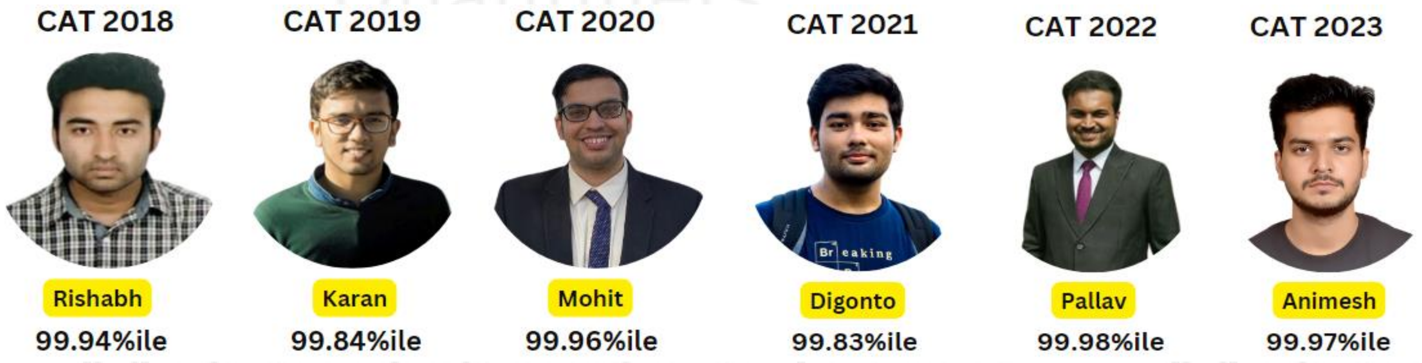
$$a = xy/(x + y)$$



$$\boxed{x = \frac{ab}{a+b}}$$

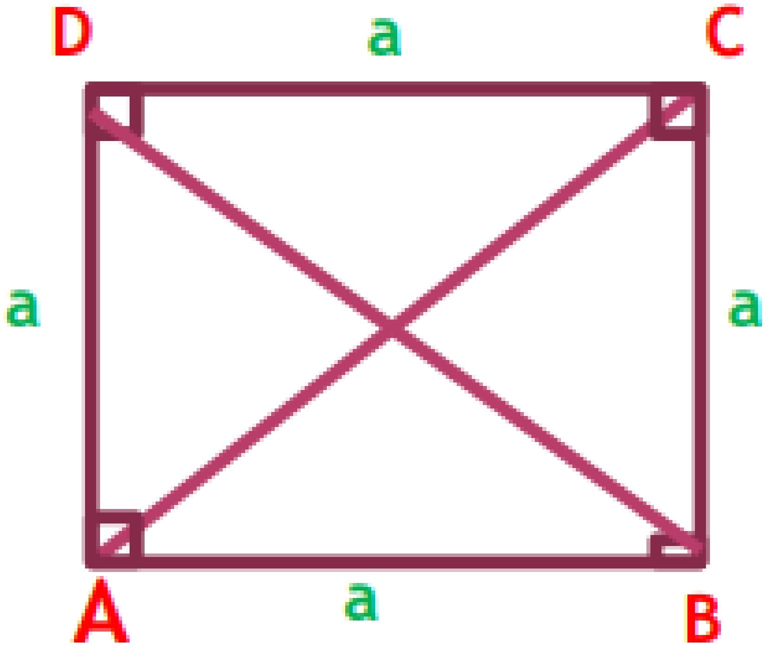


$$\boxed{y = \frac{abc}{a.a + b.b + ab}}$$



Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

SQUARE



- Diagonal bisect each other at 90 degree.
- Diagonals divide the square in four equal triangle.

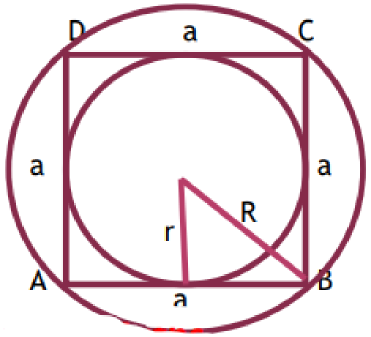
INCIRCLE & CIRCUMCIRCLE OF SQUARE

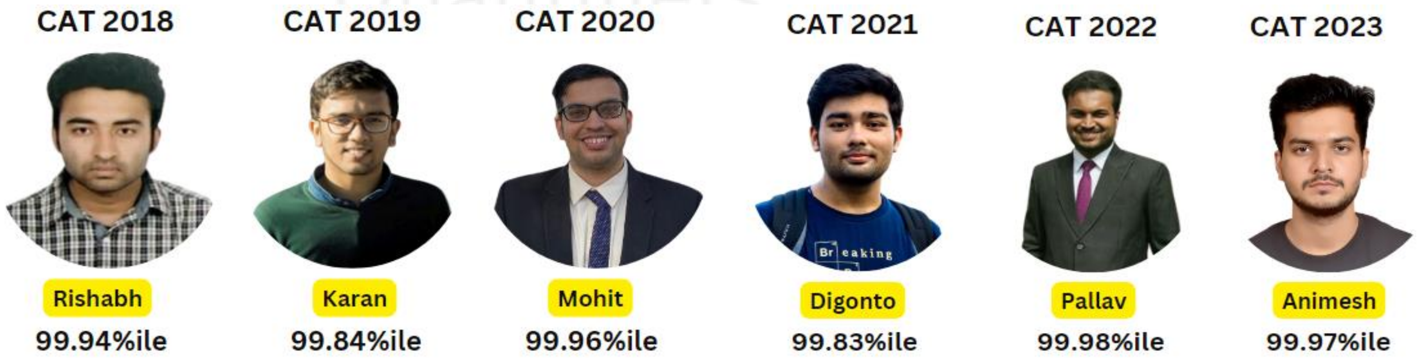
$$r = \frac{a}{2}$$

$$R = \frac{a}{\sqrt{2}}$$

$$area \frac{circumcircle}{incircle} = \frac{2}{1}$$

$$\frac{R}{r} = \frac{\sqrt{2}}{1}$$

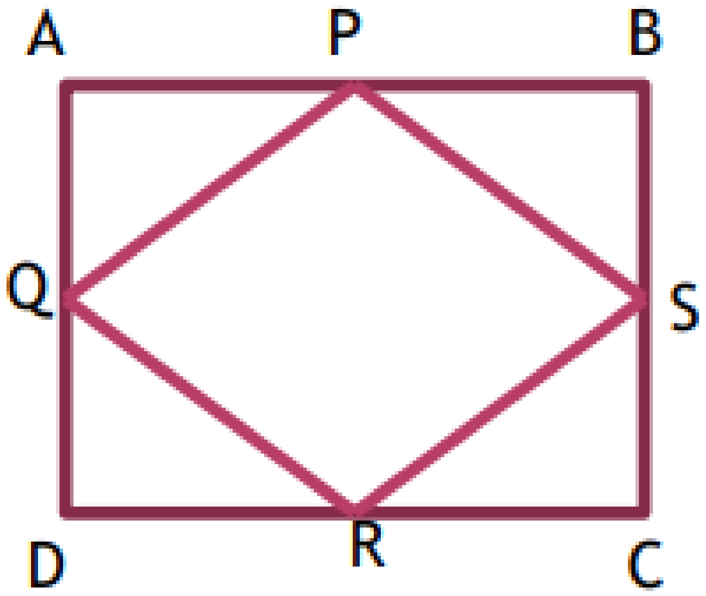


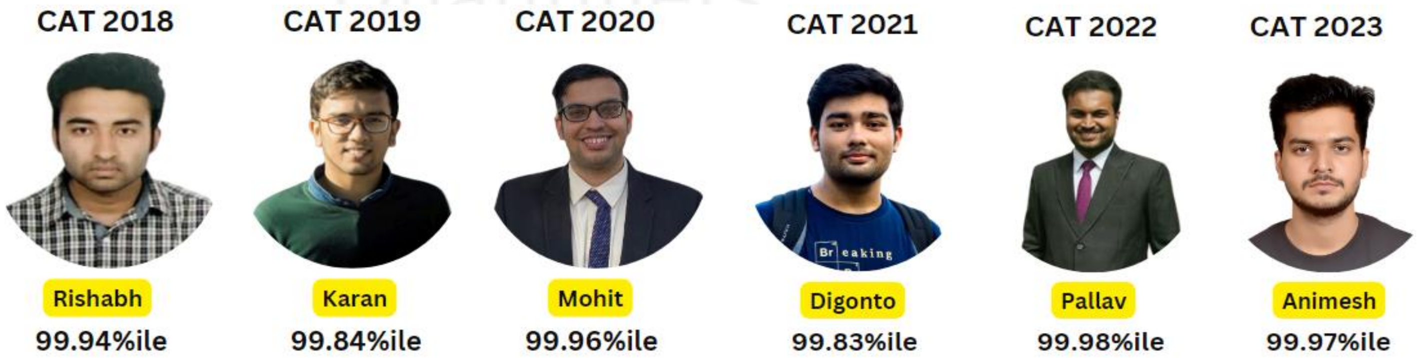


MIDPOINT OF SQUARE

$$area\ \frac{ABCD}{PQRS} = \frac{2}{1}$$

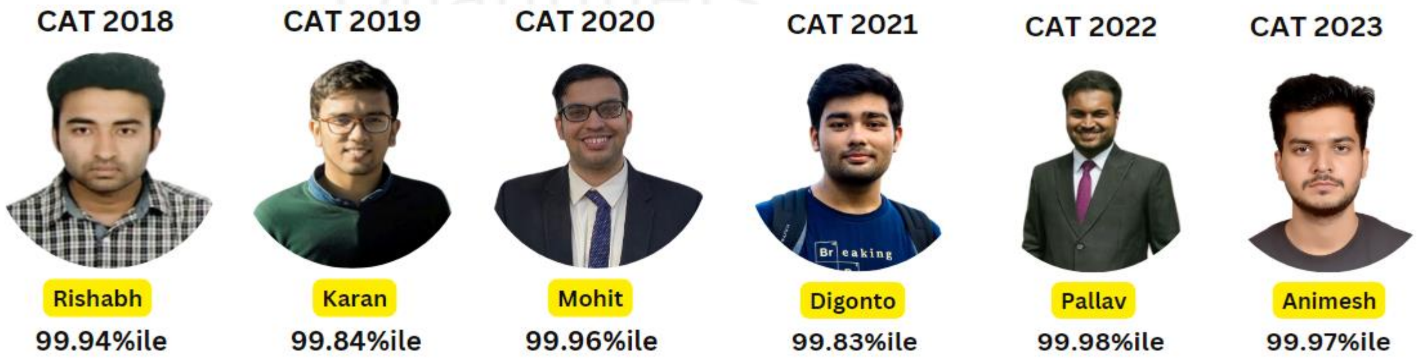
When mid point of square ABCD is met then a new square PQRS is made.





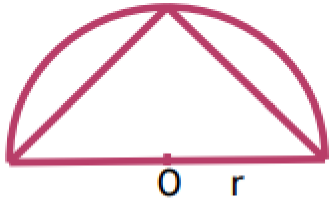
IMPORTANT CONCEPT

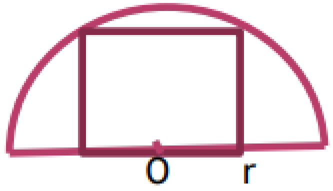
$$area ratio = 8\pi : 16 : 4\pi : 8 : 2\pi : 4 : \pi : 2$$

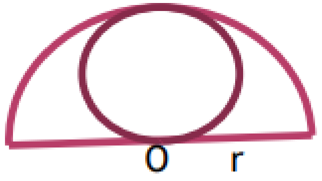


SEMICIRCLE CONCEPT

A semicircle of radius r



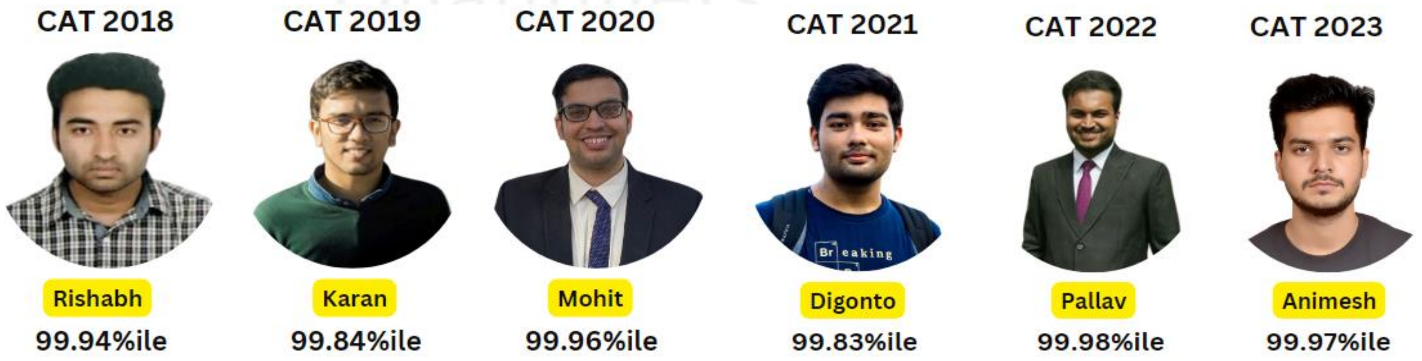




Area=r^2

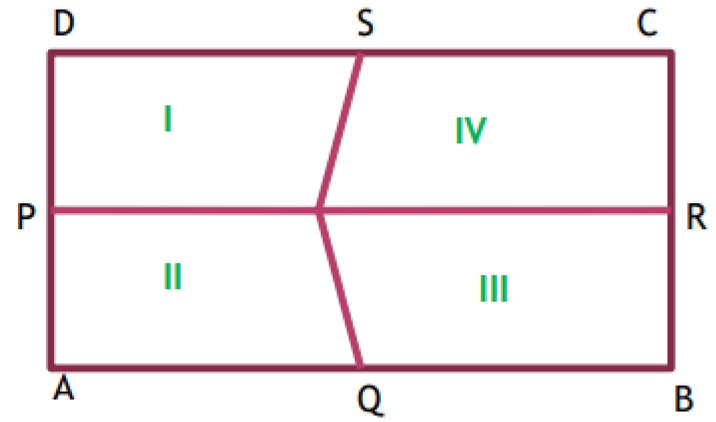
$$area = \frac{4}{5}(r \times r)$$

$$area = \frac{\pi}{4}(r \times r)$$



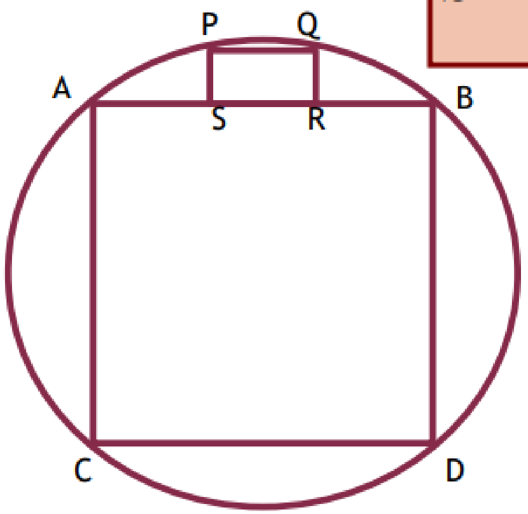
SQUARE/RECTANGLE

When midpoint of square or rectangle met at one point, then



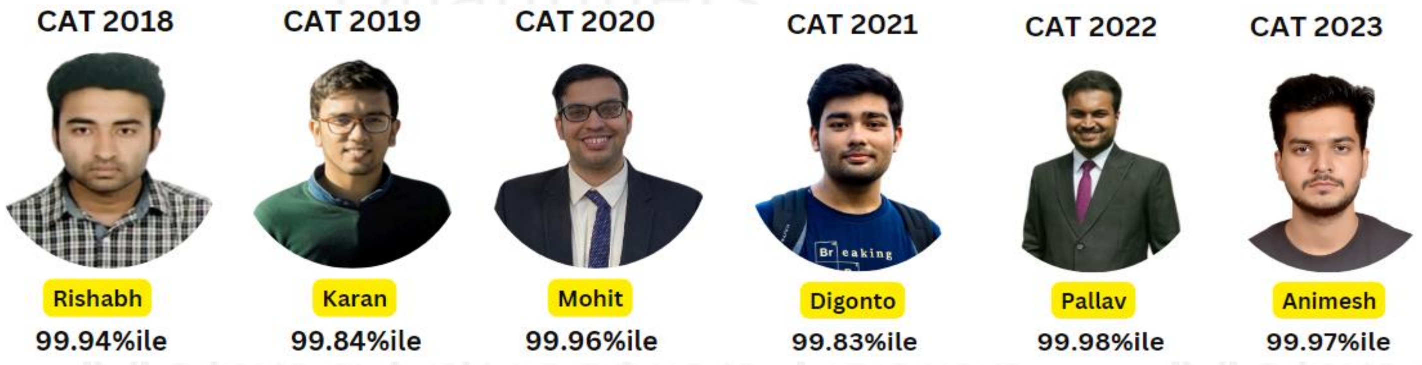
$$I + III = II + IV$$

CONCEPT OF SQUARE



$$SIDE\,\frac{PQRS}{ABCD} = \frac{1}{5}$$

PQRS & ABCD is square in circle.

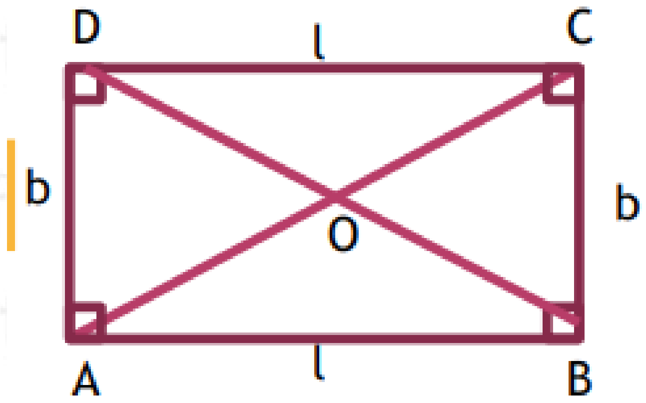


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

RECTANGLE

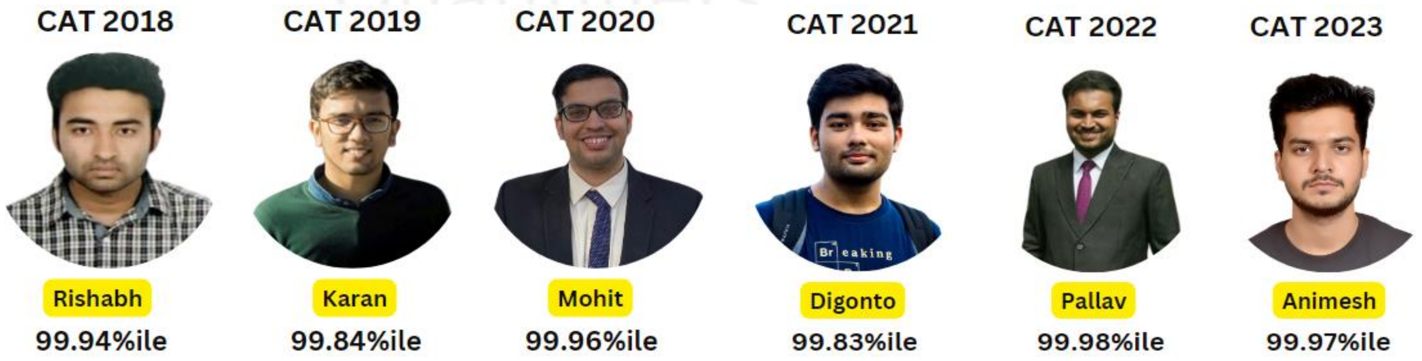
perimeter = 2(l+b)
area-= l*b

- A rectangle has four sides.
- Opposite sides are equal.
- All four angles are 90 degree.
- Diagonal do not bisect vertex angle.
- Opposite triangles made by diagonal of rectangle are congruent.
- Diagonal bisect each other But not at 90 degree.



$$diagonal = AC = d = \sqrt{l \times l + b \times b}$$

$$AO = BO = CO = DO = \frac{AC}{2} = \frac{BD}{2}$$

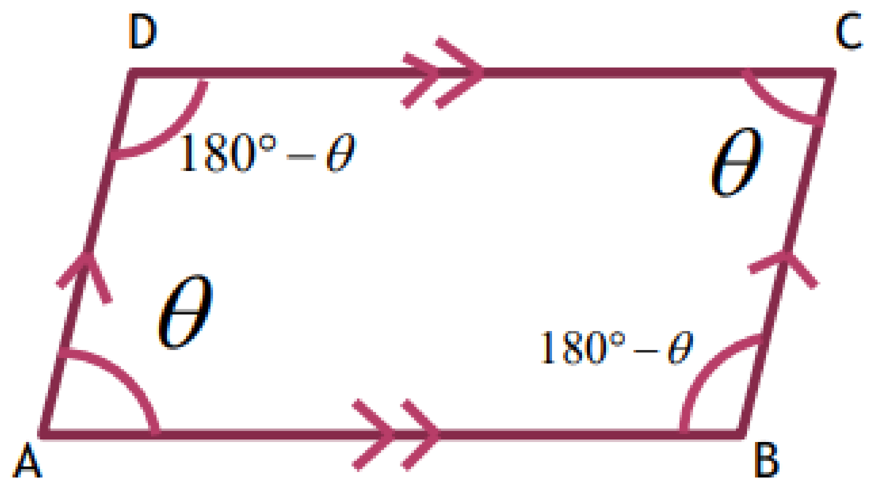


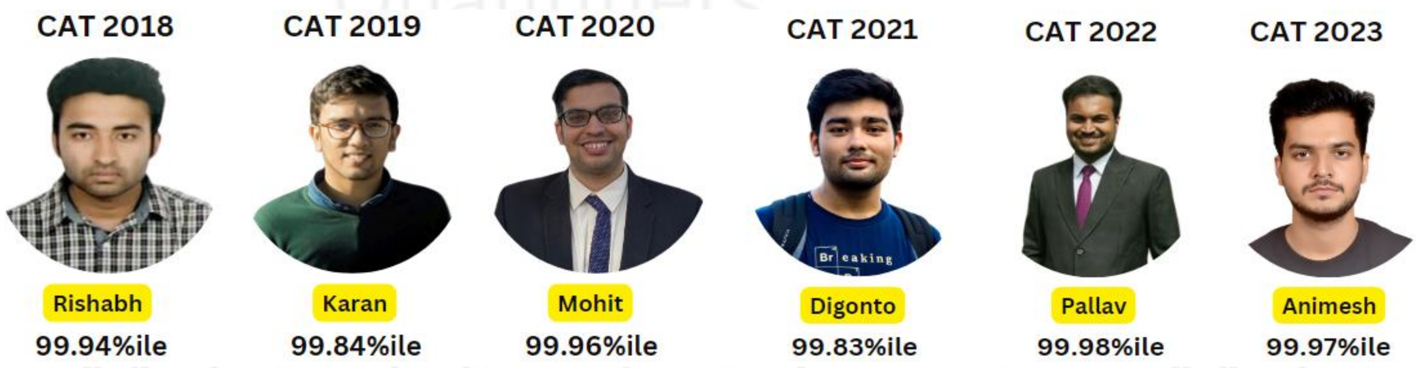
PARALLELOGRAM

- Opposite sides of a parallelogram is parallel.
- Opposite angles are equal.
- Sum of adjacent angle of parallelogram is 180 degree.

AB || CD

$$\angle A = \angle C$$
$$\angle B = \angle D$$





PARALLELOGRAM

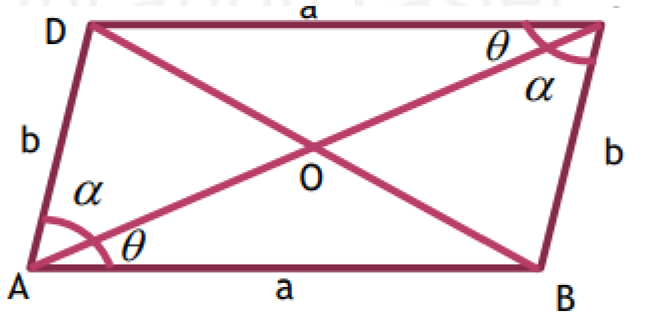
Opposite sides of a parallelogram is equal.

Diagonal do not bisect vertex angle

$$\Delta ADC \cong \Delta ABC$$

$$AB = CD$$

$$AD = BC$$



$$\alpha \neq \theta$$

$$AC \neq BD$$

$$AO = CO = \frac{AC}{2}$$

$$BO = DO = \frac{BD}{2}$$

$$\Delta AOB \cong \Delta COD$$

$$\Delta AOD \cong \Delta COB$$

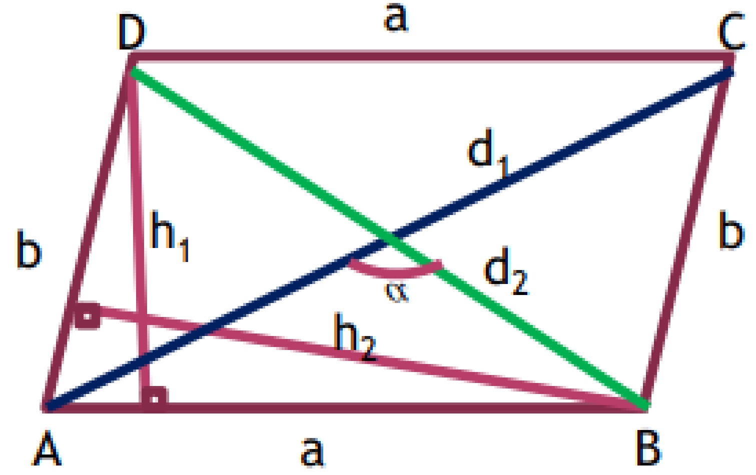
AREA OF PARALLELOGRAM

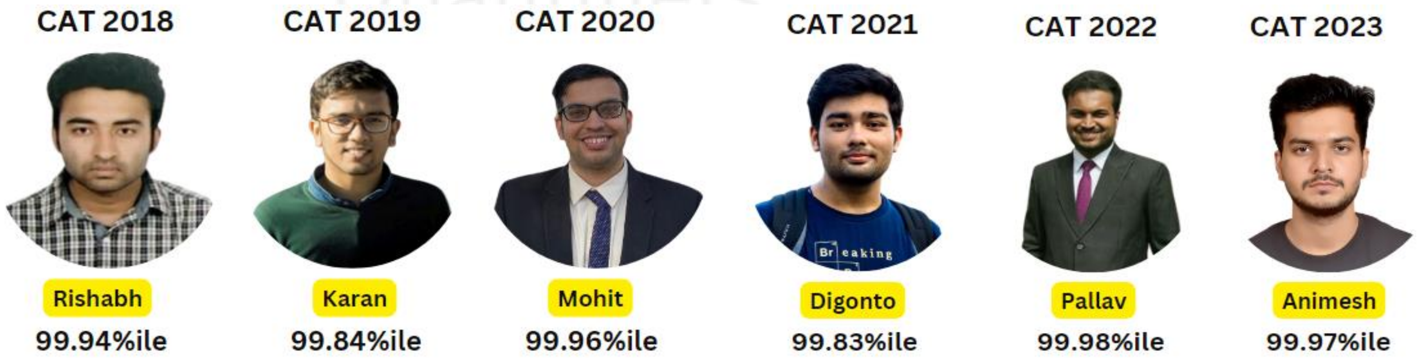
$$area = 2 \times area\Delta ADB = 2 \times area\Delta BCD = ab\sin\theta$$

$$Area = base \times height$$

$$\frac{a}{b} = \frac{h_2}{h_1}$$

$$area = \frac{1}{2} d_1 \times d_2 \sin\alpha$$

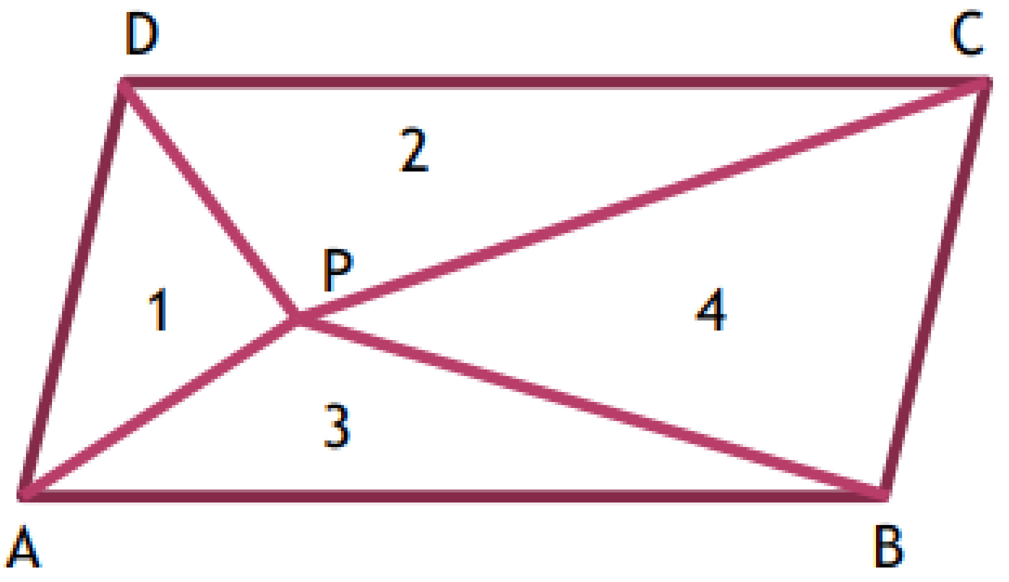




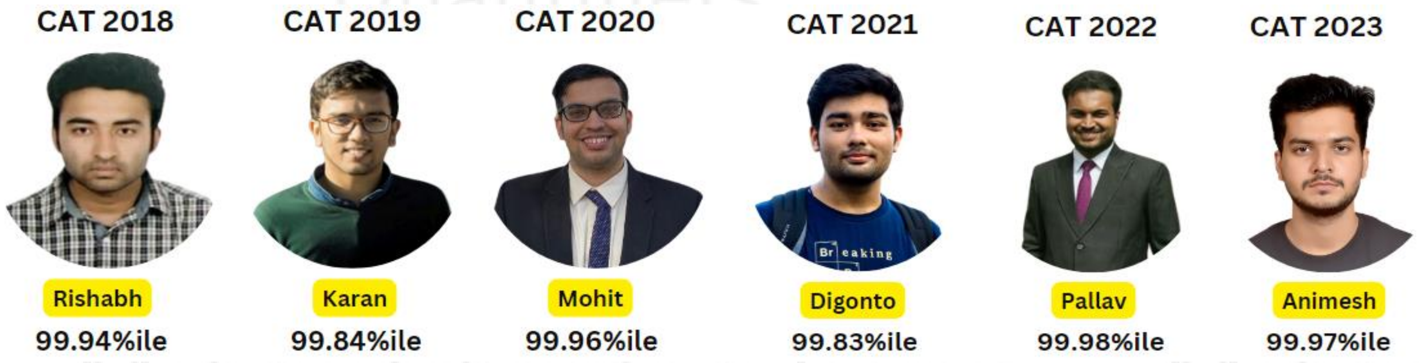
CONCEPT OF PARALLELOGRAM

P is any point inside the parallelogram ABCD.

Point P is met with all four vertex, then four triangle are formed.



$$area\Delta APB + area\Delta CPD = area\Delta APD + area\Delta BPC = \frac{1}{2} area \parallel gm$$

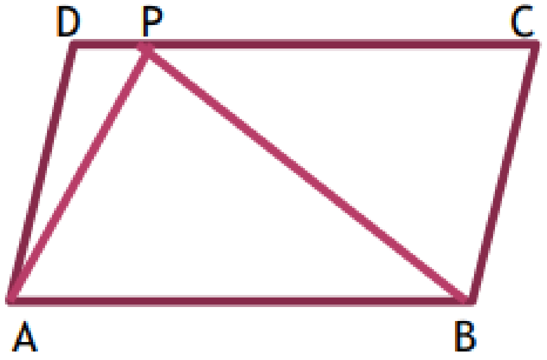


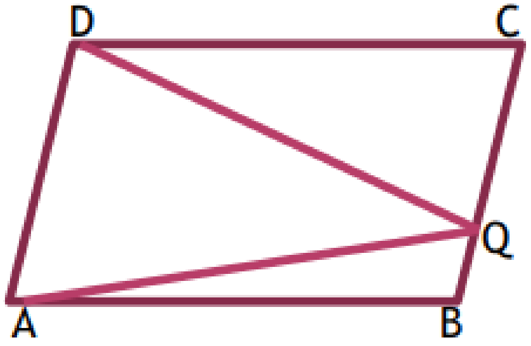
TRAINGLE IN PARALLELOGRAM

We have two parallelogram with same area.

A triangle is inscribed in these parallelogram.

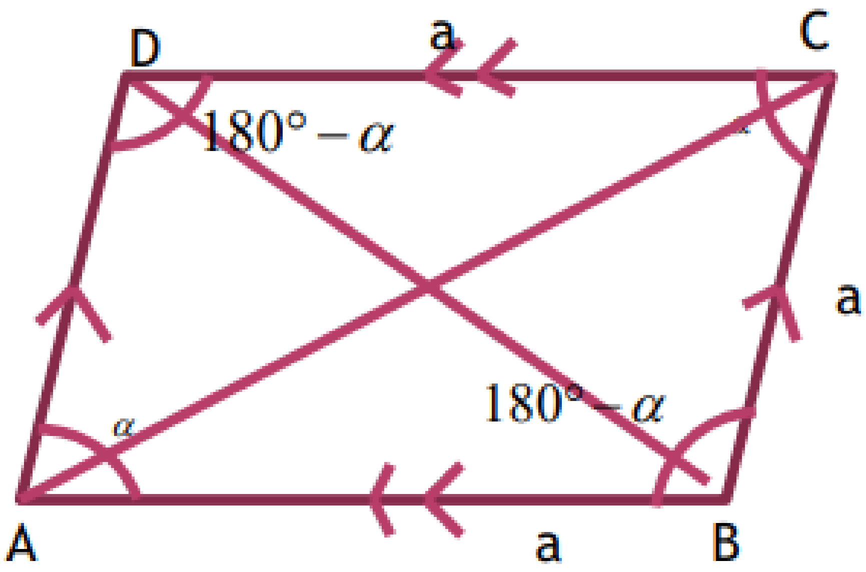
$$\frac{ar(\Delta APB)}{ar(ABCD)} = \frac{1}{2}$$





RHOMBUS

- All sides are equal.
- Opposite sides are parallel.
- Opposite angles are equal.
- Sum of adjacent angle are 180 degree.



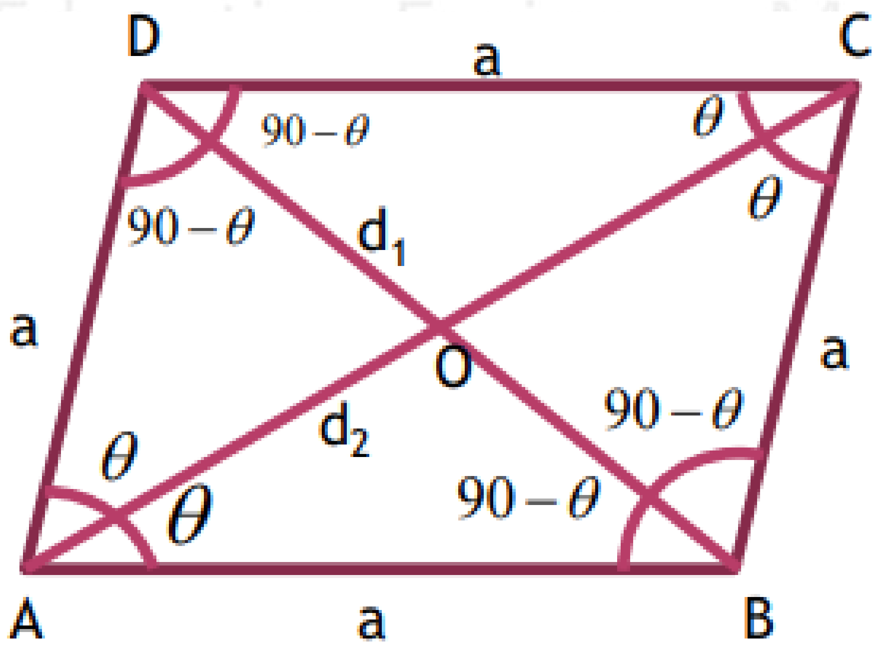
$$\angle A = \angle C$$
$$\angle B = \angle D$$

$$Perimeter = 4a$$

$$AC \neq BD$$

DIAGONAL & AREA OF RHOMBUS

- Diagonal bisect vertex angle.
- Diagonal bisect each other at 90 degree.

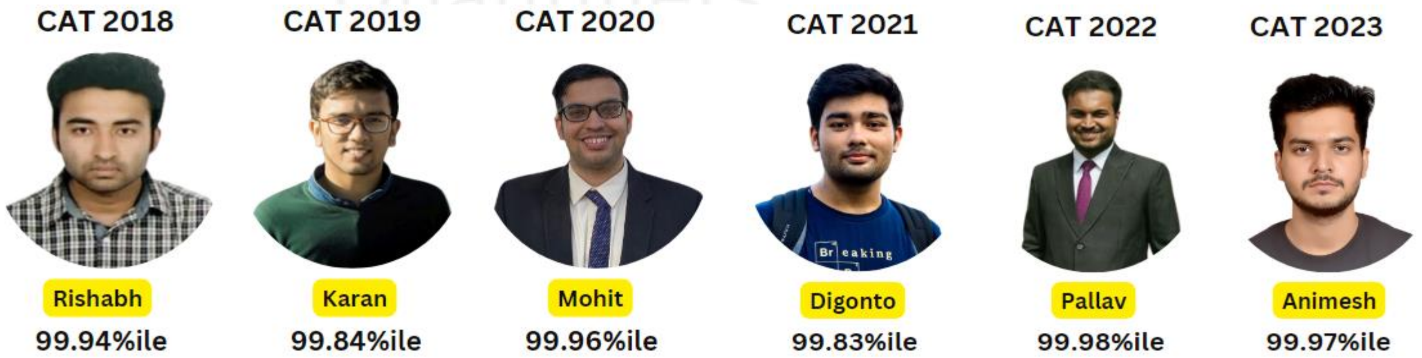


$$AO = CO = \frac{AC}{2}$$

$$BO = DO = \frac{BD}{2}$$

$$d_1^2 + d_2^2 = 4a^2$$

$$area = \frac{1}{2} d_1 d_2$$



IMPORTANT CONCEPT

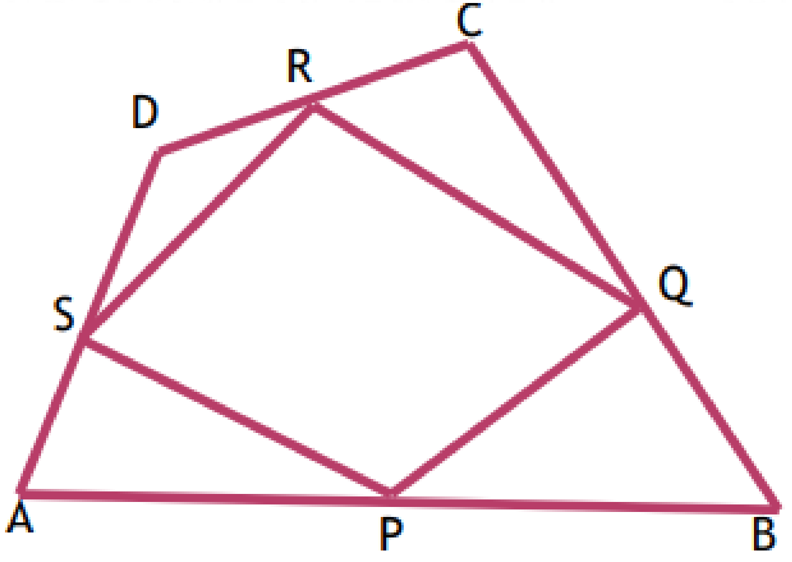
$$RS \parallel PQ \parallel AC$$

$$RS = PQ = \frac{AC}{2}$$

$$QR \parallel PS \parallel BD$$

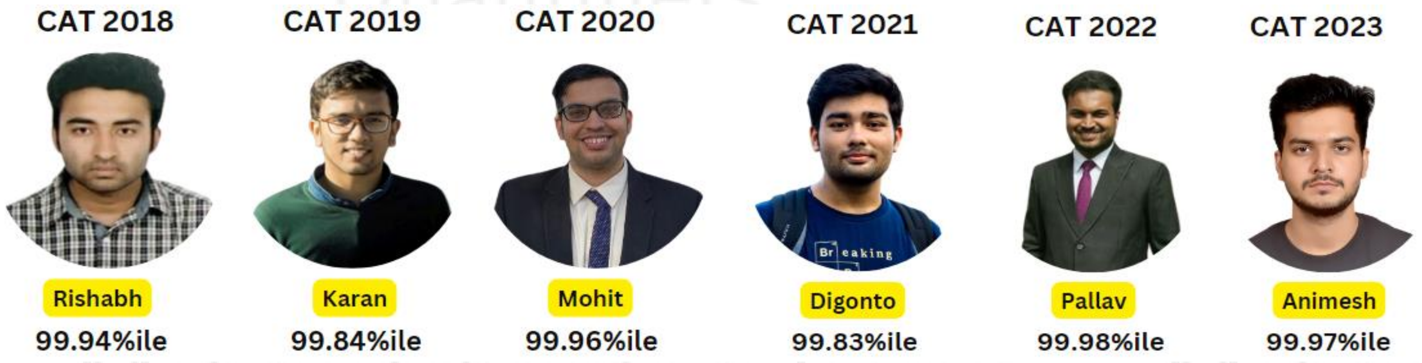
$$QR = PS = \frac{BD}{2}$$

PQRS is ||gm



IMPORTANT CHART

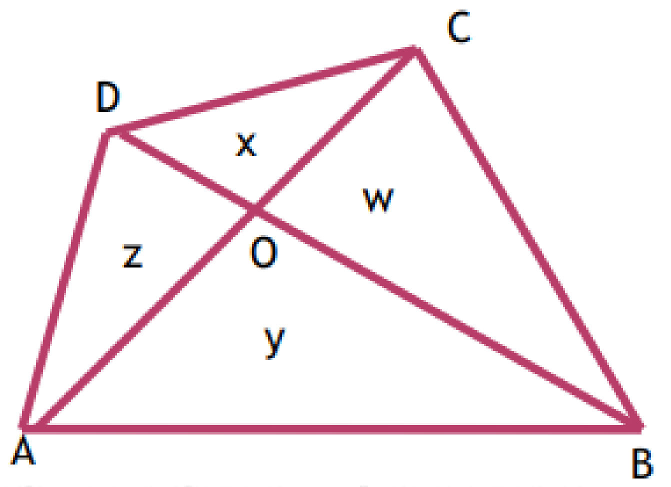
Quadrilateral	From meeting midpoint
Scalene quadrilateral	Parallelogram
Parallelogram	Parallelogram
Square	Square
Rectangle	Rhombus
Rhombus	Rectangle

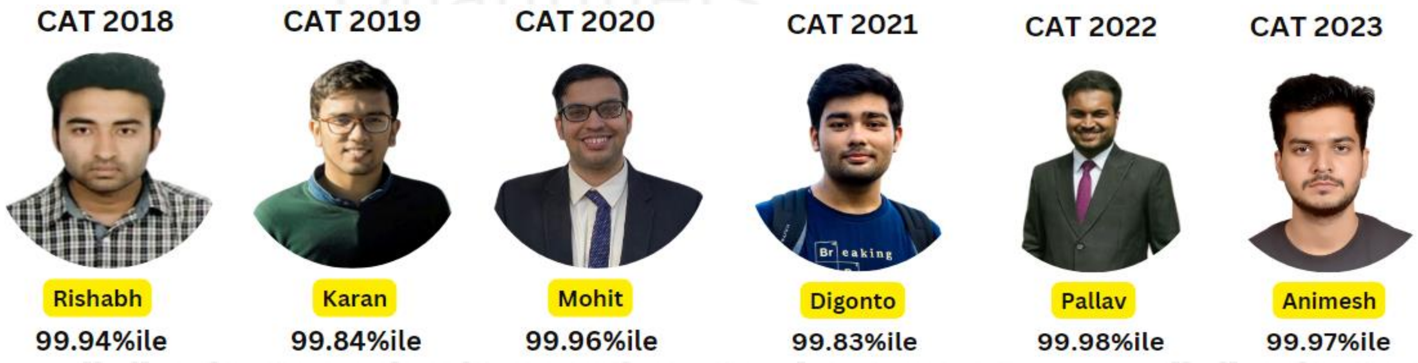


IN ANY QUADILATERAL

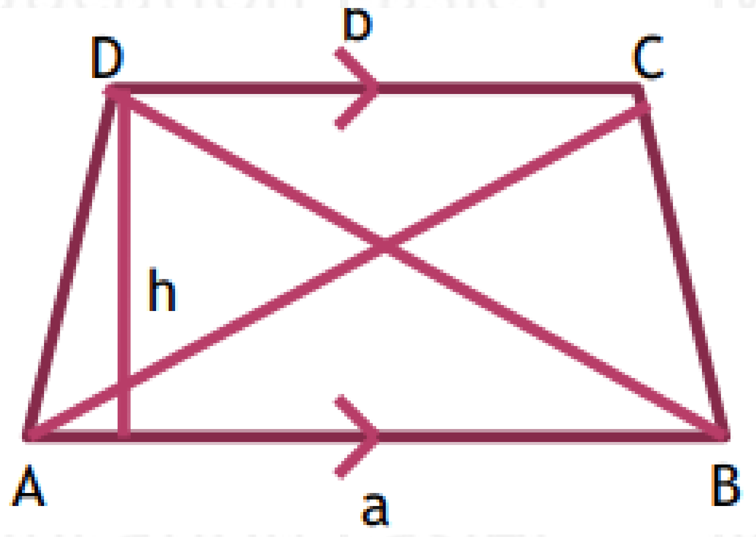
x, y, z and w are area of triangles

X*y=z*w





TRAPEZIUM



One pair of opposite side are parallel.

AB||CD

Area = 1/2 *sum of parallel lines* × *disance between the*

$$A = \frac{1}{2}(a+b) \times h$$

$$AC \neq BD$$

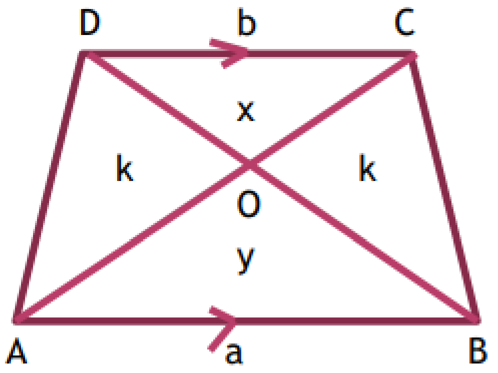
CONCEPT RELATED TO TRAPEZIUM

$$\Delta AOB \approx \Delta COD$$

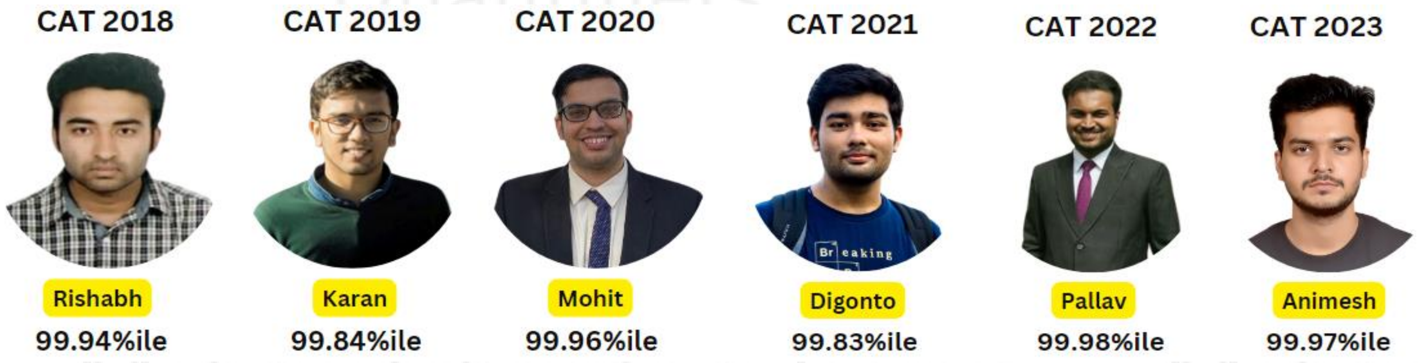
$$ar\Delta AOD = ar\Delta BOC$$

$$ar\Delta AOD = ar\Delta BOC$$

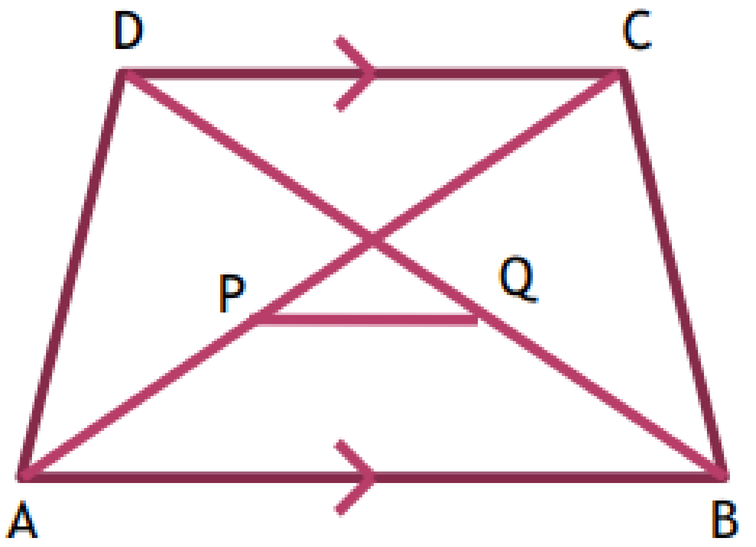
$$k = \sqrt{xy}$$



$$\Delta AOB : \Delta COD = (a \times a) : (b \times b)$$

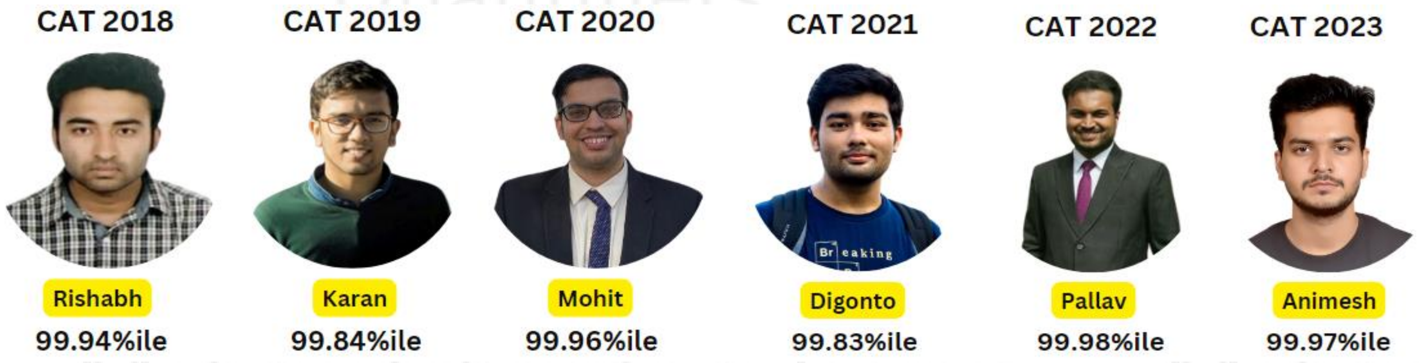


MIDPOINT OF DIAGONAL OF TRAPEZIUM



P and Q are midpoint of diagonal AC and BD.

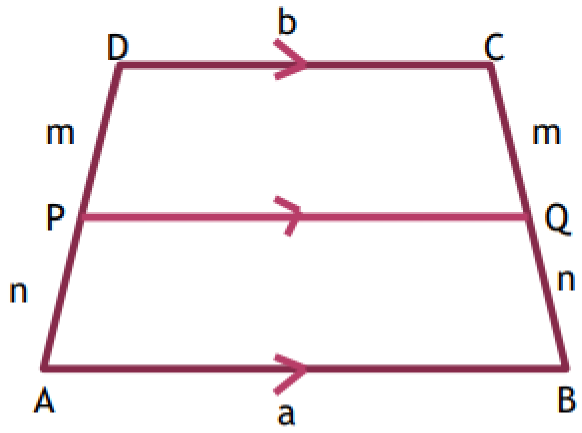
$$\boxed{PQ = \frac{a-b}{2}}$$

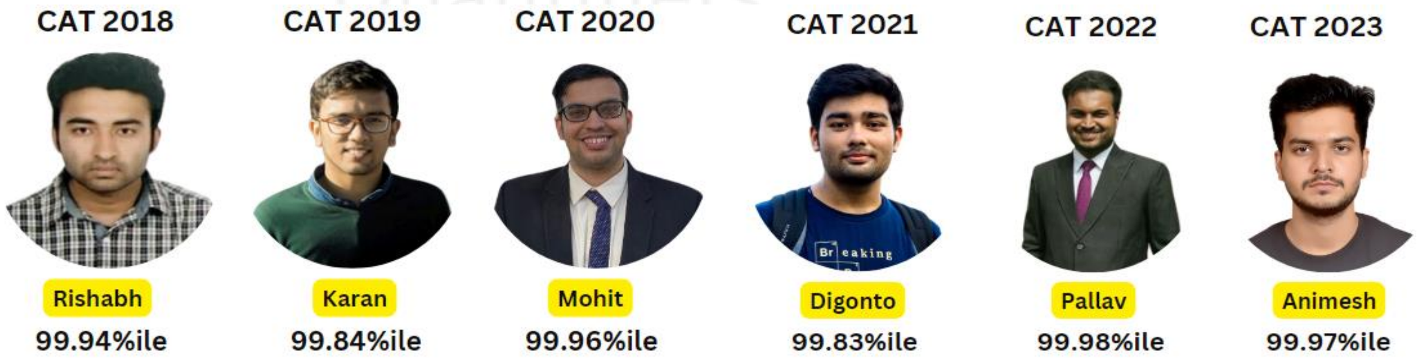


IMPORTANT CONCEPT OF TRAPEZIUM

PQ is such that PQ||AB||CD

$$PQ = \frac{ma+nb}{m+n}$$

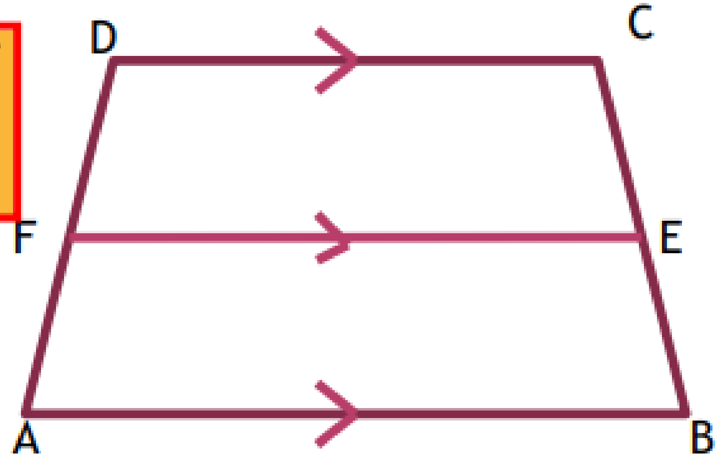


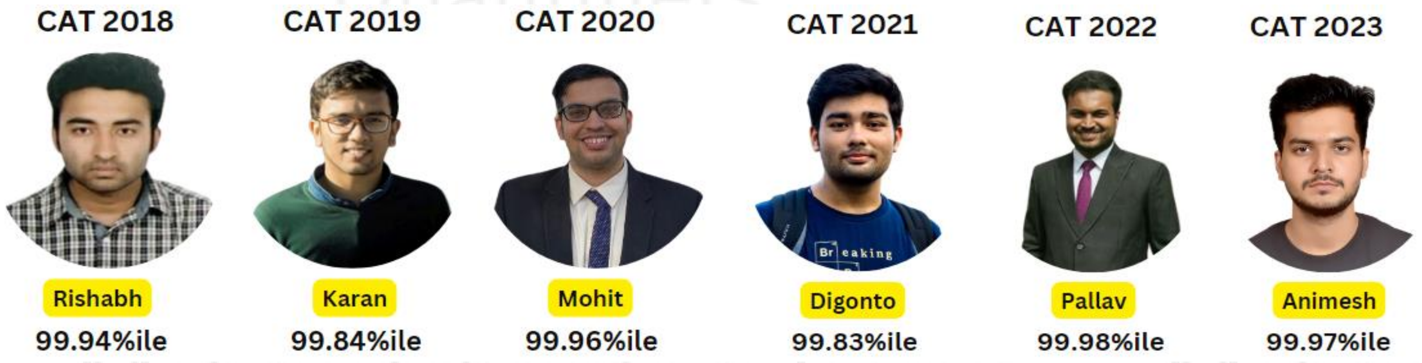


TRAPEZIUM

EF is drawn such that AB||CD||EF and ar(EFBA) = ar(EFCD)

$$\boxed{EF = \sqrt{\frac{a \times a + b \times b}{2}}}$$



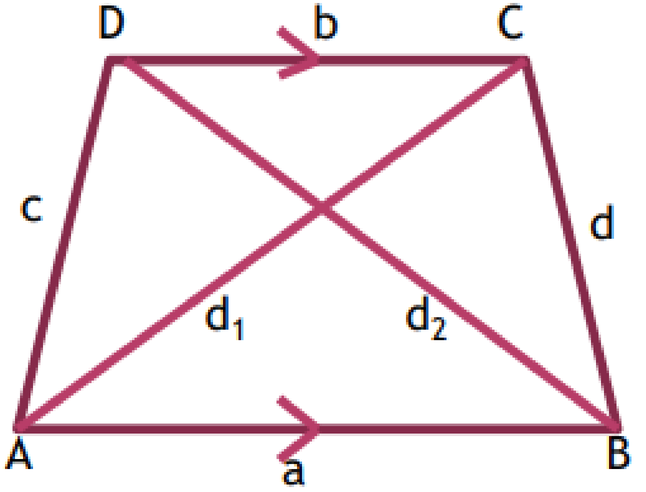


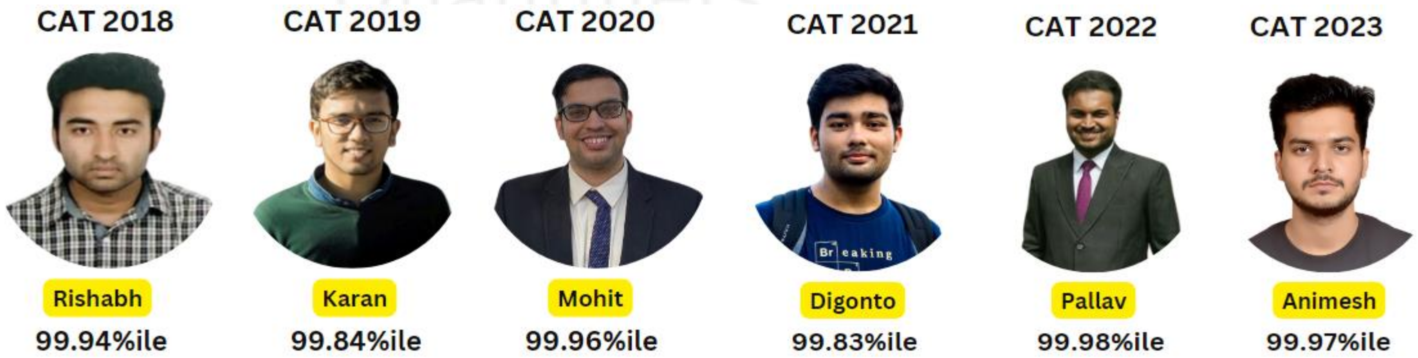
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

DIAGONAL OF TRAPEZIUM

d1 and d2 are diagonal of trapezium.

$$d_1^2+d_2^2=c^2 + d^2 +2ab$$





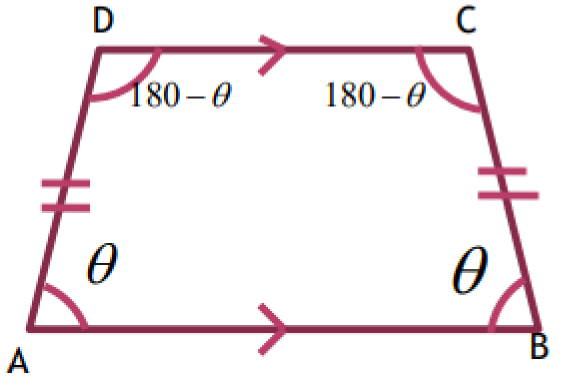
ISOSCELES TRAPEZIUM

If AB||CD and AD = BC, then ABCD is isosceles trapezium.

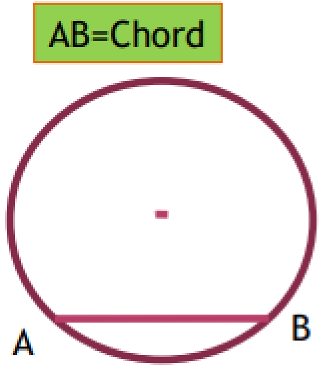
Each isosceles trapezium is a cyclic quadrilateral.

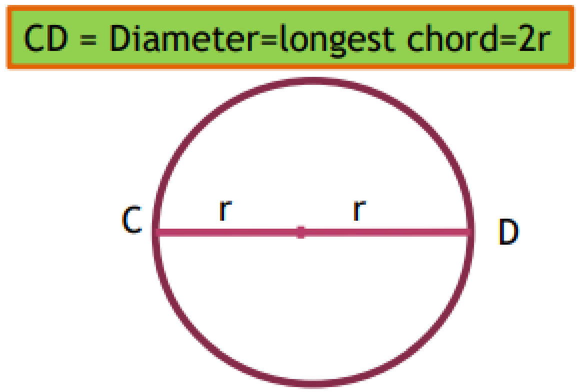
Trapezium inside a circle is always isosceles trapezium

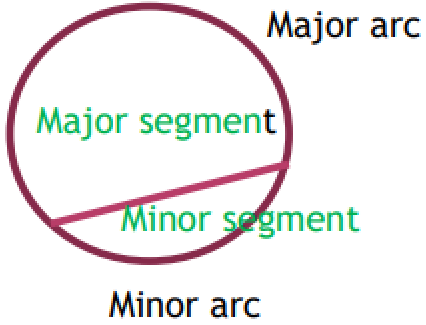
$$\angle A = \angle B, \angle C = \angle D$$
$$\angle A + \angle C = \angle B + \angle D = 180^\circ$$
$$AC = BD$$

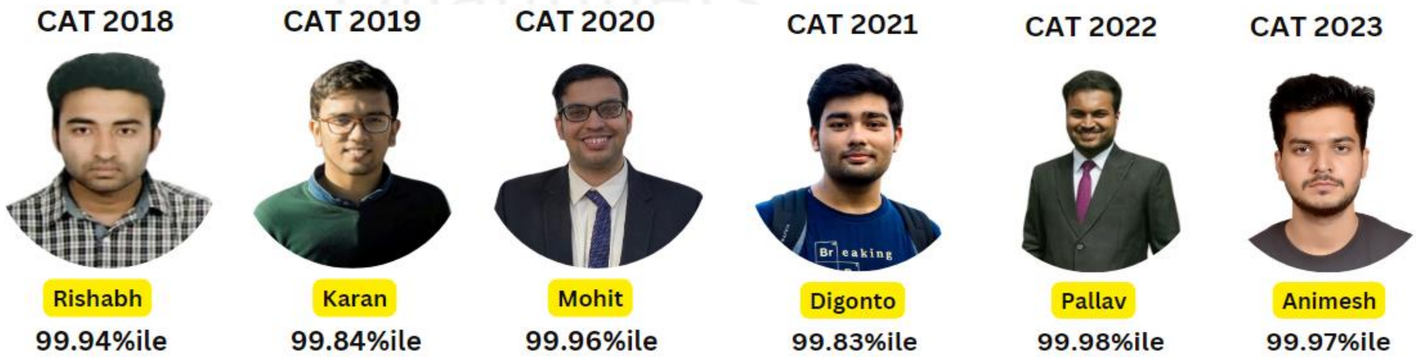


CHORD AND LONGEST CHORD







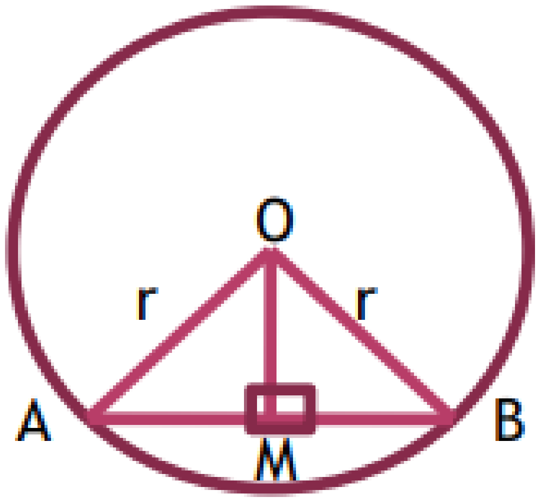


Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

CONCEPT OF CIRCLE

If AM = MB, then find OM perpendicular to AB

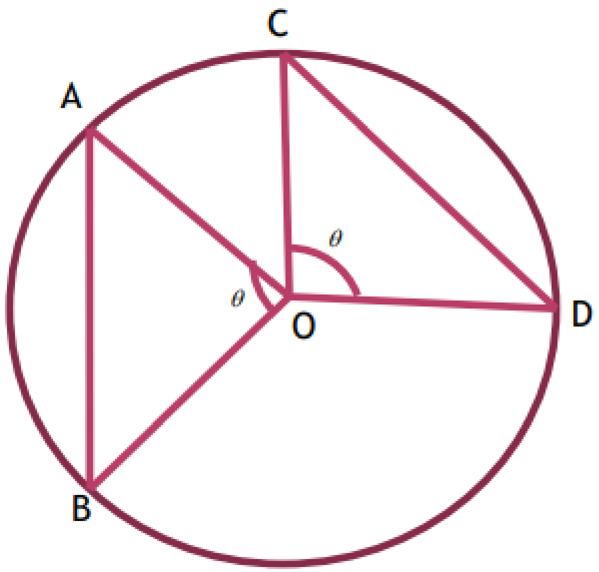
If OM perpendicular to AB, then AM = MB.



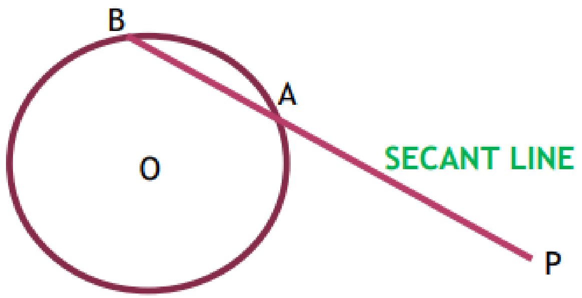
CIRCLE

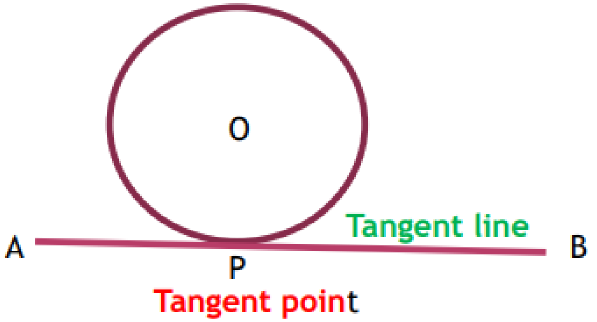
If AB = CD, then

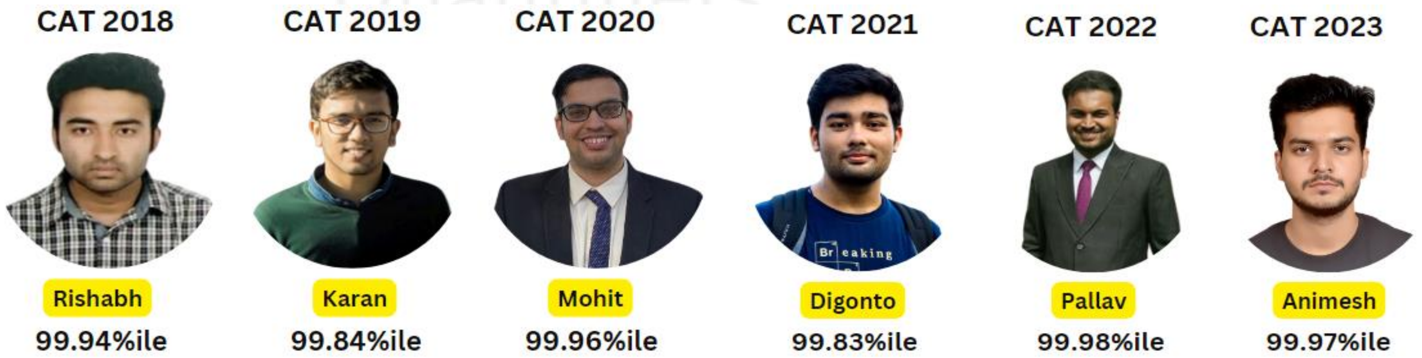
$\angle AOB = \angle COD$



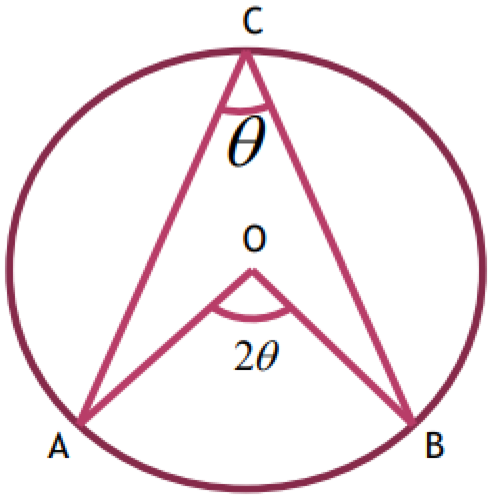
SECANT AND TANGENT

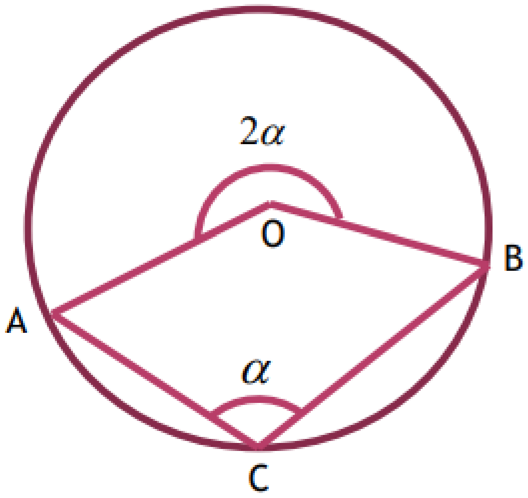


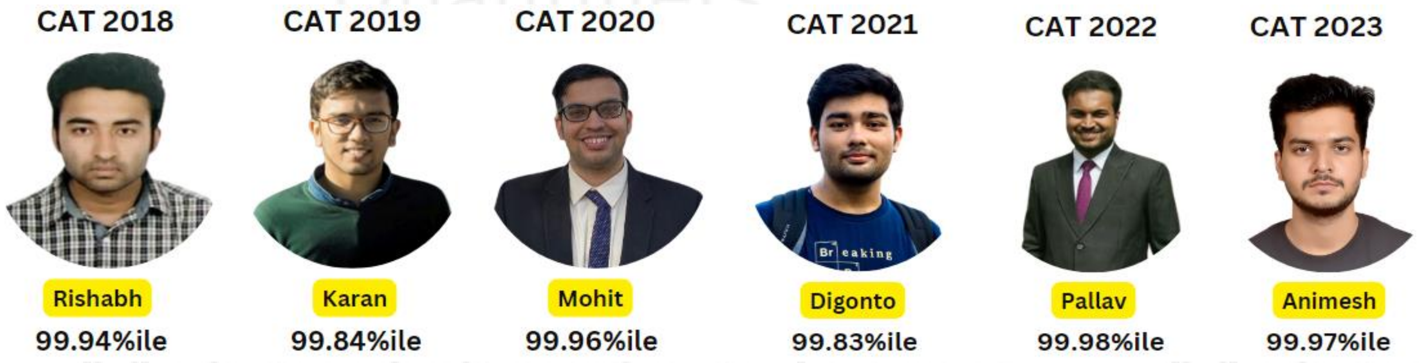




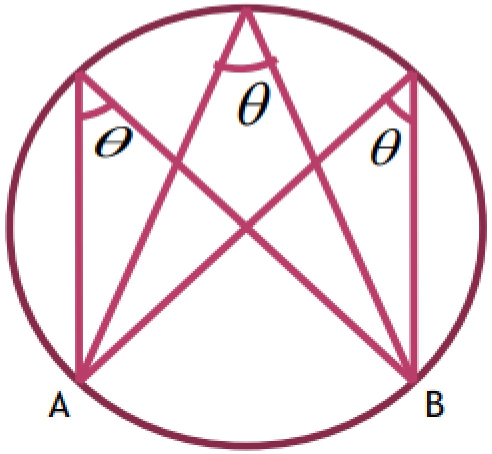
ANGLE RELATED IN CIRCLE

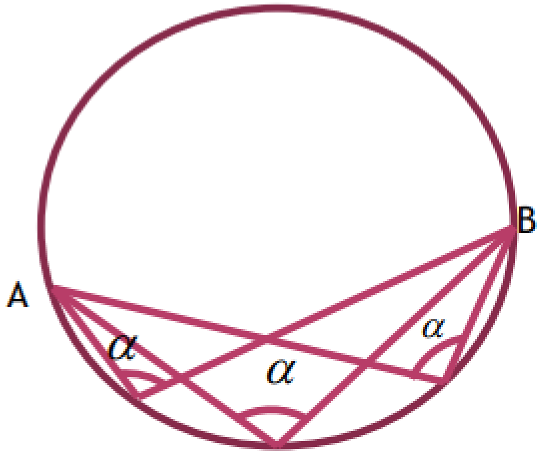




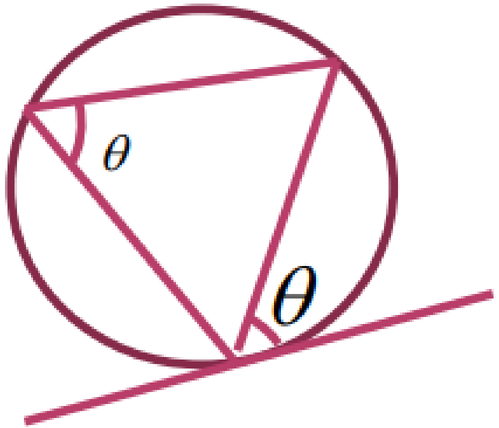


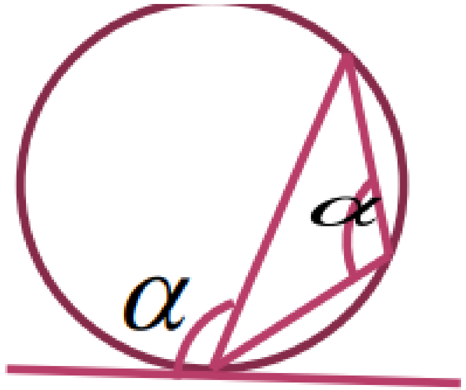
ANGLES FROM SAME ARC

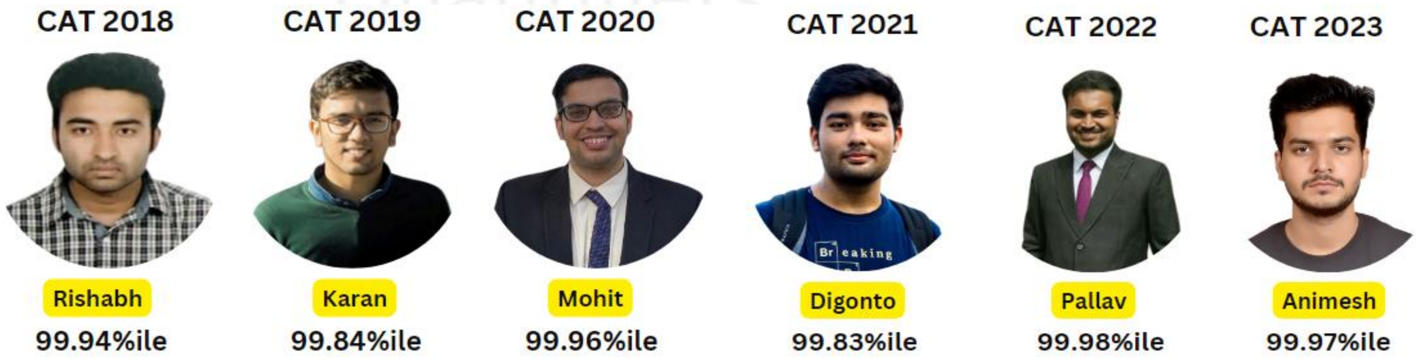




ALTERNATE SEGMENT THEOREM





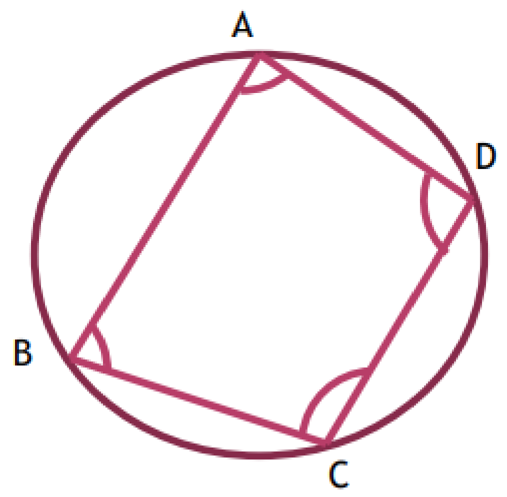


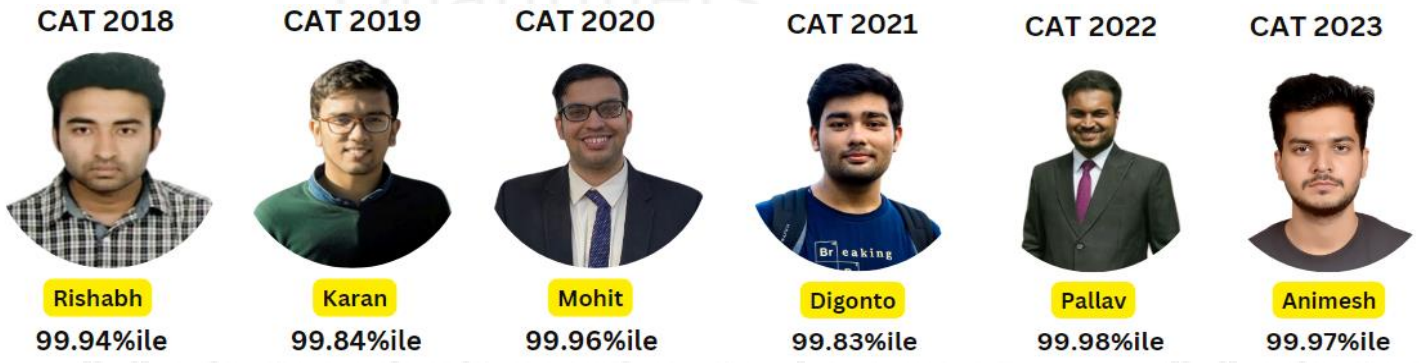
Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

CYCLIC QUADRILATERAL

$$\angle A + \angle C = 180$$

$$\angle B + \angle D = 180$$





Riya - IIM Rh

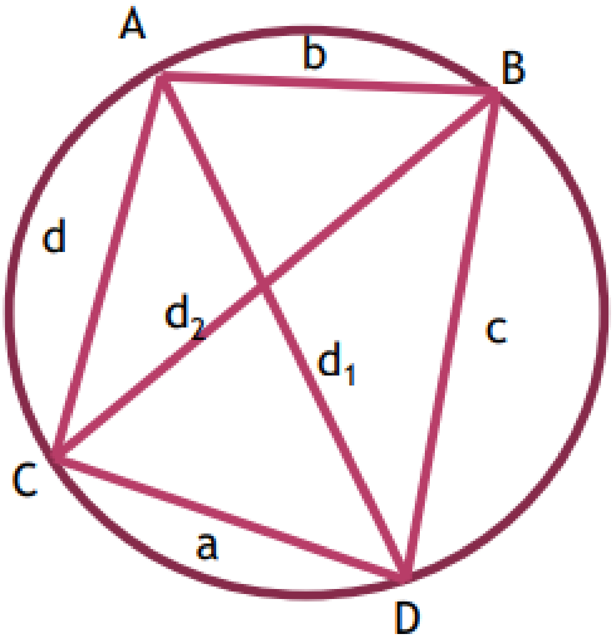
**Riya made it to
IIM Rohtak with
Quantifiers CAT Academy
So can you!!!!**

Niharika - MDI Gurgaon

**Niharika made it to
MDI Gurgaon with Quantifiers
CAT Academy
So can you!!!**

PTOLMEY'S THEOREM

$$ab + cd = d_1 d_2$$



CAT 2018	CAT 2019	CAT 2020	CAT 2021	CAT 2022	CAT 2023
Rishabh	Karan	Mohit	Digonto	Pallav	Animesh
99.94%ile	99.84%ile	99.96%ile	99.83%ile	99.98%ile	99.97%ile

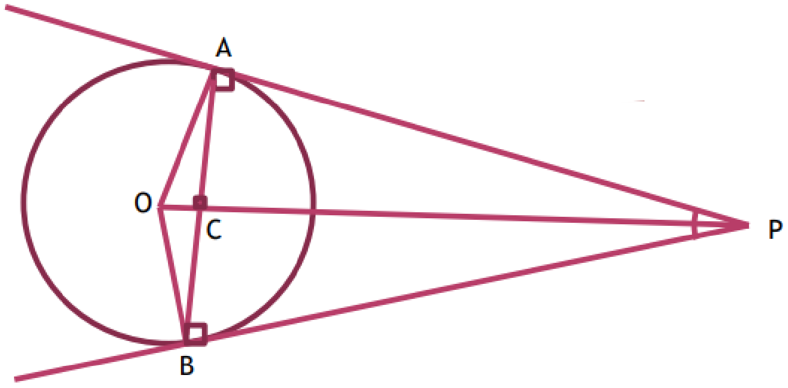
TANGENT OF CIRCLE

$\Delta OAP \cong \Delta OBP$

$PA = PB$

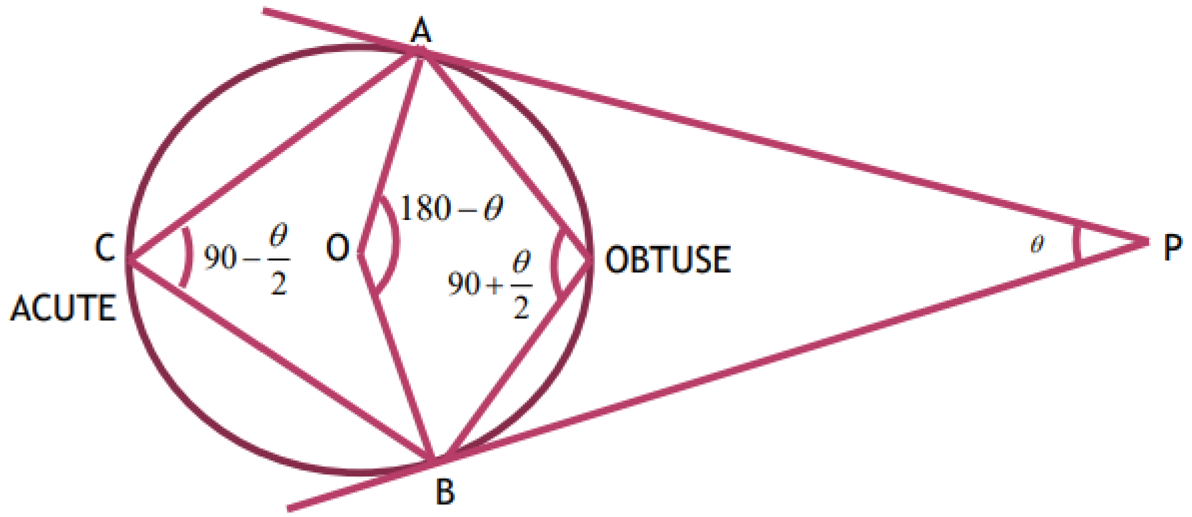
$AC = BC = \dfrac{AB}{2}$

$AC \perp OP$



OAPB is cyclic quadrilateral

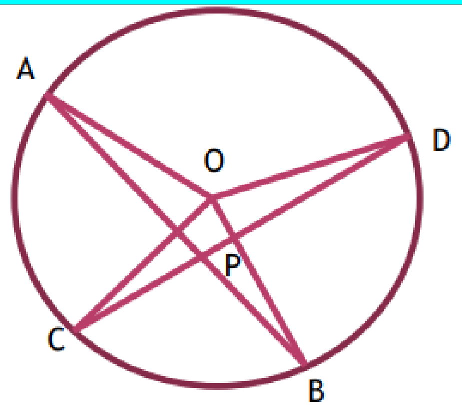
TANGENT CONCEPTS

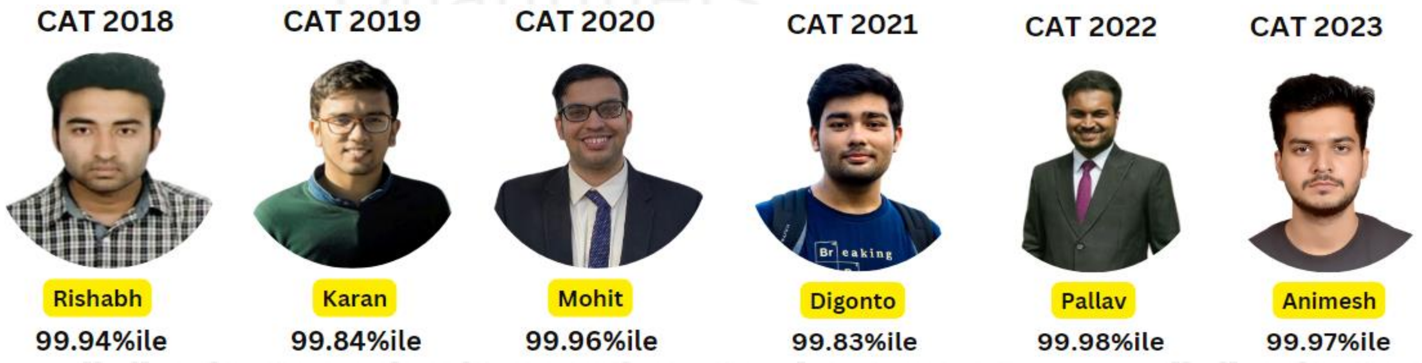


Important formula of Circle

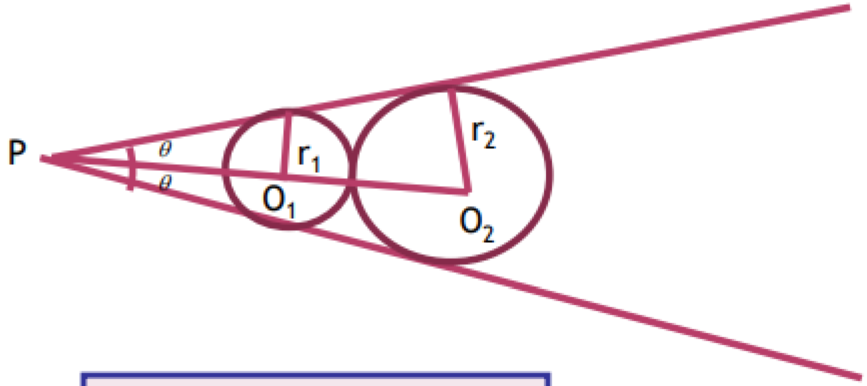
$$\angle APD = \angle BPC = \frac{\angle AOD + \angle BOC}{2}$$

$$\angle APC = 180° - \angle APD$$

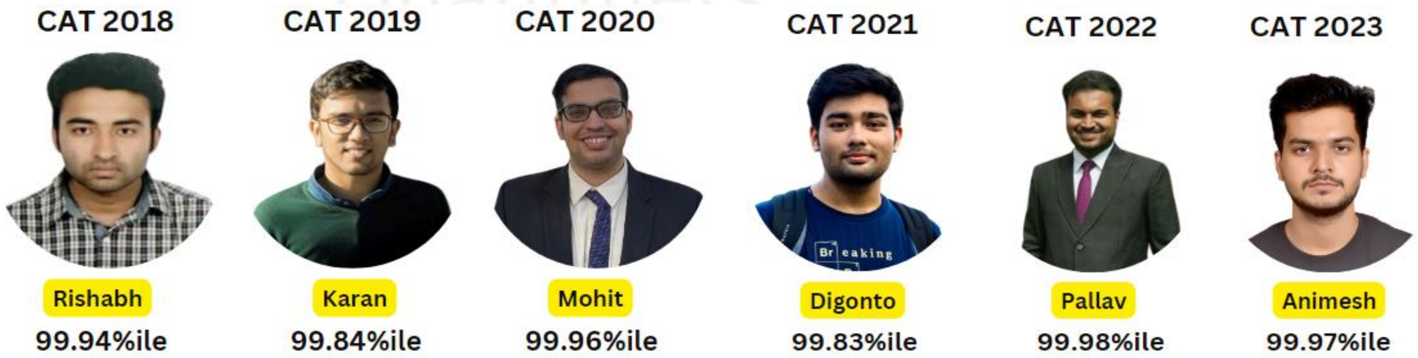




TANGENT

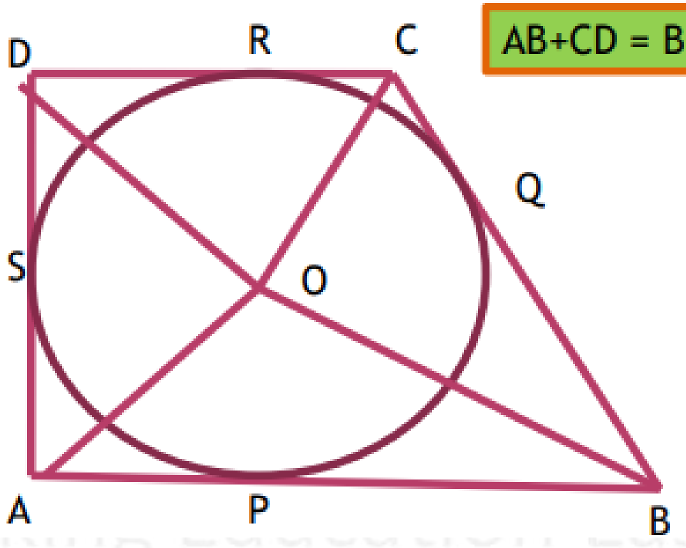


$$\boxed{\frac{r_2}{r_1} = \frac{1+\sin\theta}{1-\sin\theta}}$$

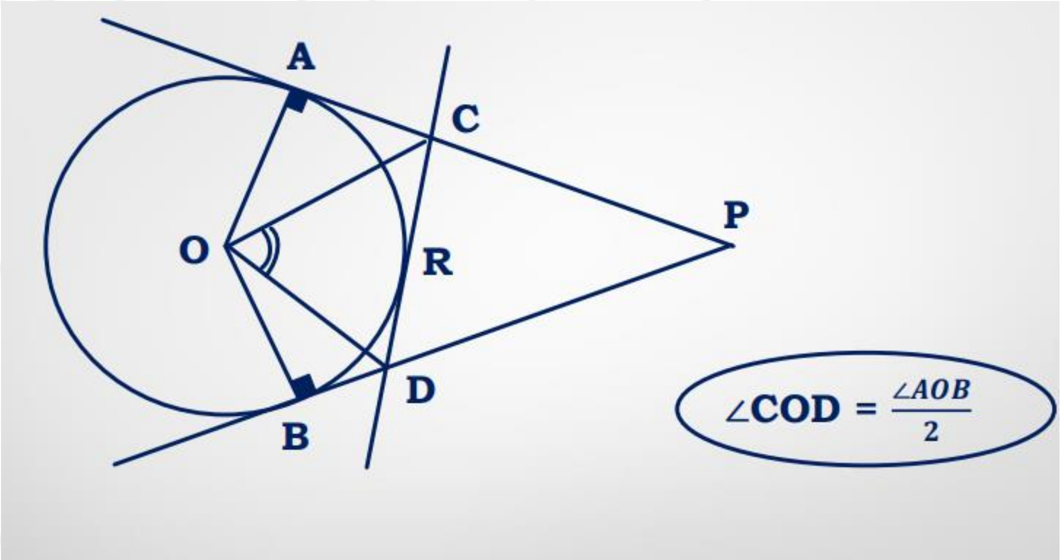


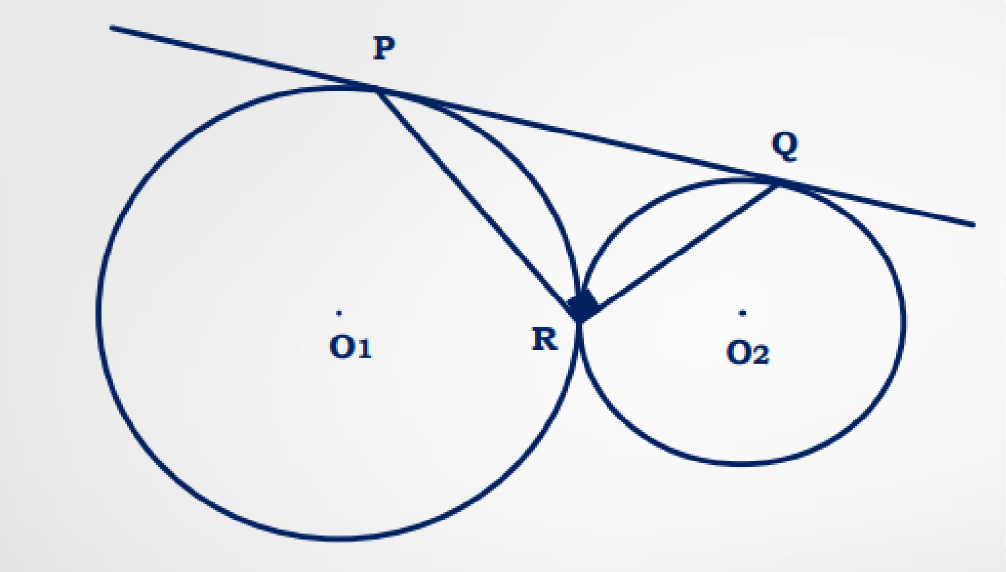
CIRCLE INSCRIBED IN A QUADILATERAL

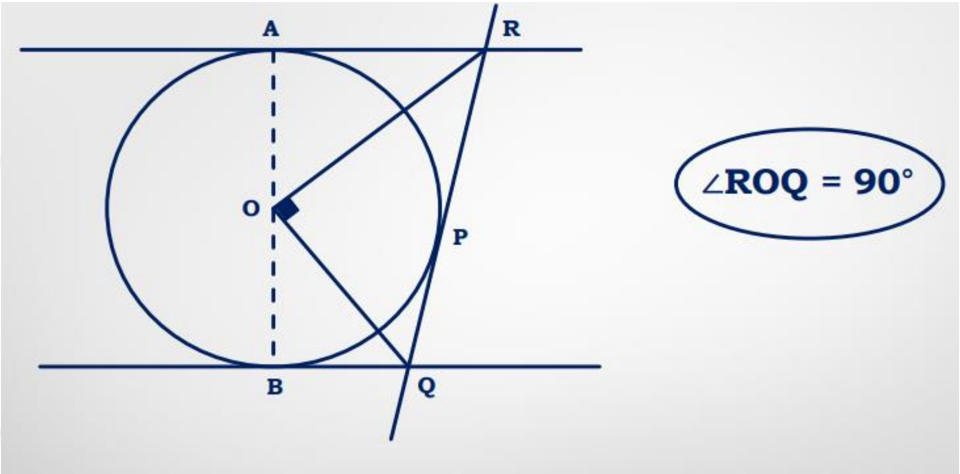
$$\angle AOB + \angle COD = 180°$$
$$\angle BOC + \angle AOD = 180°$$

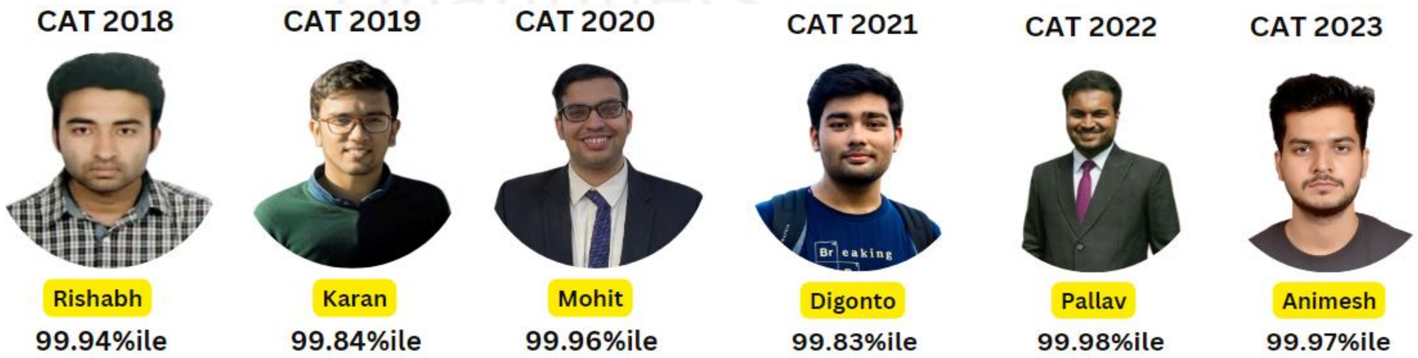


AB+CD = BC +AD

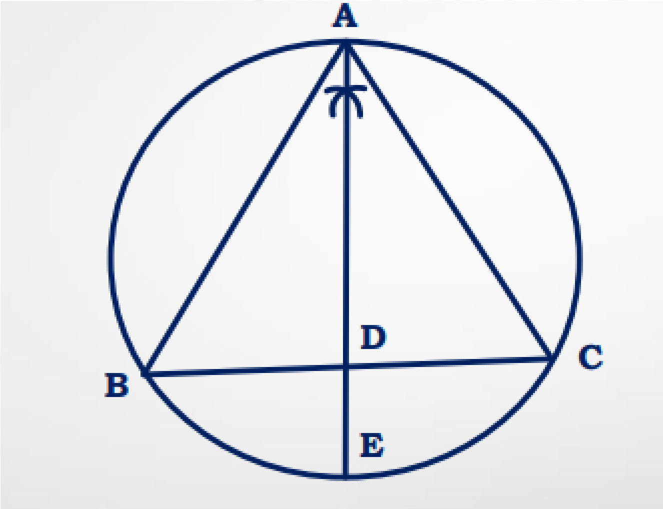

A
C
O
R
P
D
B
$\angle COD = \frac{\angle AOB}{2}$


P
Q
O_1
R
O_2

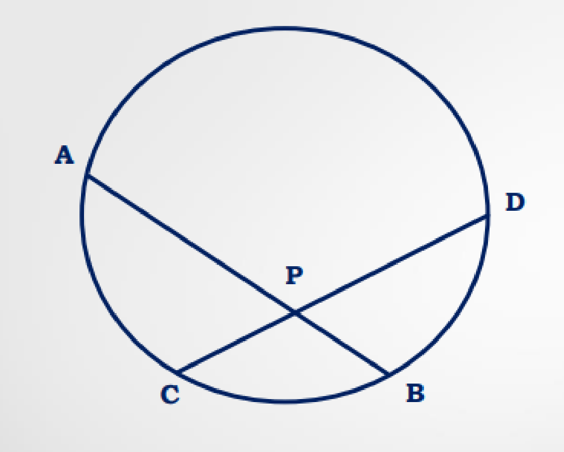

A
R
O
P
B
Q
$\angle ROQ = 90°$


CAT 2018
Rishabh
99.94%ile
CAT 2019
Karan
99.84%ile
CAT 2020
Mohit
99.96%ile
CAT 2021
Digonto
99.83%ile
CAT 2022
Pallav
99.98%ile
CAT 2023
Animesh
99.97%ile

$AB \times AC + AE \times DE = AE^2$

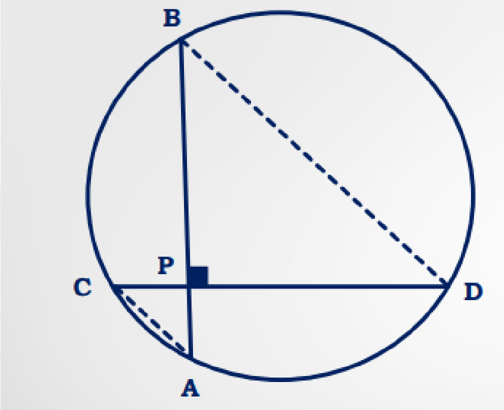


$PA \times PB = PC \times PD$

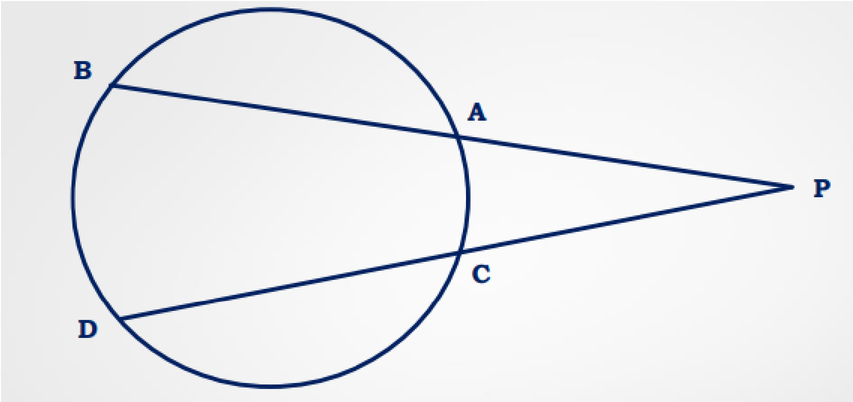


If AC = x and BD = y Then,

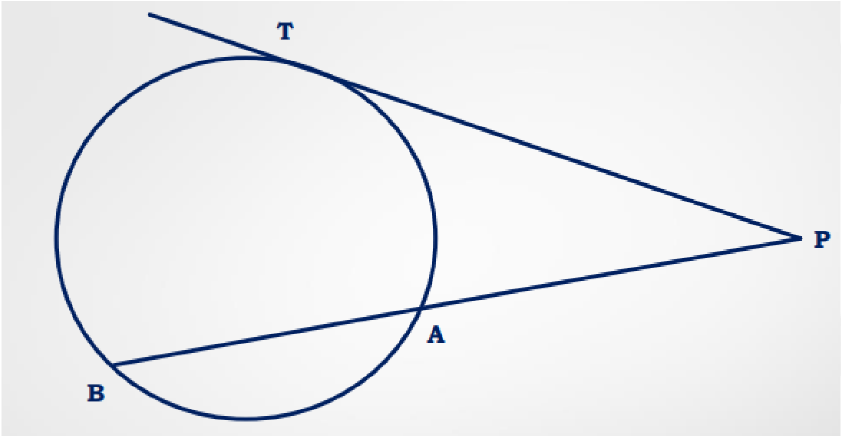
Radius of circle (r) $= \frac{\sqrt{x^2+y^2}}{2}$

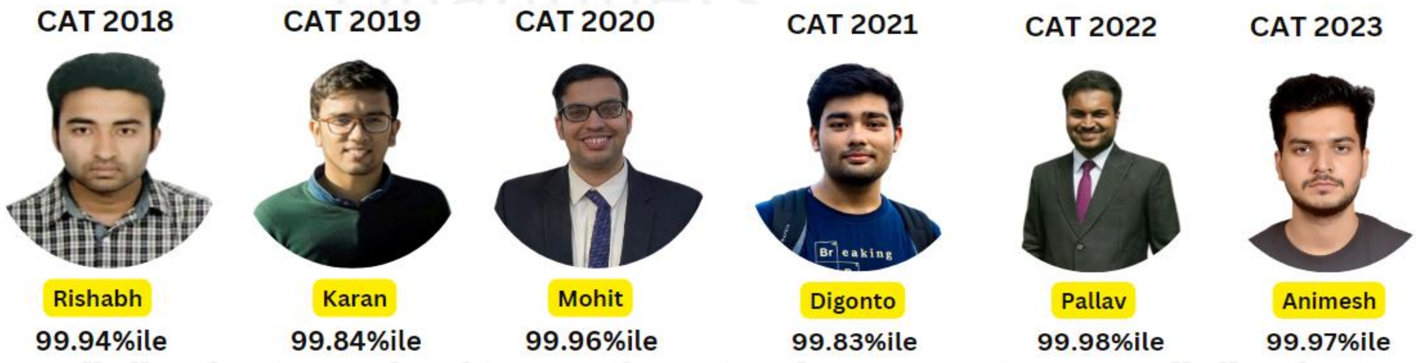


PA × PB = PC × PD



PT^2 = PA × PB





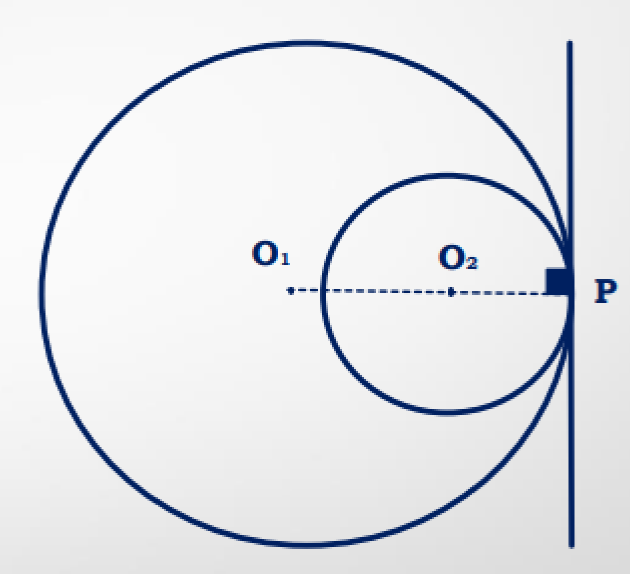
Maximum common tangent of both the circles is 1

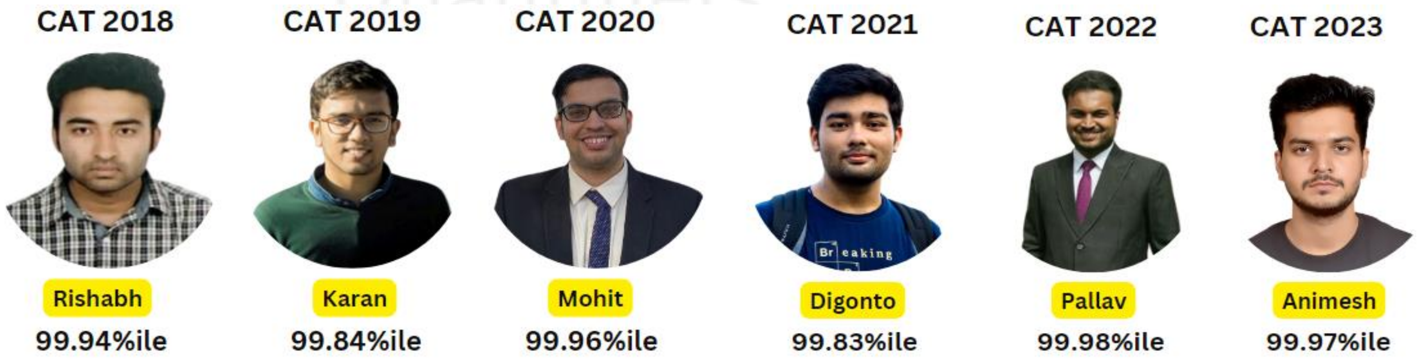
O_1O_2P will always be a straight line

$O_1P = r_1$

$O_2P = r_2$

$O_1O_2 = d = r_1 - r_2$



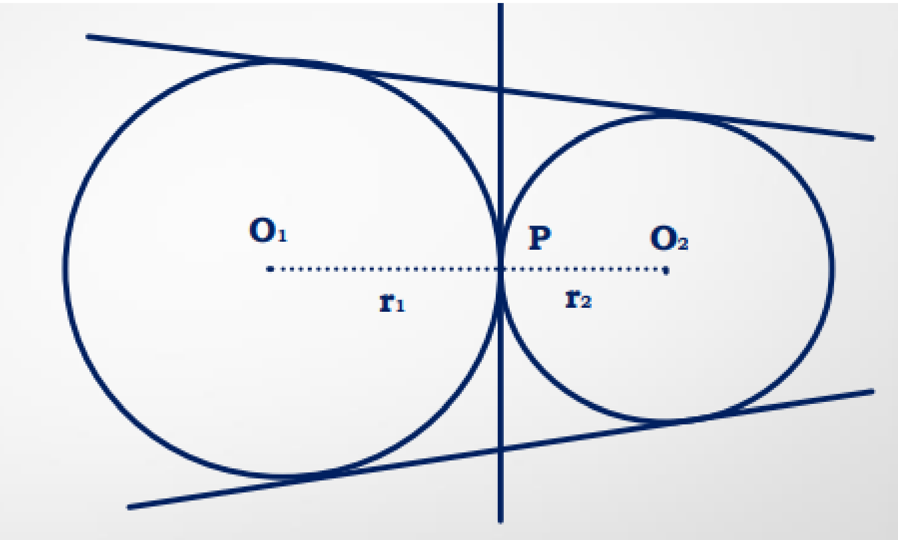


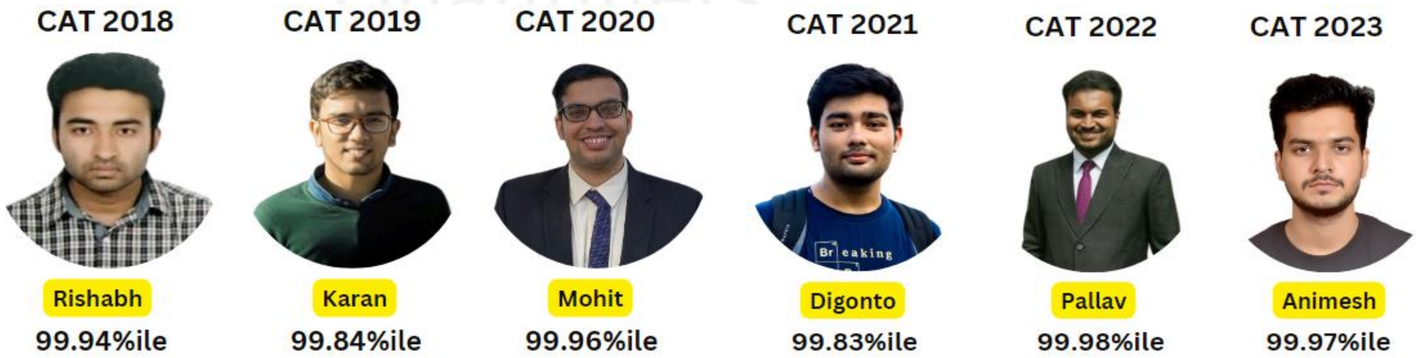
Maximum common tangent of both the circles is 3

O_1PO_2 will always be a straight line

$$O_1O_2 = O_1P + O_2P$$

$$d = r_1 + r_2$$





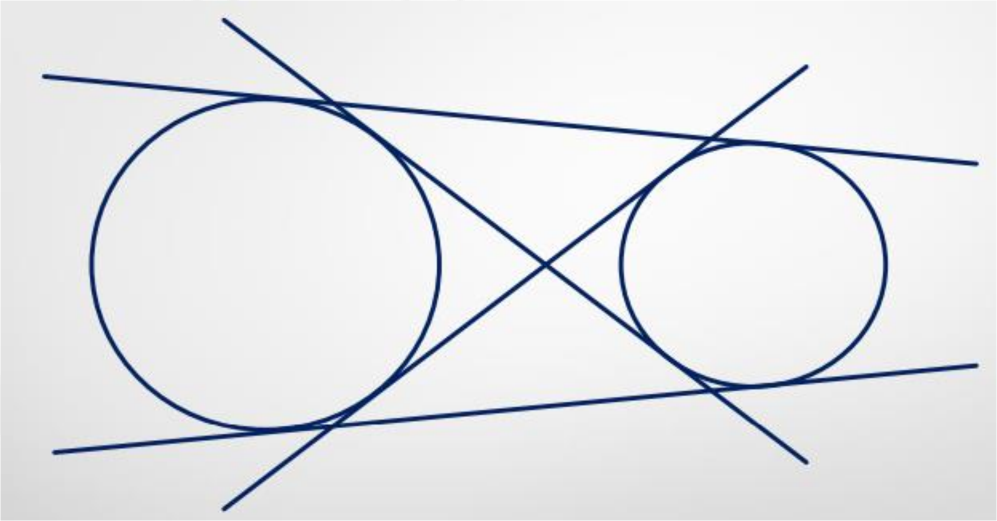
Akshita- MDI Gurgaon

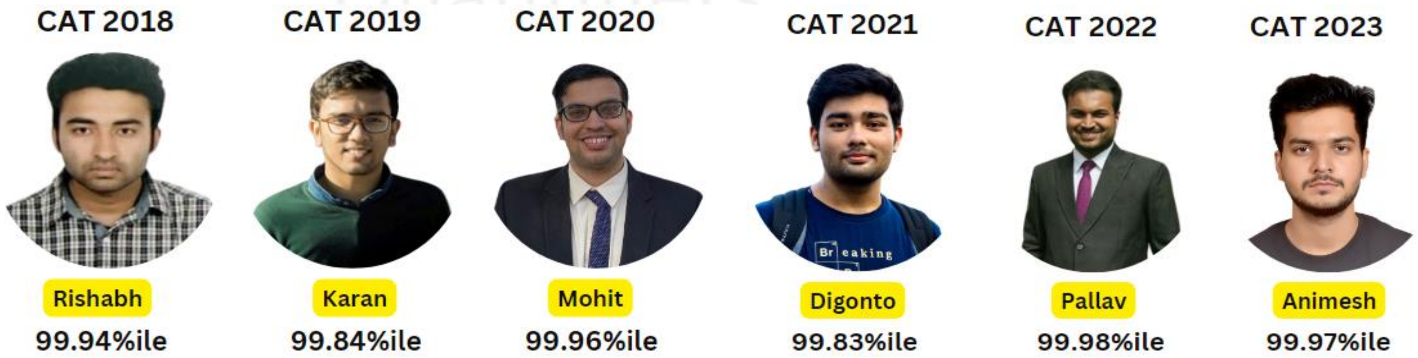
Akshita (98.85%ile) made it to MDI Gurgaon
So can you!!!!

Manali - IIM C

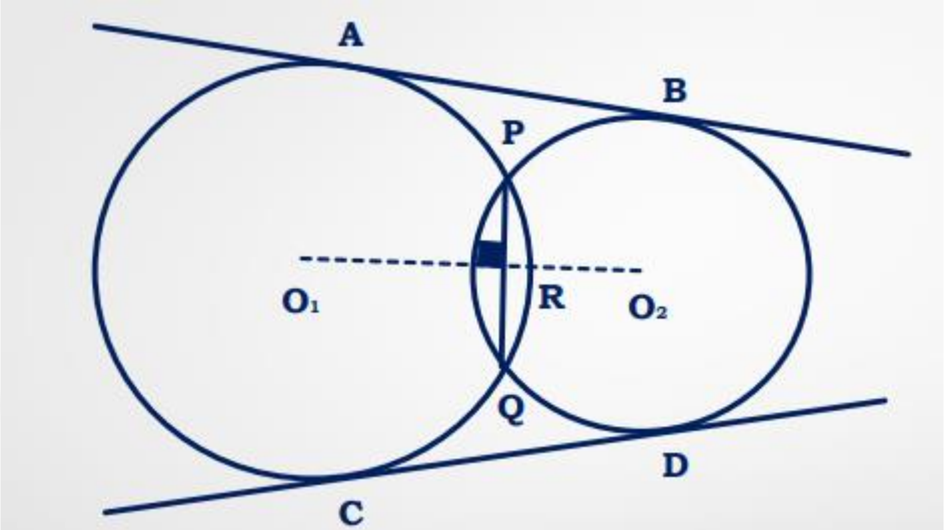
Manali (99.6%ile) made it to IIM Calcutta with Quantifiers CAT Academy
So can you!!!!

Maximum common tangent of both the circles is 4





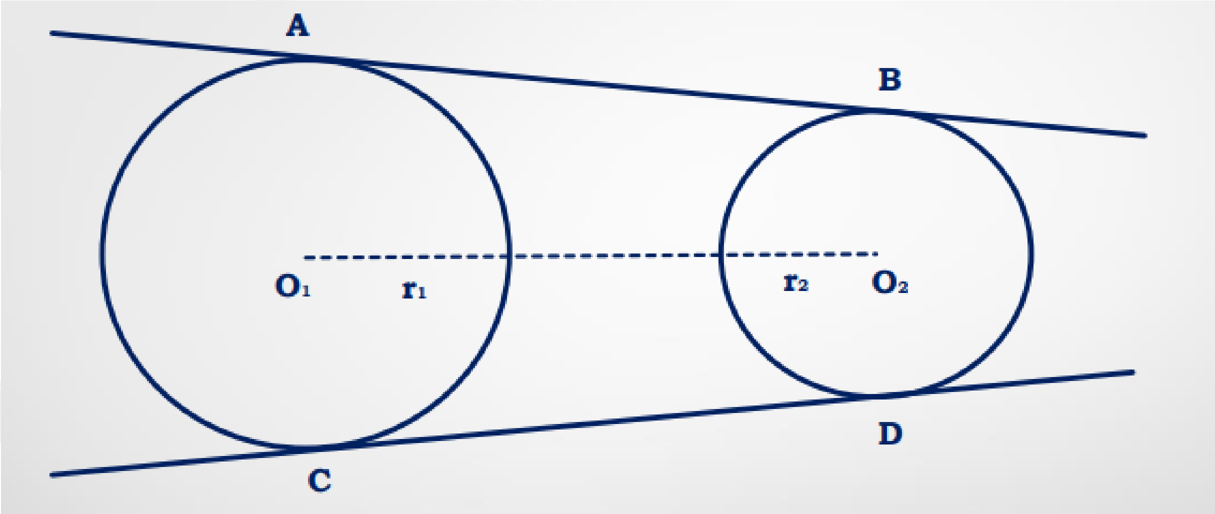
Maximum common tangent of both the circles is 2



AB and CD are direct common tangent
PQ = Common Chord

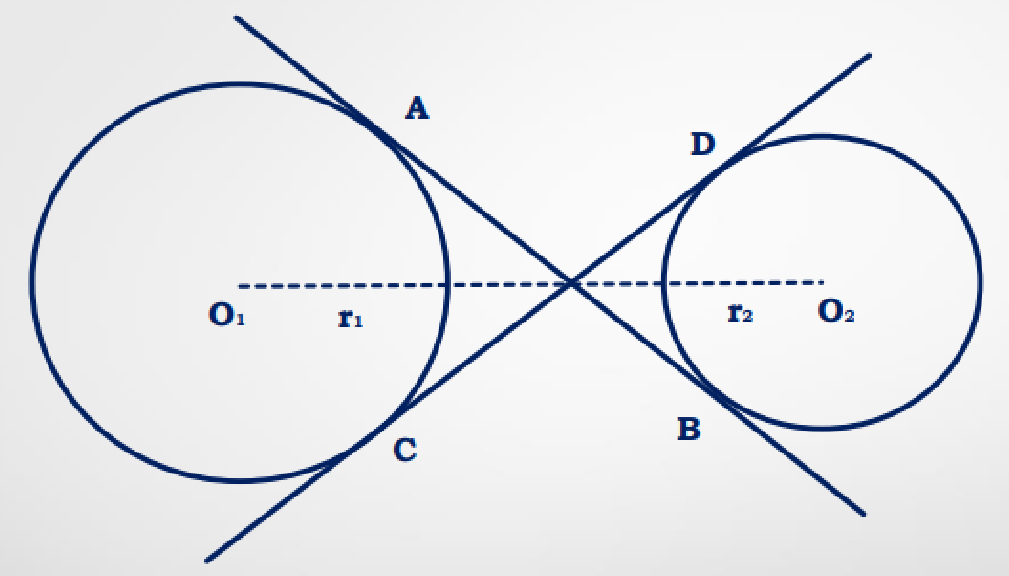
PR = QR = $PQ/2$

DIRECT COMMON TANGENT



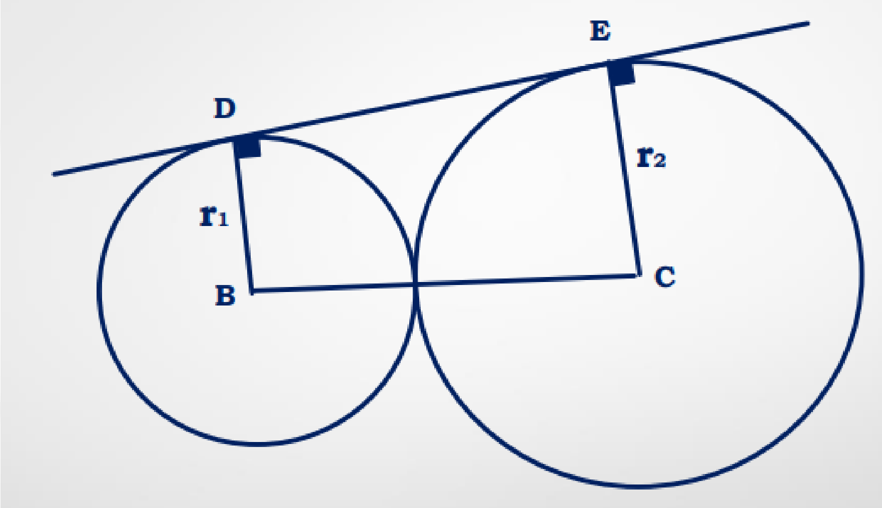
$$AB = CD = \sqrt{d^2 - (r_1 - r_2)^2}$$

TRANSVERSE COMMON TANGENT

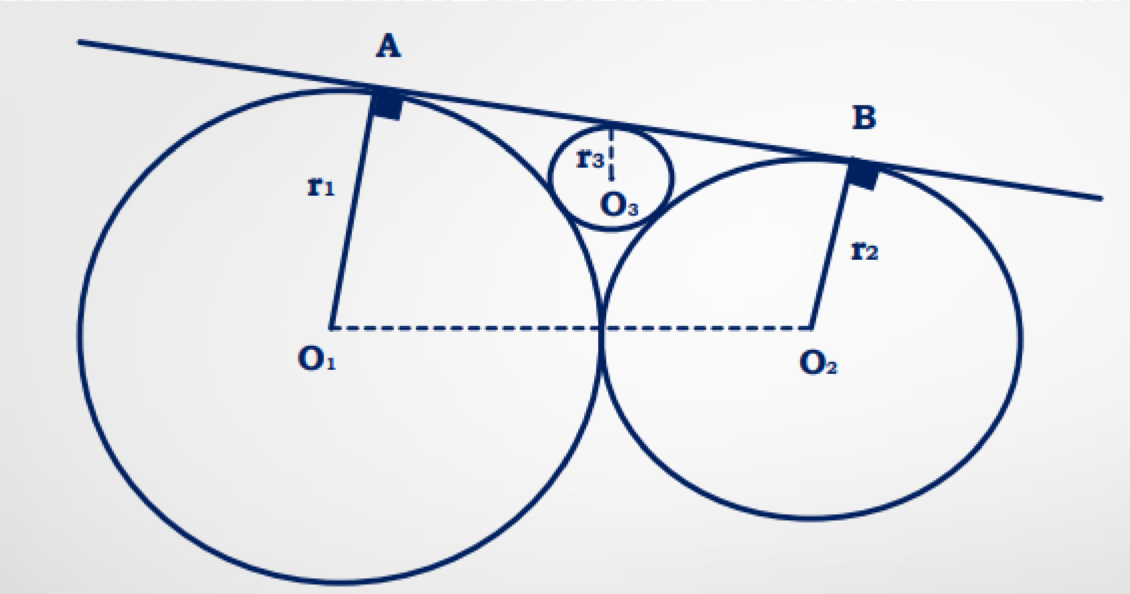


$$AB = CD = \sqrt{d^2 - (r_1 + r_2)^2}$$

DIRECT COMMON TANGENT



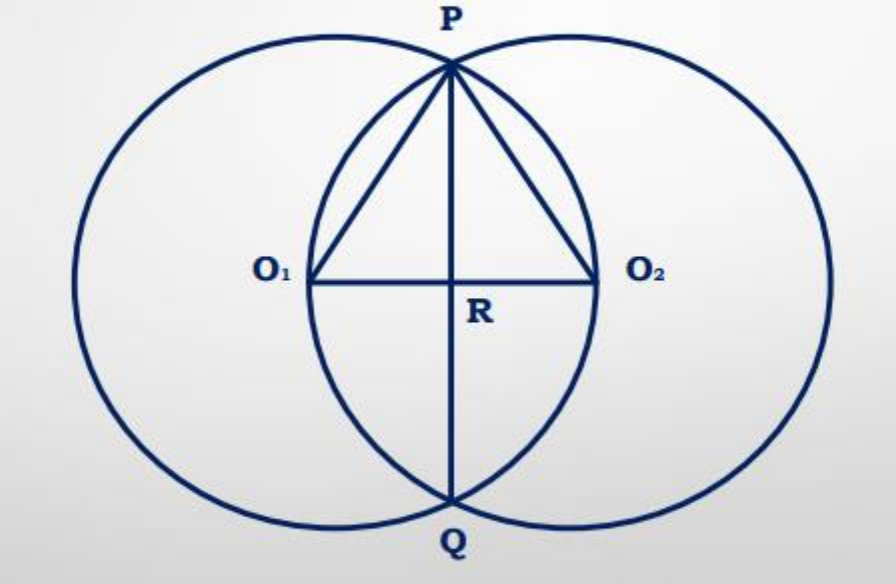
$$DE = 2\sqrt{r_1 r_2}$$

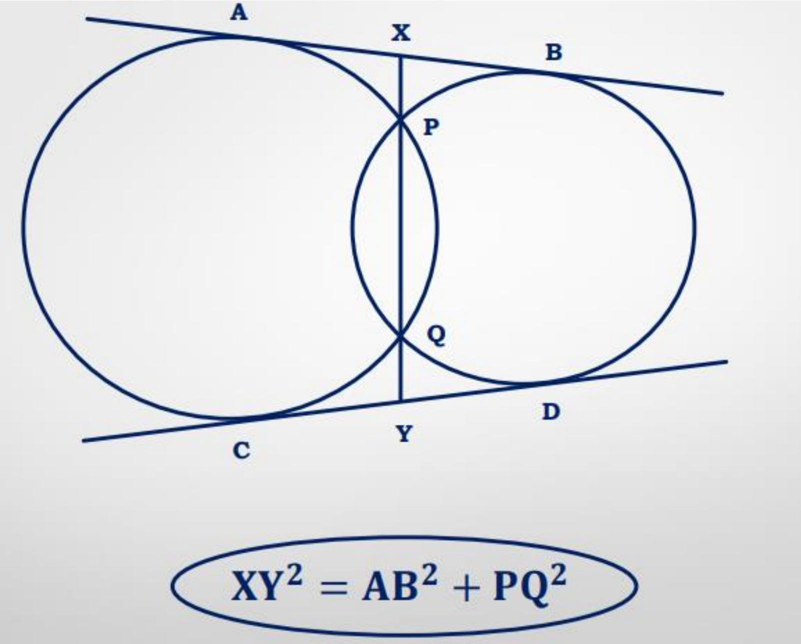


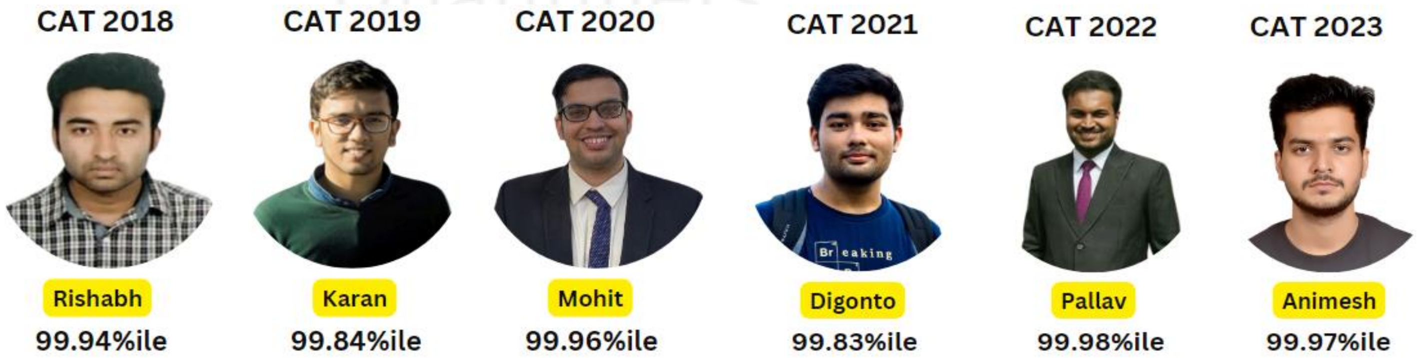
$$\frac{1}{\sqrt{r}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$$

COMMON CHORD

When two circles pass through each others centers (r1 = r2 = r3)
Then, Common Chord = $\sqrt{3}r$







Quantifiers CAT Academy - NEW CAT BATCHES STARTING SOON || Call/WA – 9988656560/7986961408

Thank you and All the best for CAT 2024

☺

Get Daily Targets on your WhatsApp
or Copy paste this URL: https://forms.gle/oSih8jQCermJq71q9

Join our WhatsApp FREE 24/7 Doubt Solving Group (if group is full, follow us on Instagram @quantifiers_cat for the new link)
Click here to join WhatsApp group
https://chat.whatsapp.com/EX8fGeD7h0zAhs9sDa5eeA